

Discipline at Work: Wage Premia, Managers, and Shirking in a Large Mexican Retailer

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Abstract

Inventory shrinkage—merchandise missing from the store—accounts for about 1.6 percent of sales for United States retailers, on the order of the industry’s net profit margin. Yet its drivers are not well understood. Partnering with a large Mexican retailer—300 stores and 25,000 workers—, I show that shrinkage is a direct, firm-wide measure of on-the-job shirking.

In line with the canonical shirking model, I study whether wage premia and managerial monitoring deter shirking as complements. Leveraging general-manager rotations across stores, I find that the arrival of a good manager reduces shrinkage by about 7 percent on average. The same arrival, however, reduces shrinkage by close to 12 percent where the premium is high—eight percentage points more than where it is low. A model in which the premium and monitoring are complements can match these facts: where the premium gives the manager’s dismissal threat something to confiscate, the same manager accomplishes more.

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I document that good managers reduce shrinkage by spending more time on the store floor, by matching their schedules to those of their workers, and by keeping a leaner stock.

1 Introduction

Inventory shrinkage—merchandise that goes missing from stores—is one of the largest sources of revenue loss in retail. In the United States it reached \$112 billion in 2022, about 1.6 percent of sales (National Retail Federation, 2023), on the order of the industry’s net profit margin; worldwide, the most recent global survey puts it at a similar 1.8 percent of sales, roughly \$100 billion a year (Tyco Retail Solutions and PlanetRetail RNG, 2018). Retailers treat shrinkage mainly as a security problem: they lock merchandise behind glass, expand video surveillance, and increase loss-prevention staffing and budgets (National Retail Federation, 2023; Selyukh, 2024). Some chains have closed stores over it. Yet the drivers of shrinkage are not well understood: despite its size, there is no systematic analysis to guide either policy or business strategy.

The evidence retailers themselves produce points inward. In the industry’s own security survey, retailers attribute more than half of their shrink—56 percent—to internal causes, their own staff and processes, against 36 percent to outside actors (National Retail Federation, 2023). The nature of the work says the same: guarding and managing inventory is the quintessential task of a supermarket job—in my setting, it appears in 92 percent of workers’ job descriptions. Yet shrinkage is rarely analyzed as a personnel issue.

Two obstacles stand in the way. The first is data: retailers rarely record shrinkage systematically across stores and over time, and where they do, it stays an accounting line, never linked to the workers and managers behind it. The second is a framework. Caring for merchandise is effort that no supervisor observes directly, and a lapse costs the worker something only if it is detected—exactly the structure of the canonical shirking model (Shapiro and Stiglitz, 1984). I therefore model shrinkage as the observable residue of on-the-job shirking. In the model, two instruments deter shirking: a wage premium, which gives the worker a rent to lose if she is caught and dismissed, and monitoring, which sets the probability that she is caught. This paper asks whether these two instruments in fact deter shrinkage and, if so, whether they work as complements or substitutes.

This paper answers these questions with administrative records from a large Mexican retailer—more than 300 stores and roughly 25,000 workers in a typical month over 2022–2024. The records measure all three ingredients at once: shrinkage audited weekly in every store, each worker’s pay relative to her local outside wage, and, from biometric time clocks, what managers and workers actually do.

Shrinkage responds to the premium: a 10 percent lower wage premium is associated with about 1.1 percent more shrinkage, and with more absenteeism and turnover. Because headquarters sets pay centrally by position, a worker’s premium varies with her local labor

market rather than with her store’s performance. Shrinkage also responds to management: leveraging the firm’s routine rotation of general managers across stores, I find that the arrival of a good manager reduces shrinkage by close to 7 percent on average. The two instruments reinforce each other: the same arrival reduces shrinkage by close to 12 percent where the premium is high—eight percentage points more than where it is low. A model in which the premium and monitoring are complements can match these facts: where the premium gives the manager’s dismissal threat something to confiscate, the same manager accomplishes more. Finally, the high-frequency data show what the managers who reduce shrinkage do: they spend more time on the store floor, match their schedules to their workers’, and keep a leaner store; they do not expand the store’s surveillance apparatus.

I present a model that adapts the canonical shirking framework to the shop floor. A worker chooses how much care to supply—the hidden effort that keeps merchandise from going missing—and care is costly. The firm pays a fixed wage by position, so the only formal incentive is dismissal: a worker caught neglecting her job loses the premium she earns over her outside option. Whether she is caught depends on the store’s monitoring, which in this setting is the general manager’s presence on the floor. The deterrent is therefore a product: the probability that a lapse is detected times the rent lost upon dismissal.

The product structure delivers the paper’s three predictions. First, a larger premium disciplines the hidden margin: shrinkage should fall as the rent rises. Second, a better monitor disciplines it too: shrinkage should fall when a better manager arrives. Third—the prediction at the center of the paper—the two instruments are complements: because they enter multiplicatively, a given manager accomplishes more where the premium is high, and a given premium accomplishes more where someone watches the floor. The model also separates margins by observability: attendance is verifiable and detects itself, so the premium moves it everywhere; shrinkage is punished only when detected, so the rent reaches it only through the monitor.

Using the firm’s administrative records and Mexico’s INEGI survey on labor and employment, I build three panels. The first is an employee-month panel covering close to 100,000 employees over 2022–2024, built from the firm’s personnel records. For each employee it records demographics (age, gender, education), the position held at each point in time together with that position’s description, activities, and performance indicators, and wages, from which I measure the premium: I match each employee to Mexico’s national labor-force survey (ENOE) and define her outside option as the average labor income in her cell of gender, education, age, municipality, and quarter; the wage premium is the difference between the wage the firm pays her and that outside wage. The panel also records the discipline margins the analysis uses: attendance stamped to the minute by biometric time clocks, shifts and

hours worked, absenteeism with the reason for each absence, suspensions, and a full history of moves across positions, areas, and stores—promotions, transfers, and separations—with the reason each occurred.

The second is a store-month panel for the firm’s more than 300 stores. It records shrinkage in both value and physical units, sales and customer counts built up from high-frequency scanner data, stock levels, each store’s format and organizational structure, and its location—including the municipality, which links the store to its local labor market, and whether it lies in the northern-border zone with its higher minimum-wage floor. Shrinkage comes from the firm’s own inventory protocol, which counts at three frequencies: a designated worker checks several times a day what is on the shelf against what should be and logs the gap; a weekly count records the difference between the stock the books expect and what is physically found; and a monthly count reconciles the daily and weekly records and provides an independent count. Merchandise that spoils or breaks is logged separately and does not enter shrinkage: the measure counts only merchandise that is actually missing.

The third is a manager-month panel. I focus on store managers—the general manager who runs each store, of whom there is one, or at most three in the largest stores. The panel records which manager runs which store in each month—the assignment histories that identify arrivals, departures, and rotations—along with her demographics, pay, and tenure, and, from the biometric time clocks, when she is physically in the store: her floor presence, including at night and on weekends, and the overlap of her schedule with her workers’ shifts.

I take the model to the data with three designs, one per prediction. The first tests the premium. Because headquarters sets wages centrally by position, a worker’s premium moves when her local outside wage moves, for reasons unrelated to her performance or her store’s. I use this variation in two ways: a two-way fixed-effects analysis that relates store premia to shrinkage within municipality and time, and a shock to the same object from Mexico’s minimum-wage policy. The country sets two minimum-wage zones, one along the northern border and one for the interior, so when the floors rise, border workers lose more of their premium than interior workers in the same position; I compare border with interior stores of the same format around each hike.

The second design tests the manager. I leverage the firm’s rotation of general managers across stores—a staffing and development policy, not a response to store performance—to estimate a fixed effect for each manager, classify as high-type those whose fixed effects imply less shrinkage, and follow shrinkage around the arrival of a high-type manager at a store. The third design tests the complementarity: I split the same arrivals by the wage environment the manager walks into, fixed at the month of arrival, and ask whether the same arrival accomplishes more where the premium is high. I also use the time-clock data to characterize

what the managers who reduce shrinkage do in the store.

Both levers move shrinkage, and in the direction the shirking model predicts. In the two-way fixed-effects gradient, a 10 percent lower premium is associated with about 1.1 percent more shrinkage, and with more absenteeism and turnover. Around the minimum-wage hikes, border shrinkage is unchanged on impact and rises by about 4.4 percent as the compression persists—the delayed timing that fits discipline eroding rather than mechanical repricing—and the rise concentrates in stores whose manager is seldom on the floor. The arrival of a high-type manager cuts shrinkage by close to 7 percent on average, with shrinkage flat in the months before arrival. The two instruments, moreover, reinforce each other: the same arrival cuts shrinkage by close to 12 percent where the premium is high, against roughly 4.6 percent where it is low—a difference of eight percentage points. And the time clocks show that what the effective managers add is attention, not apparatus: they spend more time on the store floor, including at night and on weekends, match their schedules to their workers’—they are more co-present—and keep a leaner store; they do not add surveillance. Shrinkage, in short, behaves not as a security problem to be locked away but as a personnel problem: it is governed by what the firm pays and by who is watching the floor.

The paper’s first contribution is the outcome itself. Despite its magnitude, shrinkage has been treated in economics as an accounting line rather than as behavior; to my knowledge, this is the first paper to characterize it systematically and to link it, worker by worker and manager by manager, to the personnel of the store.

Second, the paper contributes to the evidence that pay premia sustain discipline within firms. The classic reference is the interplant test of Cappelli and Chauvin (1991), who relate the premium a manufacturer pays across plants to disciplinary dismissals; the closest recent work is Emanuel and Harrington (2024), who show that a voluntary firm-wide minimum wage halves worker departures, reduces absenteeism, and raises warehouse productivity. The present study adds variation: pay is set centrally while local outside wages move, and two statutory minimum-wage schedules move the premium by law, so the premium is observed both rising and falling, across more than three hundred stores, against the same audited outcome.

Third—and closest to this paper—Coviello et al. (2022) establish a wage-by-monitoring complementarity: a minimum-wage increase raises the productivity of bound workers, and by more where monitoring is intensive. This paper differs in how monitoring is measured. Their proxy is a staffing quantity—the store-month supervisor-to-worker ratio, which, as the authors note, “captures the extensive margin of monitoring but not the intensive margin (supervisor effort)”; here the monitor is an individual manager whose arrivals and departures I observe. Rotation converts monitoring into discrete, dated, within-store events rather than

gradual changes in a staffing ratio; the time-clock records measure the intensive margin directly—the manager’s hours, location, and overlap with her workers; and the manager becomes an object of direct interest, since how managers complement a firm’s other incentive instruments is a central question of personnel economics.

Finally, the paper adds an outcome to the literature that recovers manager effects from rotation and asks what good managers do (Bertrand and Schoar, 2003; Lazear et al., 2015; Minni, 2026; Adhvaryu et al., 2026). The novelty lies in the margin and its interpretation: the manager’s effect is estimated on a margin no supervisor observes directly, decomposed into concrete managerial conduct, and shown to depend on the wage policy under which the manager operates.

2 Data and Institutional Background

2.1 The Firm and the Data

I partner with a large Mexican retailer with a nationwide presence. The firm operates more than three hundred stores across the country and employs around twenty-five thousand workers in any given month. Central headquarters standardizes stores tightly within format, setting the physical layout, the wage attached to each position, the human-resources and personnel policies, and the operating routines that govern each location.

Table 1 summarizes the firm’s scale and the structure of the data. The worker- and store-level panels that anchor the results span January 2022 through June 2024.

Table 1: Company summary

Category	Description
Country	Mexico
Sector	Retail
Number of workers	around 100,000
Number of stores	300+
Stores outside border zone	200+
Stores inside border zone	100+
Data periodicity	Scanner: second. Personnel: minute. Inventory: week.
Period	2022–2024

Data access provided through a research partnership. All analyses use anonymized records; store identifiers and exact locations are confidential.

The data combine three granular sources. The first is the firm’s personnel and scheduling system: it records worker demographics, hiring, promotions, absence, and turnover at the employee level, timestamps each worker’s entry and breaks to the minute, and carries the store’s organizational structure and work schedules. From it I build a worker- and manager-store panel and follow workers and managers over time. The second is point-of-sale scanner data, which capture sales and client counts at the second level and aggregate to the store week and month. The third is the firm’s weekly inventory audit, which measures inventory and inventory shrinkage in both value and physical units; because shrinkage is the paper’s central outcome, I describe the audit where I introduce shrinkage below.

The resulting panels cover roughly one hundred thousand distinct workers over the full period.

2.2 Stores and Workforce

The average store employs 67.3 workers, sells about USD 683,000 per month, serves 47,123 clients, and loses about 2.53 percent of sales to inventory shrinkage. Table 2 reports the distribution of sales, activity, and headcount at the store level.

Table 2: Store sales and activity (store-month)

Variable	Mean	Median	Min	Max
Sales (USD)	682,726	550,828	136,785	2,087,354
Sales (units)	426,469	347,440	95,543	1,197,007
Clients (count)	47,123	42,156	3,608	118,945
Workers per store	67.3	54	1	262

Store-month observations (more than 300 stores, January 2022–June 2024). Monetary values are in USD, converted at MXN 18.5/USD, the approximate average interbank rate over the period. Inventory shrinkage is reported separately in Table 4.

The personnel hierarchy has three layers: a single general manager per store, a middle layer of area sub-managers and supervisors, and the general workforce.

2.3 Store Managers

The general manager is the pivotal personnel actor in each store. She hires and dismisses workers, assigns and reassigns them across occupational areas, promotes and moves them within the store, sets the schedule, and oversees inventory—and she is the store’s monitor, supervising through her presence on the floor and the hours she overlaps with frontline workers. Because pay is fixed centrally by position, none of these levers runs through compensation—the manager can neither raise nor cut a worker’s wage—so her influence operates through selection, allocation, scheduling, and supervision.

The firm rotates general managers across stores routinely, and the rotation is administrative in origin. In the firm’s own description, reassignment has two drivers. The first is coverage: a store is staffed by more than one manager—up to three in the largest formats—and when a general manager retires, resigns, or falls ill, headquarters fills the vacancy by drawing a manager from a store that has more than one. Who moves is therefore determined by where a slot opens and where spare management sits, not by the performance of the store that receives her. The second is training: for younger managers, running different stores alongside different colleagues is a built-in stage of the development track, which moves them across store formats and locations on no fixed schedule. The resulting churn is substantial.

The typical store is run by about three different general managers over the two-and-a-half-year panel—a spell on the order of a year each—and more than nine in ten stores change general manager at least once during the period; on the manager’s side, over half of general managers run at least two different stores in the sample and a quarter run three or more (Table 9). This churn is the variation that identifies the portable manager effects I estimate later; the resulting manager–store panel matches more than five hundred general managers to the network over 2022–2024 and covers more than ninety percent of store-months. Hiring and separation, the allocation and internal mobility of workers, and presence-based supervision are the responsibilities I return to when examining what managers do.

2.4 Wage Premia and Shirking Measures

Two constructs anchor the analysis: the wage premium, which measures the rent the firm pays over the outside option, and a set of shirking measures, of which inventory shrinkage is the central one.

Wage premia. I define the wage premium as the difference between a worker’s wage inside the firm and the wage the same worker would command outside it,

$$W_{it}^p = w_{it} - w_{it}^m, \tag{1}$$

where w_{it} is worker i ’s firm wage in month t and w_{it}^m is her outside wage. I measure the outside wage by matching each worker to a reference worker profile in the ENOE, the national labor-force survey administered by INEGI, which covers both formal and informal workers. The match cascades through progressively coarser cells. Each worker is first matched, within her full cell—same gender, education, age, municipality, and quarter—to the average labor income of employed ENOE respondents in that cell; the full cell covers about 86 percent of worker-months. Where the full cell is empty in the survey, the match falls back one dimension at a time: relaxing gender, then age, then municipality accounts for a further 0.7, 0.5, and 1.2 percent of matches (so about one percent are matched at the state rather than the municipality level), and education—the most predictive dimension given the firm’s own wage structure, and therefore relaxed last—accounts for the remaining 12 percent. The quarter and the state are never dropped, and 99 percent of worker-months obtain an outside wage. For each worker I then compare what she earns in the firm to what a comparable worker earns in her local labor market, following the efficiency-wage logic of Cappelli and Chauvin (1991).

The firm pays a substantial premium over the local market almost everywhere: on average

a worker earns about 33 percent above her estimated outside wage (median 20 percent), roughly USD 93 per month. Against the statutory minimum wage the gap is larger—about 49 percent for the median worker. Table 3 reports the distribution of the firm wage, the outside wage, and the premium.

Table 3: Wages and the wage premium (worker-month)

	Mean	Median
Firm wage (USD)	476	428
Outside wage, ENOE (USD)	382	364
Wage premium (USD)	93	69
Wage premium (% of outside)	33.3	19.5
Wage premium (multiple)	1.33	1.20

Worker-month observations. The firm wage is monthly contractual pay; the outside wage is the ENOE cell wage matched on gender, education, age, municipality, and state. The wage premium in levels is the difference between the two; the multiple is the ratio of the firm wage to the outside wage. USD at MXN 18.5/USD. The mean multiple exceeds the ratio of means because the distribution is right-skewed among workers with low outside options.

Shirking. To gauge shirking and effort I use a set of personnel outcomes standard in the literature: following Cappelli and Chauvin (1991), turnover—the monthly rate at which workers leave the store, split into voluntary quits and firings; following Coviello et al. (2022), absenteeism—in days and as an indicator for any absence, separating non-justified from justified; and, from the scheduling system, monthly shifts and daily worked hours. As in other large retailers, tenure is short and absence and turnover are high: median tenure is 26 months, 45 percent of a store’s workers are absent in a given month (35 percent without justification), and about 6 percent separate each month.

The paper’s central and most novel measure is inventory shrinkage—the audited value of merchandise the firm records as owned but cannot find on the shelf.¹ The firm’s weekly inventory audit measures it in both value and physical units: it compares the book stock the firm’s system expects a store to hold against a physical count of what is on the shelf, and records the gap. The firm counts at three frequencies—a daily reconciliation of book against physical stock and fuller counts each week and each month—so losses are caught close to when they occur; at the close of each month it reconciles the flow identity (inflows = sales + shrinkage), so recorded losses tie out against measured stock movement. To my

¹Merchandise that is spoiled or damaged but still identifiable is logged in a separate register and is not counted as shrinkage, so the measure captures stock that is simply missing.

knowledge this is the first store-level administrative panel of shrinkage available for research. Shrinkage is a first-order concern for retailers: in the United States it reached \$112.1 billion in 2022, about 1.6 percent of sales (National Retail Federation, 2023), a magnitude that rivals what retailers keep as profit. In this firm it averages 2.53 percent of sales, so a store that eliminated its shrinkage would raise profits by about that much.

Interpreting shrinkage as shirking rests on an assumption about where the losses come from. Shrinkage responds to two behaviors of the workforce: the attention workers devote to preventing losses on the floor—watching merchandise, handling it with care, flagging problems as they appear—and the losses workers themselves originate. Effort moves both margins in the same direction: high effort prevents shrinkage, while low effort stops preventing it or, at the extreme, causes it.

The data support this reading. Figure 1 shows how store shrinkage covaries with three families of standardized store characteristics—its workforce and organization, its operations, and its local environment—each entered one at a time, so the associations are directly comparable.² The personnel and organizational measures show the largest, most systematic associations—stores whose workforces are less experienced, less stable, and more often absent without justification carry more shrinkage (standardized associations of about 0.17 to 0.23). Customer traffic enters weakly (about 0.10), and the environmental measures—commercial robbery, homicide, and perceived insecurity—cluster near zero (all below 0.03 in absolute value). The signals the data volunteer most strongly are about people and organization, which is why the paper treats shrinkage as a personnel outcome—the retail analogue of the shirking margin in efficiency-wage theory (Shapiro and Stiglitz, 1984; Cappelli and Chauvin, 1991).

²Each marker is the standardized coefficient (with 95% confidence interval) from a store-level regression of average shrinkage-to-sales on the indicated covariate, one at a time, across the more than three hundred stores; the personnel gradients survive conditioning on store format and municipality. Environment measures are municipal averages of SESNSP commercial robbery and homicide and of ENSU perceived insecurity.

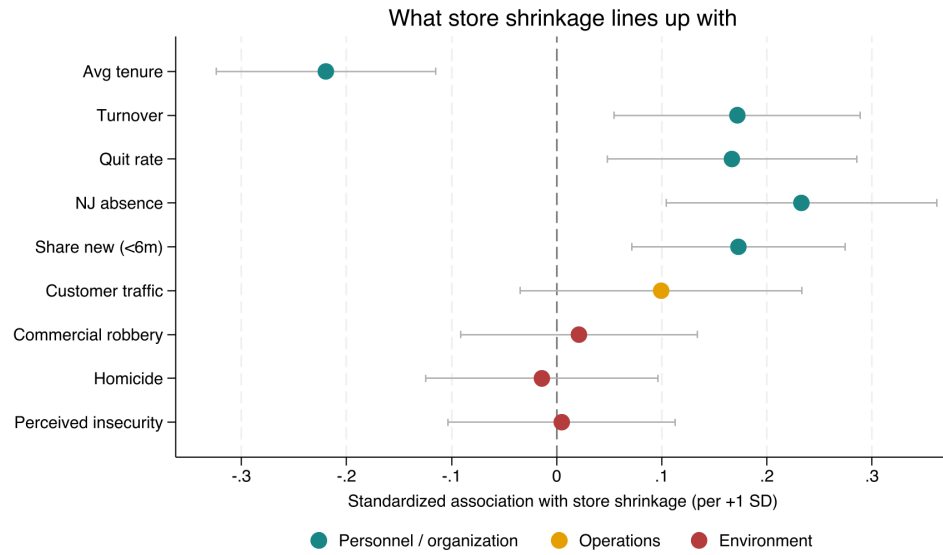


Figure 1: What store shrinkage lines up with. Standardized association of average store shrinkage with workforce, operations, and environment measures (store-level, more than 300 stores); horizontal bars are 95% confidence intervals.

Summary statistics for shrinkage and the remaining shirking measures, at the store and worker levels, appear in Table 4.

Table 4: Shirking measures at the store and worker levels

Variable	Mean	Median	Min	Max
Store level (store-month)				
<i>Inventory shrinkage</i>				
Shrinkage-to-sales (%)	2.53	2.45	0.10	10.52
Shrinkage (USD)	16,931	12,450	807	67,329
Shrinkage (units)	6,440	5,012	47	21,497
Shrinkage per worker (USD)	242	230	16	1,219
<i>Absence (store total, days per month)</i>				
Absence days, total	126.3	101	0	626
Non-justified absence days	58.2	46	0	313
Justified absence days	28.3	20	0	414
<i>Absence (store share of workers)</i>				
Any absence (share)	0.453	0.451	0	1.00
Any non-justified (share)	0.350	0.348	0	1.00
Any justified (share)	0.081	0.061	0	0.91
<i>Turnover (% of workforce per month)</i>				
Turnover rate, total	6.20	5.48	0	31.25
Turnover rate, voluntary	5.28	4.64	0	31.25
Worker level (worker-month)				
<i>Absence (days per month)</i>				
Absence days, total	1.88	0	0	31
Non-justified absence days	0.86	0	0	28
Justified absence days	0.42	0	0	31
<i>Absence (indicator)</i>				
Any absence (= 1)	0.462	0	0	1
Any non-justified (= 1)	0.361	0	0	1
Any justified (= 1)	0.081	0	0	1
<i>Effort</i>				
Monthly shifts	21.33	25	1	56
Daily worked hours	8.26	7.92	0	24

Store statistics are computed at the store-month level (more than 300 stores); worker statistics at the worker-month level (roughly 590,000 worker-months), January 2022 through June 2024. USD at MXN 18.5/USD. At the store level, absence days are summed across the store’s workers, the absence *shares* are the fraction of the store’s workers with at least one absence of that type in the month, and turnover rates are separations per hundred workers. At the worker level, absence days are per-worker monthly counts and the absence indicators equal one if the worker had at least one absence of that type. *Non-justified* absence is absence without an approved justification; *justified* absence is documented leave; vacation days are reported separately and excluded here. Effort margins (shifts, hours) are recorded by the scheduling system.

3 Theoretical Motivation

I develop a model that adapts the efficiency wage framework of Shapiro and Stiglitz (1984) and Coviello et al. (2022) to this setting. Three features of the firm shape the approach. First, all workers earn a fixed salary; there is no pay-for-performance. Second, workers can always find outside employment at the prevailing local market wage, so the relevant outside option is not unemployment but the market wage itself. Third, the threat of termination is the only formal incentive mechanism for preventing shrinkage. Together, these features imply that the disciplining device is the *wage premium*—the gap between the firm’s salary and what the worker can earn in the market—not the wage level per se.

The model has one further job, and it organizes the entire paper. In the no-shirking logic the expected penalty for shirking is a *product*: the probability that shirking is detected times the rent the worker forfeits if dismissed. The paper measures the two factors of that product separately: Section 4 measures the rent, and Sections 5–6 let a statute move it—the January minimum-wage hikes compress the border premium, first at the store and then in the worker-level payroll, where the timing is visible worker by worker: each worker’s statutory raise arrives on impact, and the store’s inventory response follows with a delay. Sections 7–9 identify the monitor, and Section 10 tests whether the two factors in fact multiply, by asking whether a good manager accomplishes more where the premium is high. Section 11 closes by joining the two designs: the manager’s presence determines where the statutory squeeze does its damage. The model delivers the interaction as its central prediction, along with three companions: a premium that no one monitors cannot discipline the shrinkage margin (the low-type mirror); the margins the worker internalizes directly—her own attendance and retention—respond to the premium under any manager; and a marginal, policy-induced movement in the rent moves the hidden margin where monitoring leaves slack and leaves the well-watched store insured.

3.1 Environment

Workers. Worker i is employed at store j under manager m . Workers are heterogeneous in their match-specific type $x_i \geq 0$, which captures comparative advantage in preventing inventory loss. In each period, worker i chooses two actions. The first is a *care level* $e_i \in [0, 1]$: effort devoted to watching and handling merchandise so that it does not go missing. Care is a hidden action—the store cannot verify it directly, and low care has consequences for the worker only if someone detects the losses it produces. Higher care reduces shrinkage: store-level shrinkage is $S_j = f(1 - \bar{e}_j)$, where $\bar{e}_j$ is the average care level in store j and f is strictly increasing. The second action is an *attendance level* $a_i \in [0, 1]$: showing up, on time,

for the shifts the schedule assigns. Attendance is *not* a hidden action: the time-clock records it to the minute, so a worker who fails on this margin is detected mechanically, whoever runs the store. This distinction between a margin that must be policed and a margin that records itself does real work below.

The cost of care is $c(x_i, e)$, satisfying $c_e > 0$ and $c_{ee} > 0$ (increasing and convex in care), with $c_x < 0$ and $c_{xe} < 0$ (high types have a lower level and lower marginal cost of care). The cost of attendance is $\kappa(a)$, increasing and convex.

Wages and outside option. Worker i receives a fixed wage

$$w_i = w^m + W_i^p, \quad (2)$$

where w^m is the prevailing local market wage—the wage the worker can obtain immediately if she leaves the firm—and $W_i^p \geq 0$ is the *wage premium* (Section 2, Equation 1). The wage is fixed within a contract period and does not depend on realized shrinkage: none of the firm’s incentives run through compensation. The model is therefore stationary and carries no time subscripts: the worker faces a standing premium W_i^p , the propositions are comparative statics in its level, and each worker-month observation in the data maps to the model evaluated at its current premium (Equation 1).³

Separation hazards and monitoring. The job ends through two distinct hazards, one per margin. On the care margin, manager m monitors with intensity $\mu_m \geq 0$, and a worker exerting care e keeps her job with probability $\pi(e; \mu_m)$, with

$$\pi_e(e; \mu_m) \geq 0 \quad (\text{more care} \Rightarrow \text{less likely to be dismissed}), \quad (3)$$

$$\pi_{e\mu}(e; \mu_m) \geq 0 \quad (\text{the closer the monitoring, the more strongly retention rewards care}) \quad (4)$$

A concrete parametrization satisfying (3)–(4) is discussed in Section 3.7.

On the attendance margin, retention is $\pi^A(a)$ with $\pi^A \geq 0$: absence is recorded and sanctioned mechanically, so this hazard does not depend on μ_m . Treating the two hazards as mutually exclusive within a period, the overall retention probability is

$$\Pi(e, a; \mu_m) = \pi(e; \mu_m) + \pi^A(a) - 1, \quad (5)$$

³Because V in Equation (10) discounts the entire future at the worker’s current premium, the mapping treats that premium as permanent: the exercise compares steady states, reading an observation with a lower premium as the stationary problem at that level rather than as a point on an anticipated transition path.

equivalently, the dismissal hazard is the sum $1 - \Pi = (1 - \pi(e; \mu_m)) + (1 - \pi^A(a))$.

3.2 Worker's Problem

Following Coviello et al. (2022), I set the problem in discrete time. Let $V^E(x_i)$ be type- x_i 's value of employment and V^A the value of the outside option. Because the worker can always find employment at the market wage w^m , the flow value of the outside option is $u^A = \frac{r}{1+r} V^A = w^m$.

Type x_i 's value function satisfies

$$V^E(x_i) = \max_{e, a \in [0,1]} \left[w_i - c(x_i, e) - \kappa(a) + \frac{1}{1+r} (\Pi(e, a; \mu_m) V^E(x_i) + (1 - \Pi(e, a; \mu_m)) V^A) \right]. \quad (6)$$

Define the *employment surplus* $V(x_i) \equiv V^E(x_i) - V^A$. Substituting (2) and $u^A = w^m$ into (6):

$$V(x_i) = \max_{e, a \in [0,1]} \left[\underbrace{W_i^p - c(x_i, e) - \kappa(a)}_{\text{flow surplus}} + \frac{\Pi(e, a; \mu_m)}{1+r} V(x_i) \right]. \quad (7)$$

The flow surplus—the gain from employment relative to the market option—equals the wage premium minus the costs of care and attendance. The market wage w^m drops out entirely: only the gap W_i^p determines incentives.

Assuming the maximization problem is strictly concave in (e, a) (Section 3.7), the first-order conditions for an interior optimum are

$$\underbrace{c_e(x_i, e^*)}_{\text{marginal cost of care}} = \underbrace{\frac{\pi_e(e^*; \mu_m)}{1+r} V(x_i)}_{\text{marginal benefit: detection avoided} \times \text{job surplus}}, \quad (8)$$

$$\underbrace{\kappa'(a^*)}_{\text{marginal cost of attendance}} = \underbrace{\frac{\pi^A(a^*)}{1+r} V(x_i)}_{\text{marginal benefit: sanction avoided} \times \text{job surplus}}. \quad (9)$$

Both conditions price the same object—the surplus $V(x_i)$ the worker stands to lose—but through different detection technologies. The care condition (8) runs through $\pi_e(e; \mu_m)$: it is the *manager's* monitoring that makes low care costly. The attendance condition (9) runs through π^A , which contains no μ_m : the time-clock does the detecting, whoever the manager is. This asymmetry is the model's second organizing prediction, and the data deliver it repeatedly.

Solving (7) at the optimum:

$$V(x_i) = \frac{(1+r)[W_i^p - c(x_i, e^*) - \kappa(a^*)]}{1+r - \Pi(e^*, a^*; \mu_m)}. \quad (10)$$

Since individual rationality requires $W_i^p > c(x_i, e^*) + \kappa(a^*)$ and $\Pi < 1+r$, we have $V(x_i) > 0$. $V(x_i)$ is *strictly increasing in* W_i^p : a larger premium means the job is more valuable relative to the market option, and the threat of termination is a stronger disciplining device.

One corollary is worth recording: so long as W_i^p exceeds $c(x_i, e^*) + \kappa(a^*)$, the surplus stays strictly positive, so a marginal erosion of the premium moves the effort margins of Proposition 1 but leaves participation intact.

3.3 The Wage Premium as the Efficiency Wage

This result is the departure from both Shapiro and Stiglitz (1984) and Coviello et al. (2022), and it fits the firm I study.⁴

In Shapiro and Stiglitz (1984), the punishment for shirking is entry into the unemployment pool; the efficiency wage must compensate workers for the risk of unemployment spells, and the relevant discipline device is the wage level relative to the expected value of unemployment. In this firm, workers face a market that always offers w^m , so unemployment per se is not part of the outside option: what dismissal destroys is the premium, not employment itself.

In Coviello et al. (2022), most workers earn performance pay, and a statutory floor raises the value of the job for those it binds, motivating effort through a job-security channel. This firm is the polar case their taxonomy sets aside (their Section II.C) because their workers have substantial commissions: a *pure efficiency wage* firm, with fixed salaries and a premium over the market paid essentially everywhere—workers earn, on average, about 1.33 times their outside wage (Section 2). In such a firm the premium is the whole of the incentive, making the setting a clean laboratory for the mechanism.

The same arithmetic shows where the premium’s empirical variation comes from. Because the contract wage is set centrally by position and moves rarely, movements in the local outside wage pass through one-for-one to the premium: a store facing a higher w^m pays a smaller W_i^p , and the discipline the firm buys with it weakens—the compression logic of Propper

⁴The treatment of monitoring and dismissal parallels Coviello et al. (2022): there, a shirking worker is detected with probability $\pi(e)$ when monitored (and with a constant probability otherwise), termination follows detection, and monitoring coverage μ is treated as exogenous—they test for endogeneity to the minimum wage, find it below detectable levels, and proceed with exogenous μ . The difference is the object: their μ is the share of workers monitored in a store; here μ_m is the intensity embodied in the manager, likewise taken as given and validated with predetermined presence measures (Section 3.7).

and Van Reenen (2010), whose regulated hospitals lose performance precisely where local outside wages erode the premium they can pay. This pass-through is the variation the two-way fixed-effects design of Section 4 exploits, across stores and within workers.

The comparative static behind that design is immediate from the worker's problem.

Proposition 1 (The premium disciplines both margins). *Let (e^*, a^*) solve (8)–(9) at an interior optimum. Then:*

- (i) *Care rises with the premium: $\partial e^*/\partial W^p > 0$; store shrinkage $S = f(1 - \bar{e})$ falls.*
- (ii) *Attendance rises with the premium: $\partial a^*/\partial W^p > 0$; absence falls, and time on the job (shifts, hours) rises.*
- (iii) *Separations fall with the premium: the dismissal hazard $1 - \Pi(e^*, a^*; \mu_m)$ is decreasing in W^p , and because the surplus V rises, voluntary quits under idiosyncratic shocks to the outside option fall as well.*

Proof. By the envelope theorem applied to (7), $\partial V/\partial W^p = (1 + r)/(1 + r - \Pi) > 0$. Differentiating (8) implicitly:

$$\frac{\partial e^*}{\partial W^p} = \frac{\frac{\partial V}{\partial W^p} \cdot \frac{\pi_e(e^*; \mu_m)}{1 + r}}{c_{ee}(x_i, e^*) - \frac{\pi_{ee}(e^*; \mu_m)}{1 + r} V(x_i)} > 0, \quad (11)$$

where the numerator is positive and the denominator is positive by strict concavity. The identical argument on (9) gives $\partial a^*/\partial W^p > 0$. Part (iii) follows because Π is increasing in both actions, and $V > 0$ increasing in W^p makes the job strictly more worth keeping. $\square$

Proposition 1 is the prediction that Sections 4 and 5 take to the data.

3.4 Manager Heterogeneity and the No-Shirking Premium

Rewriting the care optimality condition as a *no-shirking condition* (NSC) in the spirit of Shapiro and Stiglitz (1984), substitute (10) into (8) and solve for the premium:

$$W_i^p \geq \underbrace{c(x_i, e^*) + \kappa(a^*) + \frac{(1 + r - \Pi(e^*, a^*; \mu_m)) c_e(x_i, e^*)}{\pi_e(e^*; \mu_m)}}_{\equiv \widehat{wp}(e^*; \mu_m)} \quad (12)$$

for the worker to find it optimal to sustain care e^* . Call $\widehat{wp}(e^*; \mu_m)$ the *no-shirking premium*: the minimum premium required to sustain care e^* under monitoring intensity μ_m .

In the binary-effort case ($e \in \{0, \bar{e}\}$, attendance suppressed), this reduces to the familiar form

$$W_i^p \geq \frac{(r+b)\bar{e}}{q} \equiv \widehat{wp}, \quad (13)$$

where b is the exogenous separation rate, $\bar{e}$ is the effort cost, and q is the detection rate.

Proposition 2 (Monitoring and the premium are substitutes as instruments). *Under higher monitoring intensity μ_m :*

- (i) *Optimal care e^* is higher at any given premium W^p .*
- (ii) *The no-shirking premium $\widehat{wp}(e^*; \mu_m)$ is lower: the firm can sustain a given level of care with a smaller premium.*

Along an iso-effort curve, monitoring intensity and the wage premium are substitutes as incentive instruments: the firm could purchase the same discipline with more of one and less of the other.

Proof. Part (i): Higher μ_m raises $\pi_e(e^*; \mu_m)$ by (4), increasing the right-hand side of (8). Since $c_{ee} > 0$, this requires e^* to rise. Part (ii): from (13), $\partial \widehat{wp} / \partial q < 0$ directly. In the continuous case, the same sign follows from (12) under standard regularity conditions, since π_e rises with μ_m . The full computation of part (i) with the equilibrium surplus responding to μ is Step 2 of the proof of Proposition 3; the sign survives unconditionally. $\square$

Proposition 2 describes the firm's menu: on paper, pay and supervision can be traded off. It is the natural null against which to read the interaction evidence—if the rent and the monitor were interchangeable deterrents, a large premium should insure the firm against a weak monitor, and a strong monitor should render the premium redundant. The next proposition shows that the worker's problem predicts neither: in *levels*, the two instruments multiply rather than substitute, and each is worthless on the shrinkage margin without the other.

3.5 The Complementarity: Predictions for the Arrival Design

Section 10 crosses the two instruments directly: it takes the good-manager arrival design and asks whether the arrival effect is larger where the firm's standing premium is high (HWP) than where it is low (LWP), separately for high- and low-type arrivals. Two results sign that design. Proposition 3 signs the shrinkage cells; Proposition 4 signs the attendance column. Each rests on one new assumption, stated first.

Notation and two assumptions. Fix a worker of type x_i and write V for the equilibrium surplus $V(x_i)$ of (10), evaluated at (W^p, μ) . Write

$$\rho \equiv 1 + r - \Pi(e^*, a^*; \mu), \quad D \equiv c_{ee}(x_i, e^*) - \frac{\pi_{ee}(e^*; \mu)}{1 + r} V, \quad D_A \equiv \kappa''(a^*) - \frac{\pi^{A''}(a^*)}{1 + r} V$$

for the effective discount denominator and the two second-order coefficients (D is the denominator of (11); $D > 0$ and $D_A > 0$ by strict concavity). Let $R = W \times [\mu_{LT}, \mu_{HT}]$ be a rectangle of premiums and monitoring intensities on which the optimum (e^*, a^*) is interior, with the smoothness the displayed derivatives require (c and h three times, κ and π^A twice continuously differentiable).

The first assumption promotes the detection technology of Section 3.7 from illustration to hypothesis, and states formally that detection spares the compliant.

Assumption 1 (Compliant-care detection). *(a) The care hazard is $\pi(e; \mu) = 1 - b - q(\mu)h(e)$ with $b \in (0, 1)$, $q > 0$, $q' > 0$, $h \geq 0$, $h' < 0$, and $h'' \geq 0$. (b) At every optimum in R , the residual exposure of compliant care is small relative to its slope:*

$$h(e^*) \pi^{A'}(a^*) \kappa'(a^*) D < (-h'(e^*)) c_e(x_i, e^*) (r + b + 1 - \pi^A(a^*)) D_A. \quad (14)$$

Part (a) implies (3)–(4), and because $\pi_{ee} = -qh'' \leq 0$, strict concavity in care needs no separate assumption. Part (b) is the operative content of “ $h \approx 0$ at compliant care”: the left side is proportional to the exposure a compliant worker still carries, $h(e^*)$; the right side to the exposure her marginal care removes, $-h'(e^*)$. The benchmark $h(e^*) = 0$ —detection spares the compliant exactly—satisfies (14) strictly.

Assumption 2 (Curvature regularity). *The effective marginal cost of care, $M(e) \equiv c_e(x_i, e)/(-h'(e))$, is log-concave on the relevant range: $(\log M)'' \leq 0$.*

Assumption 2 disciplines the third-order terms (c_{eee} , h''') that the cross-partial of an optimum drags in and that the standard assumptions leave unsigned. It holds strictly whenever c_e and $-h'$ have constant elasticities in e —for instance, quadratic care cost with linear exposure.

Proposition 3 (The premium and the monitor are complements). *Let $e^*(W^p, \mu)$ solve (8) on R under Assumption 1(a). Then:*

(i) **(One product does all the work.)** *Optimal care is a strictly increasing transform of the expected-penalty product—the detection rate times the discounted rent:*

$$M(e^*) = q(\mu) \frac{V}{1 + r}, \quad M' > 0, \quad (15)$$

and its two slopes are the comparative statics of Propositions 1(i) and 2(i):

$$\frac{\partial e^*}{\partial W^p} = \frac{\pi_e(e^*; \mu)}{\rho D} > 0 \quad \text{and} \quad \frac{\partial e^*}{\partial \mu} = \frac{q'(\mu)(-h'(e^*))(r+b+1-\pi^A(a^*))}{(1+r)\rho D} V > 0. \quad (16)$$

The first is (11): the premium moves the rent factor. The second makes Proposition 2(i) exact with the equilibrium surplus responding to μ : the monitor moves the detection factor, and the sign requires no smallness of $h(e^*)$.

(ii) **(Local complementarity.)** Under Assumptions 1–2, each factor raises the other's grip:

$$\frac{\partial^2 e^*}{\partial W^p \partial \mu} > 0 \quad \text{on } R. \quad (17)$$

(iii) **(The premium amplifies the monitor.)** For every $W^p \in W$, the effect of installing a better monitor, $\Delta_\mu e^*(W^p) \equiv e^*(W^p, \mu_{HT}) - e^*(W^p, \mu_{LT})$, is strictly positive, and it is strictly increasing in W^p : for $W_L^p < W_H^p$ in W ,

$$\Delta_\mu e^*(W_H^p) - \Delta_\mu e^*(W_L^p) = \int_{W_L^p}^{W_H^p} \int_{\mu_{LT}}^{\mu_{HT}} \frac{\partial^2 e^*}{\partial W^p \partial \mu} d\mu dW^p > 0. \quad (18)$$

Proof. Derivatives of V are the envelope derivatives of (7), as in the proof of Proposition 1. The right-hand side of (7) is a contraction in V with modulus $\Pi/(1+r) < 1$, so V is well defined; a perturbation of μ moves the maximand directly only through Π , the induced movements of (e^*, a^*) contributing nothing at first order, so

$$\frac{\partial V}{\partial W^p} = \frac{1+r}{\rho} > 0, \quad \frac{\partial V}{\partial \mu} = \frac{\pi_\mu(e^*; \mu)}{\rho} V = -\frac{q'(\mu)h(e^*)}{\rho} V \leq 0. \quad (19)$$

A fact used repeatedly: by (5), $\rho = r + b + q(\mu)h(e^*) + (1 - \pi^A(a^*))$, so

$$\rho - q(\mu)h(e^*) = r + b + 1 - \pi^A(a^*) > 0. \quad (20)$$

Step 1: the product form. Under Assumption 1(a), $\pi_e(e; \mu) = q(\mu)(-h'(e))$, so (8) is (15) with $M \equiv c_e/(-h')$. Differentiating,

$$M'(e) = \frac{c_{ee}}{-h'} + \frac{c_e h''}{(h')^2} > 0,$$

so (15) has at most one solution; and since the derivative of the maximand of (7) in e equals $(-h')[qV/(1+r) - M(e)]$ —positive to the left of the crossing, negative to the right—the

care problem is single-peaked and e^* is its unique maximum. At the optimum, substituting $c_e = q(-h')V/(1+r)$ into M' gives $M'(e^*) = [c_{ee} + qh''V/(1+r)]/(-h') = D/(-h')$.

Step 2: the two slopes. Differentiate (15) in W^p : $M'(e^*) \partial e^*/\partial W^p = qV_{W^p}/(1+r) = q/\rho$ by (19), so $\partial e^*/\partial W^p = q(-h')/(\rho D) = \pi_e/(\rho D)$, which is (11). Differentiate in μ :

$$M'(e^*) \frac{\partial e^*}{\partial \mu} = \frac{q'V + qV_\mu}{1+r} = \frac{q'V}{1+r} \left(1 - \frac{qh(e^*)}{\rho}\right) = \frac{q'V}{1+r} \cdot \frac{r+b+1-\pi^A(a^*)}{\rho} > 0,$$

by (19) and (20); dividing by $M'(e^*) = D/(-h')$ gives (16). The computation makes explicit what Proposition 2(i) holds fixed: a stricter monitor also trims the surplus of a worker who still carries exposure ($V_\mu \leq 0$), and (20) shows the deterrence channel dominates unconditionally.

Step 3: the attendance responses. Implicit differentiation of (9) gives, for $\theta \in \{W^p, \mu\}$,

$$\frac{\partial a^*}{\partial \theta} = \frac{\pi^{A'}(a^*)}{(1+r)D_A} \frac{\partial V}{\partial \theta}, \quad (21)$$

so $\partial a^*/\partial W^p = \pi^{A'}/(\rho D_A) > 0$ (Proposition 1(ii)) and $\partial a^*/\partial \mu = -\pi^{A'}q'h(e^*)V/(\rho(1+r)D_A) \leq 0$.

Step 4: the cross-partial. Both sides of (15) are positive; take logarithms and write $\ell \equiv \log M$, $v \equiv \log V$:

$$\ell(e^*) = \log q(\mu) + v(W^p, \mu) - \log(1+r). \quad (22)$$

Differentiating in W^p gives $\ell'(e^*) e_{W^p}^* = v_{W^p}$, with $\ell'(e^*) = M'(e^*)/M(e^*) = D/c_e > 0$ and $v_{W^p} = V_{W^p}/V = (1+r)/(\rho V) > 0$. Differentiating that identity in μ and solving,

$$\frac{\partial^2 e^*}{\partial W^p \partial \mu} = \frac{v_{W^p \mu} - \ell''(e^*) e_\mu^* e_{W^p}^*}{\ell'(e^*)}. \quad (23)$$

The second numerator term is nonnegative: $\ell'' \leq 0$ by Assumption 2 and $e_\mu^*, e_{W^p}^* > 0$ by Step 2. For the first, differentiate $v_{W^p} = (1+r)/(\rho V)$ totally in μ , using $d\rho/d\mu = -(\pi_e e_\mu^* + \pi^{A'} a_\mu^* + \pi_\mu)$ and $\rho V_\mu = \pi_\mu V$ from (19): the π_μ terms cancel, leaving

$$v_{W^p \mu} = \frac{(1+r)\beta}{\rho^2 V}, \quad \beta \equiv \pi_e(e^*; \mu) e_\mu^* + \pi^{A'}(a^*) a_\mu^*. \quad (24)$$

β is the induced-retention response: the movement in Π produced by the worker's behavioral answer to a stricter monitor, over and above the mechanical effect π_μ . Substituting the closed

forms of Steps 2–3 and reducing with (8) ($q(-h')V/(1+r) = c_e$) and (9) ($\pi^A V/(1+r) = \kappa'$),

$$\beta = \frac{q'(\mu)}{\rho D D_A} \left[(-h'(e^*)) c_e (r + b + 1 - \pi^A(a^*)) D_A - h(e^*) \pi^A \kappa' D \right],$$

which is strictly positive if and only if (14) holds. Hence $v_{W^p\mu} > 0$, and (23) delivers (17). By Young's theorem the same inequality reads in the other direction: the premium's grip on care, $\partial e^*/\partial W^p$, is increasing in μ .

Step 5: integration. $\Delta_\mu e^*(W^p) = \int_{\mu_{LT}}^{\mu_{HT}} \partial e^*/\partial \mu d\mu > 0$ by (16), and (18) follows from the fundamental theorem of calculus applied twice, the integrand in (17) being continuous and strictly positive on the compact rectangle. $\square$

Proposition 4 (Attendance is manager-invariant). *Let (e^*, a^*) solve (8)–(9) on R under Assumption 1(a), with $\pi^A(a^*) > 0$. Then:*

(i) (**Exact decomposition.**) *At every $(W^p, \mu) \in R$,*

$$\frac{\partial a^*}{\partial \mu} = \frac{\pi_\mu(e^*; \mu) V}{1+r} \cdot \frac{\partial a^*}{\partial W^p} = - \frac{q'(\mu) h(e^*) V}{1+r} \cdot \frac{\partial a^*}{\partial W^p}, \quad (25)$$

while on the care margin

$$\frac{\partial e^*}{\partial \mu} = \frac{q'(\mu) V}{1+r} \cdot \frac{r+b+1-\pi^A(a^*)}{q(\mu)} \cdot \frac{\partial e^*}{\partial W^p}. \quad (26)$$

Each margin's response to the manager is its response to the premium, rescaled by the worker's detection stake. For care the stake is the deterrence slack $(r+b+1-\pi^A)/q > 0$, bounded away from zero however compliant the worker; for attendance the stake is $-h(e^)$, the residual exposure of compliant care itself.*

(ii) (**Sign and bound.**) *The manager's total effect on attendance is nonpositive and proportional to residual exposure: there is $\tilde{\mu} \in (\mu_{LT}, \mu_{HT})$ with*

$$\Delta_\mu a^*(W^p) \equiv a^*(W^p, \mu_{HT}) - a^*(W^p, \mu_{LT}) = -(\mu_{HT} - \mu_{LT}) \left[q' h(e^*) \frac{V}{1+r} \frac{\partial a^*}{\partial W^p} \right]_{\mu=\tilde{\mu}} \leq 0, \quad (27)$$

so that, with $\overline{H} \equiv \sup_R q'(\mu) h(e^)$, $|\Delta_\mu a^*(W^p)| \leq (\mu_{HT} - \mu_{LT}) \overline{H} \sup_R (V/(1+r)) \partial a^*/\partial W^p$ and $|\log V(W^p, \mu_{HT}) - \log V(W^p, \mu_{LT})| \leq (\mu_{HT} - \mu_{LT}) \overline{H}/(r+b)$.*

(iii) (**The invariance benchmark.**) *As residual exposure vanishes ($\overline{H} \rightarrow 0$, the remaining primitives fixed), $\Delta_\mu a^*(W^p) \rightarrow 0$ and $V(W^p, \mu_{HT})/V(W^p, \mu_{LT}) \rightarrow 1$ at every premium, while $\Delta_\mu e^*(W^p)$ stays bounded away from zero: by (16) the care response*

runs through the slope $-h'(e^)$, not the level $h(e^*)$. The two margins separate exactly in the limit where detection spares the compliant.*

Proof. (i) By (21), the ratio of the two attendance responses is the ratio of the two surplus responses: $(\partial a^*/\partial \mu)/(\partial a^*/\partial W^p) = V_\mu/V_{W^p} = (\pi_\mu V/\rho)(\rho/(1+r)) = \pi_\mu V/(1+r)$, using (19); under Assumption 1(a), $\pi_\mu = -q'h(e^*)$. This is (25). Dividing (16) by (11) gives (26). (ii) Integrate (25) over $[\mu_{LT}, \mu_{HT}]$; the integrand is continuous on the compact interval, so the mean value theorem for integrals gives (27), and the displayed bound follows. For the surplus: $\partial \log V/\partial \mu = \pi_\mu/\rho$, and $\rho \geq r+b$ by (20), so $|\partial \log V/\partial \mu| \leq \bar{H}/(r+b)$; integrating over $[\mu_{LT}, \mu_{HT}]$ gives the bound on $\log V$. (iii) Immediate from (ii) and from (16), whose right-hand side involves $h(e^*)$ only through ρ , which remains bounded. $\square$

Remark 1 (One first-order condition, three propositions). *Propositions 1, 2(i), 3, and 4 are four readings of (15). The care condition prices the product $q(\mu)V/(1+r)$ —the probability of detection times the rent, the object the opening of this section announced. Proposition 1(i) differentiates it in the rent; Proposition 2(i) in the detection rate; Proposition 3(ii) is the cross-partial of the same condition, and (24) names the mechanism: a peso of premium is worth its annuity over the expected life of the job, $(1+r)/\rho$ per peso, and a stricter monitor—acting only through the worker’s induced behavior, the mechanical π_μ terms cancelling—raises retention, lengthens that life, and so makes the same peso discipline harder. There is no tension with Proposition 2(ii): the substitutes reading describes the firm’s iso-care menu (12), the complements reading the worker’s response surface—one condition, two questions. Proposition 4 is the same logic where the detection factor is absent: (9) contains μ_m only through V , and V moves with the manager only by the exposure a compliant worker still carries.*

Proposition 3 is the lens through which to read the evidence of Section 10, cell by cell. Part (iii) predicts the high-type pattern: the arrival of a good manager cuts shrinkage roughly three times more in high-premium environments—the monitor and the rent are the two factors of one product, each scaling the other’s productivity, the same reinforcement Coviello et al. (2022) find from the opposite direction when a minimum-wage rent raises productivity only in intensely monitored stores. Proposition 4 predicts the column the design holds fixed: non-justified absence does not respond to arrivals of either type in either environment—while the same outcome responds robustly to the premium itself in Section 4. The attendance nulls are not a failure of the model; they are its signature, because attendance is the margin whose detection the manager does not supply. The model even signs the admissible deviation: through the surplus, a stricter monitor could only *raise* absence, and only in proportion

to the exposure a compliant worker still carries (25)—a magnitude the design should not distinguish from zero.

3.6 Embodied Monitoring: Actions versus Returns

The model treats μ_m as an attribute of the *person*, not a technology of the store, and the distinction has teeth. Monitoring in this firm is not an apparatus the manager can purchase—loss prevention staff, security headcount, and the supervisory share do not move on arrival—and it is not a reorganization she leaves behind. It is her presence on the floor: high-type managers work more shifts, overlap more of their workers’ hours, and their co-presence disciplines punctuality shift by shift (Section 9). Three implications follow, and each is testable in the designs of Sections 8–10.

First, *portability*: if μ_m travels with the person, then a store’s shrinkage should break on the exact month a high- μ manager walks in, with flat pre-trends—the signature the arrival event studies of Section 8 display.

Second, *actions flat, returns scaled*: because μ_m is embodied, the manager brings the same monitoring to every store she runs—nothing in her problem re-optimizes her presence against the local rent. The model therefore predicts that the manager’s *actions* (her floor presence, her internal reallocation, her inventory tightening) should not vary systematically with the store’s premium, while the worker’s *response* to those actions—and hence the manager’s measured return—should scale with W^p , by Proposition 3(iii). Section 10.3 adjudicates this prediction: the complementarity is *intrinsic* to the threat, not a story about managers working harder where wages are high. Had the actions scaled with the premium, the complementarity would be behavioral and the model’s multiplicative structure would be the wrong reading; the data side with the model.

Third, *reversion on departure*: what presence produces, departure removes. When a high-type manager leaves, μ falls back, the care condition (8) relaxes, and shrinkage should climb back toward its pre-arrival level—while any structural change the manager made to the store itself (a leaner inventory position) has no reason to revert. The arrival–departure asymmetry of Section 8 finds precisely this split: the shrinkage gain erodes when she goes, the inventory position does not move.

The squeeze design sharpens the point to its finest. As a moderator of the compression response, the manager’s measured presence is predetermined, passes the pre-trend checks, and carries the interaction (Section 11); her estimated fixed effect—the statistical label—fails those checks at the border and cannot serve as the moderator (Appendix G). What insures a store against an erosion of the rent is what the manager does on the floor, not what the

econometrics says she is.

The reader can therefore hold one sentence through the paper: *the firm pays a rent and employs a monitor; the rent disciplines every margin the worker internalizes, the monitor makes the rent effective on the margin she must police, and policy that moves the rent redistributes discipline across stores according to who is watching.*

3.7 Discussion of Assumptions

The model maintains four assumptions worth discussing; none is load-bearing for the sign of the predictions. Throughout, workers are risk neutral: flow utility is linear in the wage.

The reemployment assumption. Immediate reemployment at the market wage is a conservative normalization, not a substantive restriction. Suppose instead—realistically for a developing economy—that a separated worker spends a search spell in informal work at $w_u < w^m$ and regains formal employment at rate λ per period. The flow value of the outside option becomes $(\lambda w^m + r w_u)/(\lambda + r)$, and everything that follows holds with the premium augmented by the constant $\delta \equiv (w^m - w_u) r / (\lambda + r) > 0$: the employment surplus remains strictly increasing in W_i^p , no proposition changes sign, and the no-shirking premium falls by δ . Calibrated with ENOE reemployment spells (about one quarter) and formal–informal wage gaps, δ is roughly three percent of the median premium, so the premium remains the disciplining device.

The detection technology. Assumption 1(a) fixes the care hazard at $\pi(e; \mu) = 1 - b - q(\mu) h(e)$: b is an exogenous separation rate, $q(\mu)$ the detection rate the manager generates ($q' > 0$), and $h(e)$ the worker’s exposure to detection ($h' < 0$, $h'' \geq 0$, with $h \approx 0$ at compliant care). It delivers (3)–(4) directly, makes the care problem concave by construction ($\pi_{ee} = -qh'' \leq 0$), and carries the property the two-margin results rest on: a worker who complies has essentially nothing to fear from a watchful manager, so monitoring changes the *return to care*, not the survival odds of the compliant. Assumption 1(b) is that property stated as an inequality, and (25) is what it buys. Two limits are worth stating plainly. First, (14) rules out detection regimes that endanger the compliant—audit technologies whose false positives grow with monitoring intensity fast enough that a stricter monitor lowers a diligent worker’s retention by more than her induced care restores it; under such a regime a good manager would tax the job’s value, and the attendance nulls would not be predicted. Second, exact invariance is a limit, not a specification: a technology with $h = 0$ on the whole range of equilibrium care would also erase π_e , and with it the care margin itself. The model therefore never delivers $\Delta_\mu a^* = 0$ identically; it delivers (27)—an attendance response bounded by

the compliant worker’s residual exposure, with the care response bounded away from zero alongside it. What the data can test is that the bound is indistinguishable from zero, and it is.

Concavity and curvature. Under Assumption 1(a) the worker’s problem is strictly concave in care by construction; strict concavity in attendance ($D_A > 0$) is assumed, with sufficient conditions parallel to Assumptions 1–2 of Coviello et al. (2022). Assumption 2 adds the one curvature condition the complementarity needs: the cross-partial of an optimum involves third derivatives (c_{eee}, h''') that the standard assumptions leave unsigned, and log-concavity of $c_e/(-h')$ is exactly the discipline they require. It holds strictly throughout the constant-elasticity family (for instance, quadratic care cost with linear exposure), and only Proposition 3(ii)–(iii) uses it.

Measuring μ_m . Empirically, μ_m is the manager’s monitoring conduct, and the paper measures it in two ways. The arrival designs classify managers by their AKM fixed effect (Section 7): managers whose shrinkage fixed effect lies below the median of their format-by-border cell (“high-type,” HT) carry higher μ_m ; those above it (“low-type,” LT) lower μ_m . Section 9 then grounds the interpretation in what the manager *does*: high-type managers work more shifts, overlap more hours with their frontline workers, and their co-presence disciplines worker punctuality shift by shift—they do not buy more monitoring, they *are* more monitoring. Presence, not the statistical label, is the operative content of μ_m : when the squeeze design pits the two against each other, the manager’s pre-shock presence carries the interaction (Section 11), while her estimated fixed effect cannot: at the border the label fails its own pre-trend test, so it is an invalid moderator there rather than a refuted one (Appendix G).

3.8 Where the Data Refine the Model

The simplest version of the model makes sharp predictions, and several do not survive contact with the data unchanged. I flag five tensions and the refinements they force.

Monitoring is embodied, not purchased. The model writes monitoring as an intensity the firm can dial up. High-type managers, however, do not buy more monitoring apparatus or reorganize the store—they add no loss-prevention, supervisory, or security staff, and the team’s composition is unchanged on arrival. They are physically present more, and their co-presence disciplines worker punctuality shift by shift. I therefore interpret μ_m as an embodied

attribute of the manager that is portable across stores and reverts when she departs, not a store-level technology the firm can set independently of who runs the store.

The complementarity is intrinsic, not behavioral. A natural reading of the interaction evidence is that good managers optimize against the rent—supplying more supervision where the premium gives supervision more to protect. The data reject it: action by action, the manager’s conduct barely responds to the premium, while her enforcement and her results respond sharply. The model accommodates this by keeping μ_m out of the manager’s choice set altogether: the complementarity lives in the worker’s first-order condition, where the detection rate multiplies the rent, not in the manager’s behavior. The refinement is a discipline on interpretation—the premium scales what supervision is *worth*, not how much of it the manager supplies.

Shirking is not one margin. A single-action model predicts that whatever disciplines shrinkage should discipline absence too. The data split them: absence responds to the premium but not to the manager, in either environment, while shrinkage responds to both and to their interaction. The refinement is the two-margin structure above—care must be detected by someone, attendance records itself—and it converts an apparent set of nulls into the model’s cleanest signature. The squeeze adds an asymmetry the same structure absorbs: the statutory raise moves attendance everywhere, yet the squeeze’s erosion, which read symmetrically should reduce it, leaves attendance at its status quo and surfaces instead as shrinkage in unwatched stores (Sections 6 and 11). The verifiable margin is disciplined tightly—most workers sit at or near full compliance, where a marginal loss of surplus has no room to move behavior—while the hidden margin is interior, and interior margins are where marginal changes in the rent do their work. Where each worker stands relative to the corner decides which margin a marginal movement in the rent moves.

Complements in levels, substitutes at the margin. Written as a no-shirking condition, the premium and monitoring look like substitute instruments—the firm could sustain the same care with more of one and less of the other. The evidence is two-signed: a good manager accomplishes more where the standing premium is high (Section 10), yet the statutory erosion of the rent damages only the stores where little monitoring is supplied, and the statutory raise converts into shrinkage discipline only where it is (Section 11). The refinement is to distinguish—with the level half formalized in Proposition 3(ii)–(iii)—the level complementarity of the detection-times-rent product from the marginal redundancy of an extra unit of rent under a manager who is already watching.

What persists and what reverts. Treating the manager as a bundle of contemporaneous monitoring cannot explain why some of what she does outlasts her. Her departure (Section 8) separates her legacy cleanly: the shrinkage reduction erodes when she leaves—consistent with monitoring as an ongoing input into the worker’s care condition—while the leaner inventory she institutionalized persists. The refinement distinguishes the manager’s *flow* input (presence, which enters π_e and leaves with her) from her *stock* interventions (inventory routines, which reset the store’s technology and stay). Only the first is μ_m ; the second is why not every arrival gain should be expected to revert.

4 Wage Premia and Shirking: Two-Way Fixed-Effects Evidence

The model in Section 3 predicts an inverse relation between the wage premium and shirking: the larger the rent a worker stands to lose if she is caught, the more care she supplies and the more reliably she shows up. This section tests that prediction in my setting and asks along which margins it holds. The outcomes, all defined in Section 2, map onto the model’s two margins: hidden *care*, which must be policed—inventory shrinkage, absenteeism, and turnover—and *attendance*, which the time clock records mechanically—monthly shifts and daily hours. A premium that disciplines shirking should lower the first group and raise the second.

Two institutional features let me measure the relation. The firm sets wages centrally by position, so firm pay is essentially fixed within position; the outside wage varies widely across Mexican municipalities and over time (Appendix Figure 18). Movements in a store’s or a worker’s premium therefore come mainly from the outside option, not from the firm’s own pay decisions. I estimate the relation at the store level here—the level of the paper’s central outcome, inventory shrinkage—and report the parallel worker-level results, which follow the same worker as her own premium moves, in Appendix A.

Identification. Because the firm fixes pay by position, a store’s premium moves when its local outside wage moves, and δ captures how shirking covaries with the rent under two conditions. The firm’s pay-setting does not respond to a store’s own discipline—ruled out by the position-level wage policy—and outside-wage movements are uncorrelated with local determinants of shirking beyond what municipality and time fixed effects absorb. The second is the demanding one: a local boom that lifts outside wages could also shift crime, staffing, or the mix of hires. Two features make it a reasonable working assumption. Municipality and year-by-quarter fixed effects absorb permanent local differences and common national shocks, so δ rests on deviations in the local wage rather than on levels; and the minimum-wage exercise of Section 5 re-estimates the same relation off statutory hikes that move the premium by law, on a known date, for reasons unrelated to any store’s conditions—sharpening this margin. I therefore read these estimates as the association the model predicts, quantified under transparent assumptions and tested more directly later.

I relate a store’s wage premium over its local labor market to each shirking and effort margin. At the store level I estimate

$$\log Y_{it} = \delta \log W_{it}^p + Z_{it}\theta + \eta_{m(i)} + \tau_t + \nu_{it}, \quad (28)$$

where Y_{it} collects store-level shrinkage, absenteeism, turnover, shifts, and hours; $\log W_{it}^p$ is the log store wage premium, the store average of the worker premium in Equation 1 across the workers of store i in quarter t ; $\eta_{m(i)}$ is a municipality fixed effect; τ_t a year-by-quarter fixed effect; and Z_{it} optionally includes log sales and store format. Standard errors are clustered at the municipality level. A negative δ on a shirking outcome, or a positive δ on an effort outcome, is consistent with the model's prediction.

A larger store premium is associated with less shirking on every margin. Table 5 reports the headline shrinkage result; Appendix A (Table 20) collects the remaining margins. A ten percent increase in the store premium is associated with:

- a 1.1% reduction in inventory shrinkage;
- a 0.36% reduction in absenteeism and a 1.3% reduction in non-justified absenteeism;
- a 1.6% reduction in total turnover and a 1.5% reduction in voluntary turnover;
- a 0.32% increase in monthly shifts and a 0.12% increase in daily worked hours.

The result is consistent across measures and specifications: shrinkage in physical units, which strips out valuation, moves with the value measure, and every store-level gradient keeps its sign and significance across store format, log sales, and alternative time fixed effects (columns 2–4; Appendix A, Table 20).

Table 5: Store-level TWFE: inventory shrinkage (Log–Log)

<i>Wage premium (Log)</i>	(1)	(2)	(3)	(4)
Coefficient	−0.113***	−0.114***	−0.045**	−0.111***
SE	(0.032)	(0.033)	(0.021)	(0.031)
Store format	No	Yes	Yes	No
Log sales	No	No	Yes	No
Municipality FE	Yes	Yes	Yes	Yes
Year×Qtr FE	Yes	Yes	Yes	No
Year + Quarter FE	No	No	No	Yes

Store-quarter level; roughly 2,400 store-quarters. Dependent mean 9.49, median 9.43 (log). Shrinkage is the value of inventory losses in USD. Standard errors clustered at the municipality level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5 Minimum-Wage Hikes at the Border: Moving the Premium by Statute

Section 4 showed that shrinkage tracks the wage premium across the firm’s stores. There, though, the premium moves with the local labor market, so a local shock that lifted outside wages could by itself have moved shrinkage too. This section moves the premium from the outside instead, on a schedule no store controls. Mexico raises its minimum wage every January, and since 2019 it maintains a separate, higher floor along the northern border. A hike raises the wage a worker is legally guaranteed *elsewhere*—her outside option—by an amount and on a date written in the *Diario Oficial*, while the firm’s pay scale, set centrally by position, does not move with it; the premium the firm pays over that option narrows for reasons external to any store. That statutory narrowing is what I call a squeeze, and it is the same object the gradient of Section 4 traced, now moved exogenously rather than merely observed.

I compare border with interior stores of the same format around the 2023 and 2024 hikes. The comparison bites because the border ladder sits far closer to its floor: the same statutory raise reaches almost the entire border store while barely touching the interior, format for format. The squeeze raises shrinkage, but with a lag: border shrinkage is flat on impact and rises about 4.4 percent as the compression persists ($p = 0.02$). Whether that damage depends on the manager who happens to be on the floor when the shock lands, I take up in Section 11, once the manager results are in hand.

5.1 Institutional Background: Two Floors, One Pay Policy

Mexico sets its minimum wage nationally and raises it every January. In 2019 the government created a second, higher minimum-wage zone along the northern border—the *Zona Libre de la Frontera Norte*—and since then the two floors have risen by the same percentage each January, so their ratio has stayed fixed at about 1.51 (Figure 2).

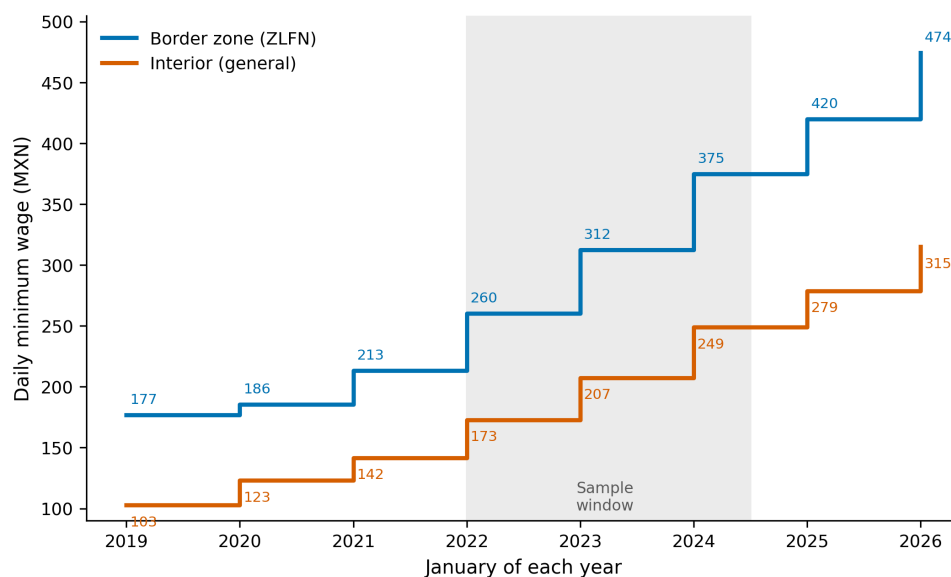


Figure 2: Daily minimum wage in Mexico, 2019–2026: the interior (general) floor and the northern-border (ZLFN) floor. The border zone was created in January 2019 at roughly 1.72 times the general floor; since 2020 both floors have risen by the same percentage each January, holding their ratio near 1.51. The shaded band marks the sample window. Source: CONASAMI.

The January 2024 increase, for example, raised both floors by the same twenty percent, from permanently different bases. About a quarter of the firm’s stores lie inside the border zone. The firm’s pay policy turned this geography into an experiment. Before the border zone existed, the firm paid each position the same wage regardless of location. In 2019 the firm stood at a comfortable distance from the floor: pay ran well above the minimum, and only a few workers held wages the new border floor could reach. The annual increases changed that from 2021 onward. As the two floors climbed from permanently different bases, the border floor began to overtake wages that the interior floor could not touch, and a geographic gap opened inside the firm: the same position came to earn more at the border than outside it—not because the firm re-priced the position, but because the floor forced its border wage upward.

What a January hike does to a border store is therefore written in the statute: the floor beneath the store’s wage ladder rises by a known amount on a known date, and the premium the firm pays over the guaranteed outside option narrows. In the model of Section 3 that premium is the rent the threat of dismissal confiscates, so narrowing it should weaken discipline, and the loss should surface in shrinkage—the margin that gives when care lapses (Proposition 1). The design below asks whether it does.

5.2 Design

I compare border with interior stores of the same format around the January 2023 and 2024 hikes, stacking each as its own experiment (Cengiz et al., 2019). The outcome is log shrinkage, measured net of sales. For store k of format f in hike h at event month τ ,

$$Y_{kh\tau} = \alpha_{k,h} + \gamma_{f(k),m} + \sum_{\tau \neq -1} \beta_{\tau} (\text{Border}_k \times \mathbf{1}[\tau]) + \varepsilon_{kh\tau}, \quad (29)$$

with store-by-hike fixed effects $\alpha_{k,h}$, format-by-calendar-month fixed effects $\gamma_{f(k),m}$, a window $\tau \in [-6, 6]$, and reference month $\tau = -1$. I cluster standard errors by store. Identification requires that, absent the hike, border stores would have tracked interior stores of their format up to seasonal norms—the assumption the flat pre-hike coefficients support—and that the timing of the shock is external to any store, which the statutory January schedule delivers. The design follows the border-comparison tradition of the minimum-wage literature (Dube et al., 2010). The statute sharpens, rather than replaces, the gradient evidence. In Section 4 the premium moves with local labor markets, so a local shock that moves both outside wages and shrinkage remains the residual threat. Here the movement in the premium is written in the *Diario Oficial*: its date is known, its size is known, and neither depends on any store’s conditions; format-by-month fixed effects absorb seasonal retail patterns, and the flat pre-hike coefficients show border and interior stores on common paths before each January.

5.3 A Delayed Increase in Shrinkage

Figure 3 reports the estimates. The pre-hike coefficients are flat; border shrinkage is quiet through impact and rises over the following two quarters. I summarize it with a three-period specification that replaces the event-time indicators with an impact phase $[0, 2]$ and a compression phase $[3, 6]$ months,

$$Y_{kh\tau} = \alpha_{k,h} + \gamma_{f(k),m} + \beta_E (\text{Border}_k \times \text{Impact}) + \beta_L (\text{Border}_k \times \text{Compression}) + \varepsilon_{kh\tau}. \quad (30)$$

Border shrinkage is unchanged on impact ($\beta_E = -0.009$, $p = 0.60$) and rises by 4.4 percent in the compression phase ($\beta_L = +0.044$, $p = 0.022$). It erodes discipline as the compression accumulates, not on the date the floor moves.

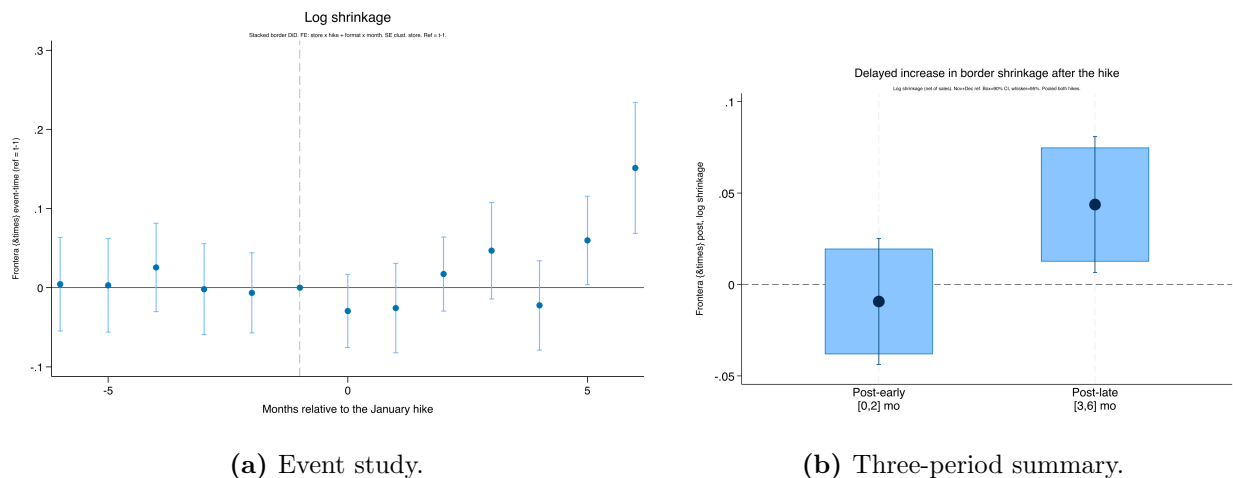


Figure 3: The delayed shrinkage response. Flat pre-trends and no effect on impact, then a +4.4 percent rise in border shrinkage over the compression phase ($p = 0.022$). Stacked across both hikes; standard errors clustered by store.

[JS: I would like to see a “first stage” for this event study in which the premium is the y-axis variable.]

5.4 Inside the Store Response: The Raise and the Squeeze

A border store, however, is not one worker, and the hike does not land on every rung of its wage ladder the same way. The store-level coefficient is one number for many unlike experiences, and reading it requires knowing what the statute did to each rung. The rest of this section opens the store response into its parts: whom the incoming floor overtakes, whom it closes in on from below, and why the difference sets the clock of the response.

Workers at and near the minimum make the squeeze concrete. Take the January 2024 hike. A border worker earning the 2023 border floor is *swept*: the statute raises her to the new floor, a twenty percent raise she did not bargain for. Her coworker one rung up, earning about 1.2 times the old floor in December—twenty percent clear of it—receives no raise, yet wakes up in January standing exactly on the new floor: her entire legal premium is gone without her wage moving at all. That is the squeeze. In the interior the same two rungs live differently: a worker at the interior floor is swept up with it, but her coworker at about 1.17 times the floor still clears the new interior floor by seventeen percent. The identical rung of the identical employer’s ladder loses far more ground at the border, because the border floor is 1.51 times higher and rises from that higher base. The store-wide picture follows rung by rung: in the November payroll before a hike, roughly one interior worker in ten still earns at least 1.6 times the incoming floor, while at the border almost none do. Virtually the entire border wage ladder lies within the floor’s reach, while the interior store, format for format, is not. Table 6 sets the firm’s ladder against the incoming floor, decile by decile, and shows

how much more of a border store the hike can reach.

Table 6: The firm’s wage ladder against the incoming minimum-wage floor

Wage decile	November 2023 (2024 floor)		November 2022 (2023 floor)	
	Interior	Border	Interior	Border
1	0.84	0.81	0.85	0.83
2	0.90	0.87	0.91	0.90
3	0.93	0.91	0.95	0.95
4	0.97	0.94	0.99	0.99
5	1.01	0.97	1.03	1.03
6	1.05	1.00	1.08	1.06
7	1.11	1.04	1.15	1.10
8	1.20	1.08	1.25	1.15
9	1.38	1.13	1.47	1.23
10	1.74	1.28	2.00	1.44
Share swept: below floor (%)	42.6	54.4	38.0	37.6
Share below $1.3\times$ floor (%)	81.6	95.7	78.4	89.4
Share below $1.6\times$ floor (%)	91.8	100.0	88.1	97.9
Median daily wage ($\times$ floor)	1.03	0.98	1.05	1.04

Full-time workers (≥ 24 shifts), non-managerial, in the November payroll before each January hike. Daily salary is the annual median of monthly pay divided by 30. Each decile cell reports the median ratio of daily salary to the minimum-wage floor arriving in January (2024: 248.93 interior, 374.89 border; 2023: 207.44, 312.41). Deciles computed within zone by year.

The raises the statute forces trace the same reach: the largest raises go to workers the floor overtakes, raises decline monotonically with the worker’s distance above it, and the gradient flattens only around 1.5 to 1.6 times the floor. The hike compresses the ladder from below, worker by worker—the first stage of the squeeze—before any behavioral response. A hike hits a border store as two things at once. For the lowest-paid workers, whom the new floor rises above, it is a *raise*: the firm must lift them to the floor, an increase they did not bargain for. For the workers just above them, whom the floor now reaches from below, it is a *squeeze*: their own wage does not move, but the wage the law guarantees them elsewhere does, so the premium that stands between them and dismissal narrows. In the model of Section 3 that premium is what deters shirking—the penalty for being caught is the probability of detection times the rent lost on dismissal—so the squeeze weakens discipline (Proposition 1); the raise’s brief strengthening on impact rests on reciprocity to an unbargained increase, logic the model does not carry (Akerlof, 1982).

The two forces run on different clocks, and the clocks explain the store-level path. A raise is a one-time event: it buys goodwill on impact, but once workers absorb it into the wage they treat as normal, its pull is spent. A squeeze is not an event but a state: in every month

that follows, the premium is smaller, the penalty for shirking is smaller, and the discipline the premium used to buy is weaker. So on impact the two roughly cancel, and the border–interior gap in shrinkage stays flat; as the raise fades and the squeeze persists, the gap opens. It opens on shrinkage in particular—the loss that surfaces when care lapses, audited weekly and accumulating with sustained neglect rather than any single bad day. The delayed rise of Figure 3 is that two-phase path: quiet on impact, rising afterward. The closest evidence that this margin responds to a minimum-wage change inside a firm is Coviello et al. (2022).

The store-level design therefore estimates the sum of the two forces, not either one alone: each border coefficient in Figure 3 adds the discipline the raise buys from the workers the floor sweeps up to the erosion it works on their coworkers above, just as the store’s mean premium adds two offsetting first stages: hike by hike, the 2023 increase compresses the border premium by about 9 pesos per day—roughly half of the statutory 17.5-peso differential among stayers—while in 2024 the border floor climbed 62 pesos and swept a far larger below-floor mass up to it, so the mechanical lift outweighs the squeeze and the mean premium rises. Separating the raise from the squeeze requires following each worker on her own rung: where she stood relative to the incoming floor in November, and how she behaves, month by month, after the floor arrives in January. Section 6 builds exactly that design.

6 The Raise and the Squeeze, Worker by Worker

Here is what the paper knows so far, and what it cannot yet see. The previous section shows that when the statute squeezes a border store’s premium, shrinkage rises months later, and its closing anatomy names the reason a store-level number cannot be the last word: the same January morning *raises* the wage of the worker below the incoming floor and *narrows* the premium of her coworker just above it, and a store-level regression averages those two experiences into one number. What is missing is the worker-level view: who inside the store responds, on which margin, and on whose clock.

The payroll supplies it, because every worker’s distance to the incoming floor is visible in November, before the floor arrives. I classify each full-time worker by the ratio of her daily salary to the January floor, and the four groups are the cast of the story. The *swept* (ratio below 1) are overtaken by the floor: the statute hands them a raise they did not bargain for. The *compressed-near* (1 to 1.3) keep their wage while the floor closes in from below—the squeezed workers of the store section, seen individually. The *compressed-far* (1.3 to 1.6) lose ground too, but from a safer distance, which makes them a built-in dose gradient. The *cushion* (1.6 and above) stand beyond the reach of the floor and of the firm’s spillover raises, and serve as the control group.⁵ I track each group through the same two-phase clock as the store design. Identification is within store-month: a worker-by-hike fixed effect absorbs everything permanent about the worker, a store-by-month-by-hike fixed effect absorbs everything that happens to her store in a given month—no store or border shock can masquerade as a group effect—and each group is measured against the cushion coworkers of the same store in the same month where the cushion exists (see the note above on its thinness at the border), with standard errors clustered by store.

6.1 Results: The Worker-Level Clock Matches the Store’s

What should each group do? The model of Section 3 makes one prediction per group; the clock is not the model’s but the premium’s, whose path each observation feeds through the comparative statics one period at a time. The swept receive a large unbargained raise,

⁵The classification is predetermined and held fixed through each event window. I set the cushion cutoff where the firm’s own post-hike raise schedule flattens out, so the control is not itself treated through internal spillovers (Autor et al., 2016; Dustmann et al., 2022); measurement details of the salary concept follow the payroll conventions of Section 2. At the border the cushion is thin by construction—the compression documented in Table 6 leaves 2.1 percent of border workers at or above 1.6 times the incoming floor in the 2023 hike, and none in the 2024 hike—so border-level group *levels* lean on the 2023 cohort, and the results I read at the border are the within-border contrasts and the dose-response, which difference out the common store-month. The thin border cushion is not a nuisance of the design but its point: no rung of the border ladder escapes the floor.

so on gift-exchange and no-shirking logic alike their discipline should *improve* from the month the raise lands—a raise is an event. The compressed lose rent every month the new floor persists, so their discipline should *erode*, slowly—a squeeze is a state; and the compressed-far, who lose more ground, should erode more. The cushion, untouched, pin the counterfactual. The margins I measure are the ones a worker controls and the time-clock verifies: the probability of an unjustified absence and unjustified absence days (the care margin the worker internalizes), suspensions (the firm’s own disciplinary response), justified absence (a placebo: certified events should not respond to wage position), and separation (the participation margin). Shrinkage does not appear here because it is a store outcome, already measured in the store section.

Read Table 7 as follows. Each row is an outcome; the columns report each group’s effect relative to its cushion coworkers, separately for the early phase (January–February, the raise’s window) and the late phase (March–June, the squeeze’s window); the rows d_{21} and d_{31} report the contrasts compressed-near–swept and compressed-far–swept within phase, which is where the squeeze shows once the common store month is netted out.

Three results organize the table, one per margin. On the worker’s care margin, the swept repay the raise: their probability of an unjustified absence falls by two percentage points and their unjustified absence days by about four percent of a day per month, against a median own raise of roughly a quarter—the reciprocity a large, unbargained raise invites (Akerlof, 1982). On the firm’s margin, the employer disciplines the swept more: suspensions rise in both phases, the firm’s response to a worker whose statutory cost just rose. And on the clock—the exercise’s central result—the compressed fall behind the swept exactly when the store section says the squeeze binds: d_{21} is statistically zero in January and February and opens to 1.3 percentage points of absence probability and 0.03 days in March through June, with the same late-phase pattern, larger still, for the compressed-far (d_{31})—the dose gradient inside the store. The raise is an event whose reciprocity arrives with it and builds through the spring; the erosion of position is a state whose behavioral cost accumulates (Card et al., 2012; Breza et al., 2018).

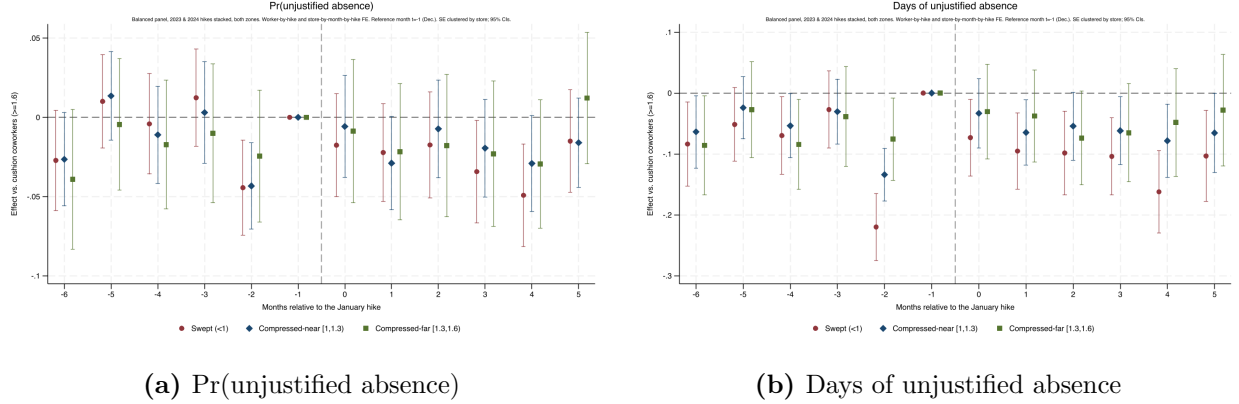
Table 7: Exposure group $\times$ phase: worker-level attendance and discipline

	Swept	Compr. near	Compr. far	d_{21}	d_{31}
<i>Panel A. Pr(unjustified absence)</i>					
Early _[0,1]	−0.011 (0.010)	−0.007 (0.009)	0.001 (0.014)	0.004 (0.007)	0.012 (0.011)
Late _[2,5]	−0.020** (0.008)	−0.007 (0.007)	0.001 (0.011)	0.013** (0.006)	0.021** (0.010)
<i>Panel B. Days of unjustified absence</i>					
Early _[0,1]	−0.009 (0.020)	0.002 (0.017)	0.018 (0.023)	0.011 (0.016)	0.027 (0.022)
Late _[2,5]	−0.041** (0.017)	−0.014 (0.015)	−0.002 (0.022)	0.028** (0.014)	0.040* (0.021)
<i>Panel C. Pr(suspension)</i>					
Early _[0,1]	0.004** (0.002)	0.003* (0.002)	0.003 (0.003)	−0.001 (0.002)	−0.001 (0.003)
Late _[2,5]	0.005*** (0.002)	0.003* (0.002)	−0.000 (0.002)	−0.002 (0.001)	−0.005** (0.002)

Balanced panel, 2023 and 2024 hikes stacked, both zones. Group effects are measured against the cushion control; d_{21} and d_{31} are the compressed-minus-swept contrasts within phase. Early = January–February; Late = March–June. Worker-by-hike and store-by-month-by-hike fixed effects; standard errors clustered by store. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 4 reads the same regression on a monthly clock: each point is an exposure-group-by-event-month coefficient, measured against cushion coworkers in the same store and month, with December as the reference month. The swept improvement arrives on impact and builds through the spring—unjustified absence days fall by 0.07 in January and by 0.16 in month four—and the compressed fall behind late: the compressed-minus-swept gap in absence days is positive in every post-hike month but is largest in months four and five (0.084 days for the near group, $p = 0.004$; 0.114 for the far, $p = 0.010$)—the dose gradient of the phase table, month by month. The monthly read also shows why the table draws its inference from pooled phases against the full pre-hike window: the November point sits below December for the swept and compressed-near groups—the payroll’s year-end seasonal anchor—while months -6 through -3 are flat, and the compressed-minus-swept contrasts, which difference out the common seasonality, show no pre-hike pattern on the absence-probability margin.

Figure 4: The minimum-wage event study, split by how much the hike cuts the worker’s premium



Notes: Monthly event-study version of Table 7, same sample and specification: balanced worker panel, 2023 and 2024 hikes stacked, both zones ($\approx 144,000$ worker-months, $\approx 12,000$ worker-hike units, 300 stores). Each point is the coefficient on an exposure-group $\times$ event-month interaction, measured against cushion coworkers (≥ 1.6 times the incoming floor) in the same store and month; the omitted month is $t = -1$ (December); groups are fixed in the pre-hike November payroll. Worker-by-hike and store-by-month-by-hike fixed effects; standard errors clustered by store; bars are 95 percent confidence intervals; the dashed vertical line marks the hike. Joint tests of the pre-hike months reject for the swept and compressed-near groups ($p < 0.01$), driven entirely by the November point—the seasonal November–December anchor—with months -6 to -3 flat; the compressed-minus-swept contrasts are clean in panel (a). This is why Table 7 draws its inference from pooled phases against the full pre-hike window.

The gap is also dose-responsive, and the dose is the border. With a border increment added to each group effect—identified within store-month, so no border-wide shock can contaminate it—the compressed–swept gap is approximately zero in the interior, where compression is mild, and 3.0 percentage points at the border (0.090 days), where the same position loses far more ground to the floor; the formal dose–response contrast is 2.2 points (0.085 days). The swept improvement, by contrast, is the same in both zones: reciprocity to an own raise is a level effect, indifferent to the dose (Table 8). The placebo behaves—justified, certified absence does not respond in any group—and the participation margin is inert: the dose–response in turnover is an informative null, as a model in which the premium still exceeds the cost of care predicts (Section 3).

Table 8: The compressed-swept gap, by zone: the border dose-response

Outcome	d_{21} (Compr. near – Swept)		Dose-response (DDD)	
	Interior	Border	$F_2 - F_1$	$F_3 - F_1$
Pr(unjustified absence)	0.008 (0.006)	0.030*** (0.011)	0.022* (0.012)	0.006 (0.019)
Days of unjustified absence	0.006 (0.013)	0.090*** (0.027)	0.085*** (0.030)	0.080* (0.044)

Group $\times$ post effects split into an interior component and a border increment; worker-by-hike, store-by-month-by-hike, and position-by-month-by-hike fixed effects; the interior and border d_{21} and the dose-response contrasts are linear combinations from a single regression; balanced panel, pooled post window; standard errors clustered by store. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7 Manager Effects on Shrinkage: AKM Decomposition

Section 4 showed that the wage premium—the rent the firm pays over the local labor market—is associated with less shirking, treating the store as the unit that collects the rent, and Sections 5–6 moved that rent by statute. But the premium disciplines shirking only if someone can catch it, and detection is the general manager’s job (Section 3): a rent that no one watches deters nothing, and a watchful manager with no rent to withhold has no threat to make. Section 4 measured the rent; the sections that follow measure the monitor, beginning with whether the person who runs the store matters for shrinkage at all.

I assemble the evidence in four steps: this section carries the first two, the sections that follow the rest.

First, I use the firm’s routine rotation of general managers across stores to separately identify manager and store fixed effects for shrinkage, following Abowd et al. (1999) (AKM), and report the decomposition. Second, I probe the two standard threats to it—endogenous mobility and limited-mobility bias—and show the manager component survives both. Third, Section 8 sorts managers into high and low types by their estimated effect and tests whether it is *portable*, tracking a store’s shrinkage when a high-type manager arrives. Fourth, Section 9 asks who the high-type managers are and what they do. The steps point the same way: the manager is a first-order determinant of shrinkage—on the order of every fixed store characteristic combined—her effect travels with her, and it works through her presence on the floor rather than a reorganization of the store. That is the empirical content of μ_m , the monitor the efficiency-wage model requires.

7.1 Identifying Manager Fixed Effects

Identification comes from the manager moves themselves. Because I observe the same manager running different stores, I can read her signature off the shrinkage that follows her from store to store; because I observe the same store under a succession of managers, I can read the store’s signature off what persists as its managers change. Without moves the two effects would be collinear—a manager who never left a single store could not be told apart from the store itself—so the rotation that might look like a nuisance is what makes the decomposition possible.

Following Abowd et al. (1999), I estimate the two-way fixed-effects model

$$Y_{jt} = \theta_{m(j,t)} + \psi_j + \delta_t + \varepsilon_{jt}, \quad (31)$$

where Y_{jt} is log shrinkage value USD in store j and month t ; $\theta_{m(j,t)}$ is a fixed effect for

the manager running store j in month t ; ψ_j is a store fixed effect absorbing permanent characteristics such as location, format, and clientele; and δ_t is a calendar-month effect absorbing seasonality and firm-wide shocks. Following Card et al. (2013), I decompose the error term as

$$\varepsilon_{jt} = \eta_{m(j,t),j} + \xi_{jt} + \nu_{jt}, \quad (32)$$

where $\eta_{m(j,t),j}$ is a manager–store match component, ξ_{jt} is a drifting store-level component, and ν_{jt} is transitory noise.

Identification. OLS estimation of Equation (31) imposes the moment conditions

$$\mathbb{E}[\theta_{m(j,t)} \varepsilon_{jt}] = 0 \quad \text{and} \quad \mathbb{E}[\psi_j \varepsilon_{jt}] = 0. \quad (33)$$

Identification therefore relies on the assignment of managers to stores being conditionally mean-independent of past, present, and future values of ε_{jt} . This assumption is weaker than random assignment: it permits sorting on the permanent components, so that better managers may systematically run better or worse stores, since both fixed effects are estimated. What it excludes is assignment based on the match component η or on transitory shocks—for instance, a manager dispatched to a store because shrinkage there is spiking, or a manager who moves only to stores where her particular style works. Such endogenous mobility would bias the estimated effects, and Section 7.4 tests its observable implications directly. Two features make exogenous mobility plausible by construction: the firm rotates general managers as a development and staffing policy—to broaden them across formats and to fill vacancies—and it does not track store shrinkage as a manager KPI or use it to place managers, so assignments do not respond to a store’s losses.

Connected sets. Manager and store effects are separately identified only within *connected sets* of stores linked by manager moves (Abowd et al., 1999, 2002). The panel contains ten connected sets; the largest covers about three-quarters of the panel’s stores, managers, and manager-months, and all variance decompositions below normalize fixed effects within their connected set. About eight percent of stores never exchange a manager with another store and drop out of the exercises shown in the next sections.

7.2 Manager Mobility

The credibility and precision of the decomposition rest on how much manager mobility the panel contains. A majority of general managers are movers: 55.0 percent run at least two different stores during the sample period, 25.4 percent run three or more, and 91.7 percent

of stores receive at least one mover. Within the largest connected set the mover share rises to 60.7 percent. The median manager is observed for thirteen months, so fixed effects are estimated from spells of meaningful length rather than one-off transitions. Table 9 collects these diagnostics.

This degree of mobility is high relative to the matched employer–employee panels on which the AKM literature was developed, in which mover shares of 12 percent (Andrews et al., 2012), 25 percent (Card et al., 2013), and 35 percent (Alvarez et al., 2018) are typical, and it is comparable to the managerial rotation documented in Adhvaryu et al. (2026). This rotation is a staffing-and-development policy, not a response to a store’s losses. The *timing* of a change is not fully orthogonal to performance—a store is somewhat more likely to change manager after a high-shrinkage month—but managers are not assigned to systematically better or worse stores (origin and destination shrinkage are statistically indistinguishable), and who moves is selected on permanent traits (seniority and pay) rather than transitory shocks. What the design requires is mean-independence of mobility from the transitory error given the fixed effects, not literal randomness.

Table 9: Manager mobility and connected sets

	General-manager panel
<i>Panel A: Coverage</i>	
Store-months with a matched general manager	> 90%
Median months observed per manager	13
<i>Panel B: Manager mobility</i>	
Share of movers (managers in ≥ 2 stores)	55.0%
Share of managers in ≥ 3 stores	25.4%
Share of stores with at least one mover	91.7%
<i>Panel C: Connected sets</i>	
Number of connected sets	10
Share of stores without movers (FE not identified)	8.3%
Largest connected set: share of the panel	$\approx 73\%$
Largest connected set: mover share	60.7%
Share of store effects identified	88%

General managers matched to monthly store outcomes, 2022–2024. A mover is a manager observed in at least two stores. “Share of the panel” is the average of the shares of stores, managers, and manager-months in the largest connected set.

7.3 Results: Managers Explain Half of the Variation in Shrinkage

Figure 5 plots the distribution of the estimated manager and store fixed effects for log shrinkage value, standardized within connected set and rescaled to a 0–100 range. Both

distributions display wide support: managers at the top of the distribution preside over systematically higher inventory losses than managers at the bottom, conditional on the store they run and the month. The dispersion in manager effects is comparable to that in store effects, a first indication that who runs the store matters on the same order of magnitude as which store is being run.

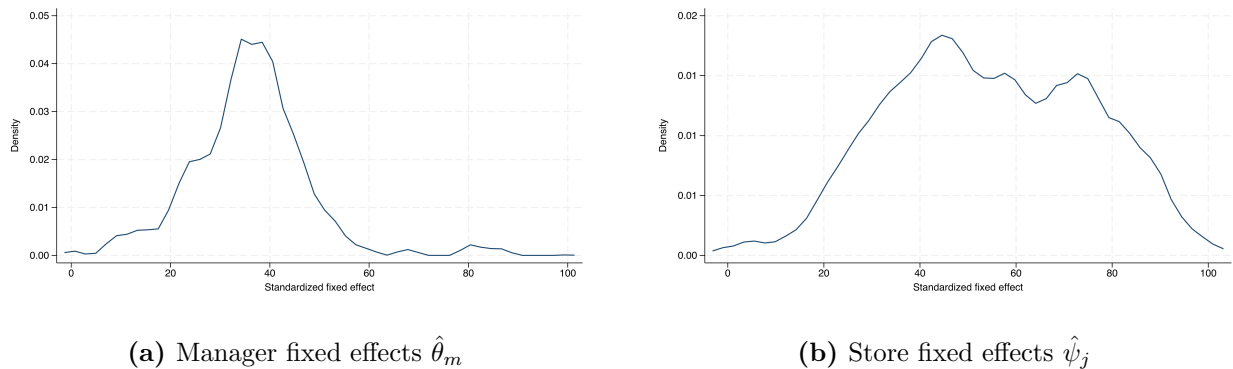


Figure 5: Distribution of estimated fixed effects, log shrinkage value

[JS: Distribution of estimated effects is not the same as an estimate of the distribution of effects.

These plots convolve real effects with sampling noise.]

Notes: Kernel densities of the fixed effects from Equation (31), standardized within connected set and rescaled to $[0, 100]$. Outcome: log monthly shrinkage value USD.

Figure 6 quantifies this comparison. I follow the decomposition $\text{Cov}(Y, \hat{X})/\text{Var}(Y)$ for each estimated component $\hat{X}$ of Equation (31), which allocates the variance of the outcome across manager effects, store effects, time effects, and the residual. Manager fixed effects account for 49.8 percent of the variance of log shrinkage value and store fixed effects for 42.4 percent, with calendar effects (2.0 percent) and the residual (5.8 percent) making up the remainder.

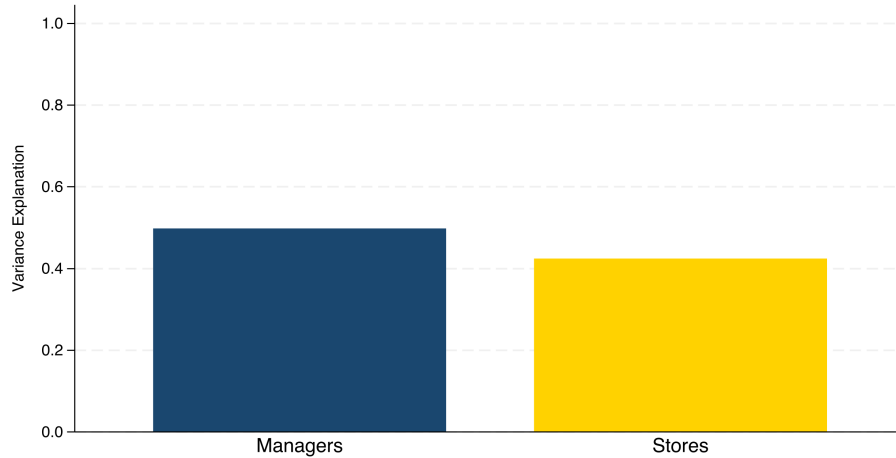


Figure 6: Explanatory power of manager and store fixed effects, log shrinkage value

Notes: Each bar reports $\text{Cov}(Y, \hat{X})/\text{Var}(Y)$ for the manager and store components of Equation (31). Time effects (2.0%) and the residual (5.8%) are omitted from the figure.

7.4 Robustness of the AKM Design

This subsection subjects the decomposition to the standard battery of AKM diagnostics, and the design passes them: managers who move between stores adjust to the destination on impact with no differential pre-trend, gains and losses from moving up and down the store-quality ladder are symmetric, and the manager component remains economically meaningful once I correct for the noise inherent in estimating fixed effects from finite mover counts. I report each diagnostic for the headline outcome and then explain the threat it addresses and how I resolve it.

Event study around manager moves. A first observable implication of exogenous mobility is that managers who move to stores with different shrinkage levels should adjust on impact, with no differential trend before the move. Following the event-study design of Card et al. (2013), I rank origin and destination stores by quartile of their average outcome, isolate managers who move across quartiles, and follow each manager across her move, tracing the shrinkage of whatever store she then runs in binned periods around it. [\[JS: I'm confused by the exercise. As time moves left to right are we looking at the same store with a different manager? Or the same manager at a different store? That seems important for interpretation and your language makes it seem like the former?\]](#) Figure 7 reports the raw outcome and two residualized versions. Three patterns support the design. First, managers moving from bottom-quartile to top-quartile stores jump to the destination level on impact, and the reverse moves fall

symmetrically. Second, within-quartile moves are flat throughout, so the jumps are not an artifact of moving itself. Third, trajectories before the move are flat within origin quartile, showing no evidence that movers were selected on a diverging shrinkage path.

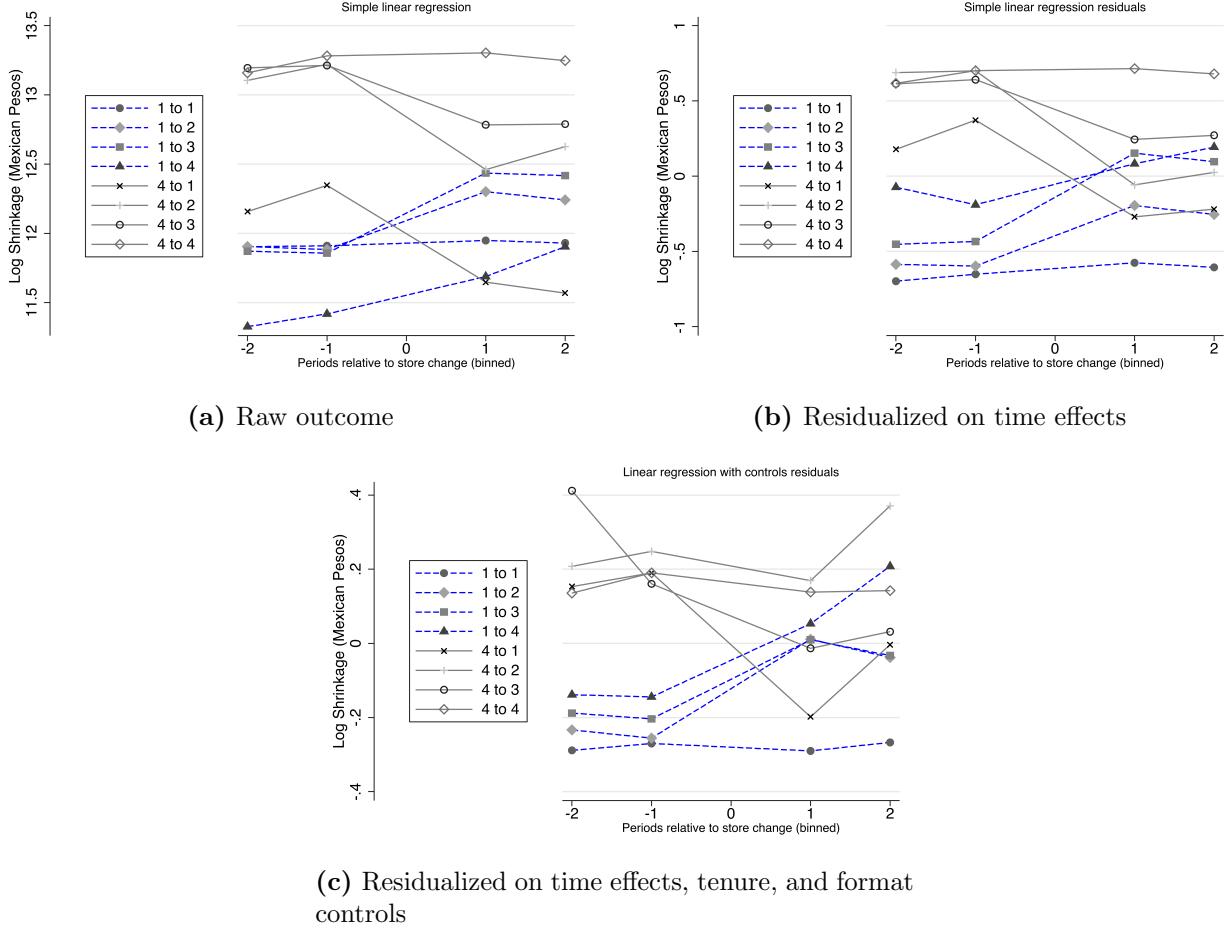


Figure 7: Event study around manager moves, by origin and destination store quartile (log shrinkage value)

[JS: These are huge effects! If a manager can impact shrinkage this much then they must have a huge impact on profits per your earlier argument. Is that true? — (sobre el panel residualizado con controles) Need to be careful how we do this, see Cattaneo(?) et al.]

Notes: Stores are ranked within connected set by quartile of average log shrinkage value. Lines labeled “ i to j ” trace movers from a quartile- i origin to a quartile- j destination in binned periods relative to the move (± 1 : one to four months; ± 2 : more than four months). Dashed lines: origin quartile 1; solid lines: origin quartile 4.

Symmetry of gains and losses. If moves are conditionally mean-independent of the match component in Equation (32), the change in residual shrinkage from moving up the store-quality ladder should be equal and opposite to the change from the reverse move; if instead managers moved to stores where their particular match is favorable, upward and downward moves would both look advantageous and the symmetry would break. Figure 8

plots the average residual change for each upward quartile pair against the corresponding downward pair, together with the -45 -degree line implied by symmetry. The points cluster around the line, and no pair deviates in the direction implied by match-based sorting. I interpret the pattern as consistent with the exogenous-mobility assumption. [\[JS: I don't follow the implementation or reasoning behind this test.\]](#)

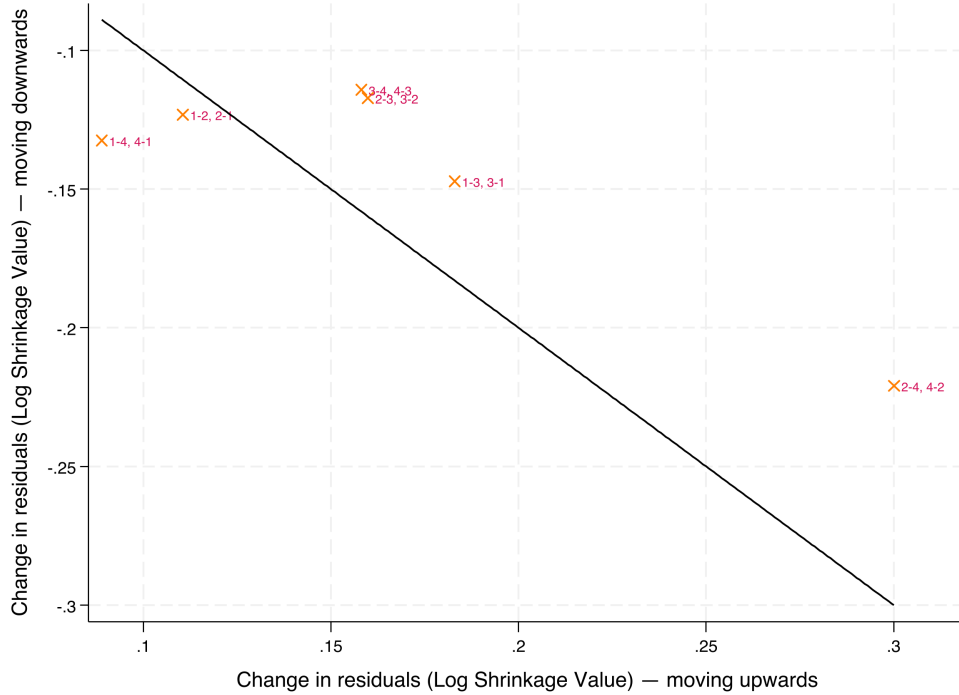


Figure 8: Symmetry of residual changes for upward and downward manager moves (log shrinkage value)

Notes: Each point is a pair of quartile transitions: the horizontal axis reports the average change in residual log shrinkage for managers moving up, the vertical axis the average change for the reverse move. Residuals are from a regression on time effects within connected set. The solid line is $y = -x$, the locus implied by symmetric gains and losses.

Limited-mobility bias. A third, purely statistical concern is that fixed effects estimated from finite mover counts are noisy, which inflates the plug-in variance components. I assess this bias directly in Appendix B, applying the homoskedastic correction of Andrews et al. (2008) and the leave-out estimator of Kline et al. (2020) on the largest connected set. [\[JS: \(ovalo grande\) Yes, this is why those earlier plots aren't quite the plots you want, I think. Much better to address these kinds of points head-on in your main results than in extensions & appendices.\]](#) The store component is essentially unaffected. The manager component shrinks under the corrections—to roughly a third of its plug-in value in the most demanding version—but it remains statistically and economically non-zero, so the manager's role in shrinkage is not a

by-product of limited mobility. And the weight of the argument does not rest on variance components in any case: the decisive evidence is the portability exercise of Section 8, which leverages the firm's rotation of managers directly and asks whether shrinkage moves when the manager moves.

8 The Portability of Manager Effects: Good-Manager Arrivals

The decomposition of Section 7 attributes a large share of the persistent variation in shrinkage to manager fixed effects. This section asks whether that effect is *portable*—whether it travels with the manager. If $\hat{\theta}_m$ is a genuine attribute of the person, a store’s shrinkage should fall the month a high-type manager arrives and not before; if it is a repackaged store characteristic or a mean-reverting shock, an arrival should move nothing on impact. Because I classify managers as high-type from the decomposition itself, this arrival test *validates* that the effect is a property of the person rather than an independent estimate of managerial quality. It passes: shrinkage falls the month a high-type manager arrives—by roughly a fifth to a quarter—and stays down, with no pre-trend. The firm therefore holds a monitor it can relocate, and placing a high-type manager in a store is the operational counterpart of raising μ_m in the model of Section 3.

8.1 Design

Classifying managers. I classify a general manager as high-type if her AKM fixed effect for log shrinkage value—estimated in Section 7—lies below the median of the manager-effect distribution within her connected set, and as low-type otherwise.

Arrival events. An arrival occurs the first time a store comes under a high-type manager for two consecutive months, and the first of those months is the event date. The two-month rule screens out temporary covers and end-of-month payroll overlaps. The estimation sample retains only ever-treated stores, so identification comes from the timing of arrivals across stores rather than from any comparison with stores that never receive a high-type manager. About one hundred sixty stores experience such an arrival during the sample period.

Specification. I estimate the event-study regression

$$Y_{jt} = \sum_{\substack{k=-8 \\ k \neq -1}}^8 \beta_k \mathbf{1}[t - E_j = k] + \varphi_j + \lambda_{m(t)} + \xi_{y(t)} + \varepsilon_{jt}, \quad (34)$$

where E_j is store j ’s arrival date, φ_j is a store fixed effect, $\lambda_{m(t)}$ and $\xi_{y(t)}$ are calendar-month and year effects, the endpoint dummies bin event times beyond ± 8 months, and the month before arrival ($k = -1$) is the omitted reference. Standard errors are clustered at the store level. The coefficients β_k trace the path of shrinkage relative to the month before arrival; the

pre-arrival coefficients test for selection, and the post-arrival coefficients measure the change. A companion difference-in-differences specification replaces the event-time dummies with a single $\text{Post}_{[0,8]}$ indicator for the first nine months under the new manager, summarizing the path in one coefficient and extending it to the other personnel outcomes.

8.2 Results: Shrinkage Falls on Impact When a Good Manager Arrives

The arrival of a high-type manager lowers store shrinkage durably. Shrinkage drops about a quarter—0.27 log points, or 24 percent—in the month the manager arrives, and stays down for the rest of the window; the eight months *before* arrival are flat and indistinguishable from zero. Figure 9 plots the path and Table 10 collapses it into a single coefficient. A fixed store characteristic cannot produce an on-impact break, and a mean-reverting shock cannot stay flat beforehand and then fall at the change, so $\hat{\theta}_m$ behaves as a portable property of the manager rather than a feature of the building.

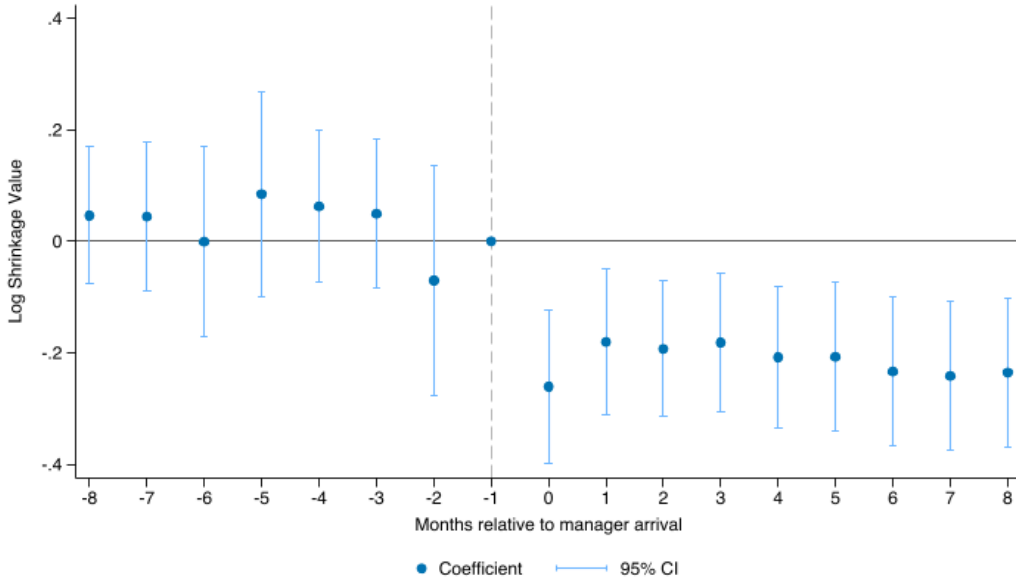


Figure 9: Event study around the arrival of a high-type manager (log shrinkage value)

Notes: Coefficients β_k from Equation (34); the month before arrival ($k = -1$) is the omitted reference. High-type = manager AKM fixed effect for log shrinkage value below the format-specific median (Section 7). Sample: ever-treated stores. Bands are 95 percent confidence intervals; standard errors clustered at the store level.

With the first nine post-arrival months pooled into one indicator, the difference-in-differences of Table 10 puts the reduction at 6.8 percent, significant at the one-percent level.

Table 10: Effect of a high-type manager arrival: difference-in-differences

	(1) Log shrink value	(2) Log shrink/ sales	(3) Log shrink/ worker	(4) Log NJ absence p.e.	(5) Log turnover p.e.
HT manager arrival ($\text{Post}_{[0,8]}$)	−0.0683*** (0.0146)	−0.0341** (0.0164)	−0.0680*** (0.0166)	−0.0020 (0.0251)	0.1738** (0.0724)
R-squared	0.932	0.822	0.737	0.554	0.386
Dep. mean	12.43	0.84	8.31	−0.24	−3.32

Notes: $\text{Post}_{[0,8]} = 1$ for event times 0 through 8. Store, calendar-month, and year fixed effects; standard errors clustered at the store level; ever-treated stores only. Per-worker (p.e.) outcomes are logs of rates. High-type classification and panel as in Figure 9. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Because arrivals are staggered across stores, the five columns of Table 11 must be read with care rather than as a range. Under staggered timing the pooled two-way fixed-effects regression presses stores treated *earlier* into service as controls for stores treated later while their shrinkage is still depressed, shrinking the estimated gap (Goodman-Bacon, 2021); the interaction-weighted estimator of Sun and Abraham (2021) (SA) is anchored instead to the never-treated stores, which are selected on manager type—the very reason I do not report the Callaway–Sant’Anna estimator. The clean-control columns strip both artifacts: restricting the *same* two-way regression to the event window $[-8, 8]$, which drops the still-treated months from the comparison set, deepens the estimate from 6.8 to 19.3 percent, in line with the dynamic-effects estimator of de Chaisemartin and D’Haultfoeuille (2020) (dCdH, 17.6 percent) and the imputation estimator of Borusyak et al. (2024) (BJS, 17.4 percent), both matching the depth of the event-study path in Figure 9. On a common window against uncontaminated controls, the estimators converge on a reduction of roughly 17–19 percent. Appendix C reports the full battery for high- and low-type arrivals.

Table 11: The high-type arrival effect across staggered-DiD estimators

Outcome	Contaminated controls		Clean controls		
	TWFE	SA	TWFE _[-8,8]	dCdH	BJS
Log Shrink Value	-0.0683*** (0.0146)	-0.0621*** (0.0174)	-0.1929*** (0.0585)	-0.1756*** (0.0628)	-0.1743*** (0.0365)
Log Shrink/Sales	-0.0341** (0.0164)	-0.0646*** (0.0170)	-0.1264** (0.0512)	-0.1480** (0.0601)	-0.1268*** (0.0417)
Log Shrink/Worker	-0.0680*** (0.0166)	-0.0566*** (0.0181)	-0.1125* (0.0636)	-0.1019 (0.0622)	-0.1070** (0.0426)
Log NJ Abs p.e.	-0.0020 (0.0251)	-0.0587** (0.0234)	-0.0102 (0.0565)	-0.0989 (0.0794)	0.0378 (0.0375)
Log Turnover p.e.	0.1738** (0.0724)	-0.0121 (0.0633)	-0.3589* (0.2063)	-0.4823** (0.1980)	-0.1792 (0.1128)

Notes: General-manager panel; treatment as in Table 10. Columns: pooled two-way fixed effects (TWFE, coincides with Table 10); Sun and Abraham (2021) (SA); the same two-way regression restricted to the event window $[-8, 8]$ (TWFE_[-8,8]), which drops still-treated months from the comparison set ($\approx 1,500$ observations, ≈ 160 stores); de Chaisemartin and D’Haultfoeuille (2020) (dCdH); and Borusyak et al. (2024) (BJS). The contaminated columns lean on comparisons that include still-treated months (TWFE) or never-treated stores selected on manager type (SA). Full detail, low-type arrivals, event-study figures, and why the Callaway–Sant’Anna estimator is not reported appear in Appendix C. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Departures. The mirror image holds. When a high-type manager *leaves* a store, shrinkage climbs back: the departure raises log shrinkage value by about 6.5 percent ($p = 0.05$), reversing roughly nine-tenths of the arrival reduction, and the event-study path in Figure 10 shows the rise building over the months after departure on top of a flat pre-departure path. Not everything reverts, though: the store’s *stock* of inventory does not climb back (an insignificant change), so what leaves with her is the ongoing discipline her presence supplies, not a restructuring of the store itself. This asymmetry anchors the mechanism in Section 9.

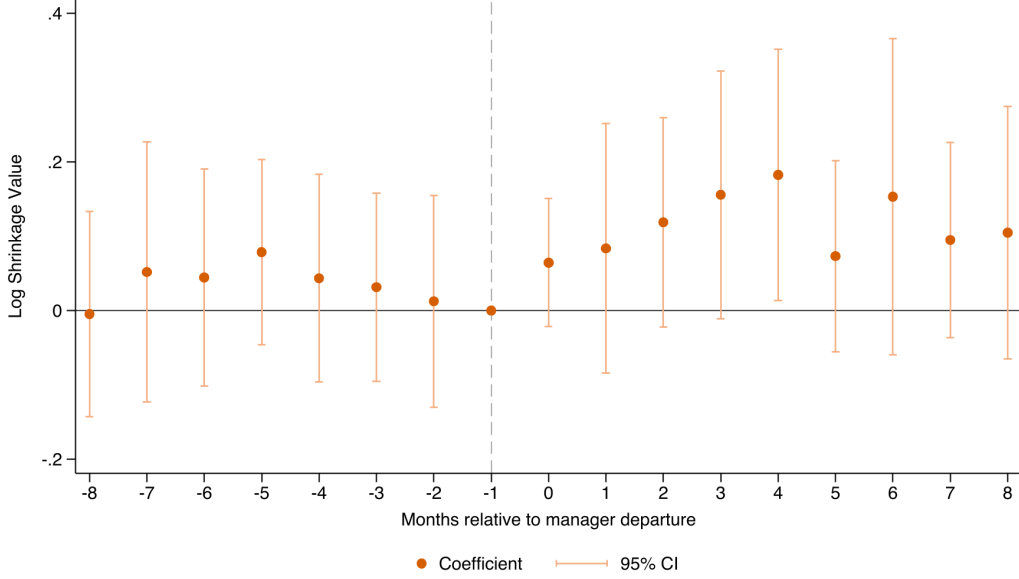


Figure 10: Event study around the departure of a high-type manager (log shrinkage value)

Notes: Mirror of Figure 9. The event is a high-type manager leaving a store (the first non-high-type month following a high-type spell); the month before departure ($k = -1$) is the omitted reference. Sample: stores that experience a departure ($\approx$ two dozen events), so individual coefficients are imprecise—the difference-in-differences summary is the sharper statement. Bands are 95 percent confidence intervals; standard errors clustered at the store level.

The arrival design establishes that manager quality is real, portable, and first-order for shrinkage, but it treats the manager as a black box: it shows *that* a high-type manager lowers shrinkage without saying who these managers are or what they do once inside the store. Section 9 opens the box. It asks whether high-type managers differ from their peers in pay, experience, or education; whether they achieve their effect by reorganizing the store or by changing their own conduct within it; and through which concrete channel—above all their presence on the floor—they cut shrinkage. Answering them turns this section’s portable fixed effect into the monitoring margin the model calls μ_m .

9 What Do Good Managers Do?

In the model, the premium deters shirking only if shirking is *detected*, so the manager’s contribution should run through monitoring—the μ_m .

This section opens up μ_m a good manager’s monitoring is her own presence on the floor—not a larger apparatus or a rebuilt team. I carry the good-manager arrival design of Section 8 to a curated set of operational and personnel outcomes and ask what, concretely, the manager changes.⁶

The answer has two parts. A good manager (i) reallocates her own time toward the floor, working more shifts and overlapping more of her workers’ hours; and (ii) redeploys the workers she inherits, rotating and promoting incumbents. In pay and schooling she is otherwise ordinary.

Presence—her hours on the floor alongside her workers—is the empirical content I give the monitoring term μ_m of the model in Section 3: the manager makes the firm’s dismissal threat credible by being there to notice, so that a high-type posting is the operational counterpart of raising μ_m .

I read the presence margin against two questions: does it switch on when a high-type manager arrives, and does it switch off when she leaves? I report the arrival effect alongside its mirror-image *departure* effect, estimated around the roughly two dozen high-type departures and read as suggestive.

9.1 Action 1: The Manager on the Floor

The one input a manager most directly controls is her own time on the floor. I measure it from the minute-level time-clock records, which log the start and end of every worked shift for every employee, the manager included. From these I construct, at the store-month level, the manager’s own presence—shifts worked, arrival time, weekend and night coverage—and, more consequentially, *co-presence*: the share of frontline worker shift-hours worked while the store’s general manager is simultaneously on the clock, computed by overlapping each worker’s shift with the manager’s daily presence interval and averaging over the store-month. I bring these measures into the arrival design of Section 8.

For a store-month outcome y_{jt} I estimate

$$y_{jt} = \beta \text{Post}_{[0,8],jt} + \varphi_j + \lambda_{m(t)} + \xi_{y(t)} + \varepsilon_{jt}, \quad (35)$$

⁶Because a high-type manager is classified from her AKM fixed effect in shrinkage (Section 7), the arrival effect on shrinkage itself is best read as a validation of that decomposition rather than as a fresh result. The mechanism outcomes I study below—presence and personnel allocation—are not derived from the shrinkage fixed effect, so they carry no such circularity.

where φ_j are store fixed effects, $\lambda_{m(t)}$ and $\xi_{y(t)}$ are calendar-month and year fixed effects, and $\text{Post}_{[0,8]}$ switches on for the first nine months following a high-type arrival, dated and screened exactly as in Section 8. Identification is within-store, before versus after the arrival, and standard errors are clustered at the store; β is the coefficient reported in each cell.

Table 12 makes the presence channel concrete. A high-type manager works 1.7 more shifts per month than the manager she replaces—roughly two additional days on the floor—while the store’s overall shift coverage, its weekend and night staffing, does not change. Because the manager is present more without the store operating differently, the frontline is supervised more directly: the share of worker-hours worked alongside the general manager rises by 3.9 percentage points, and the share of worker shifts that overlap the manager at all rises by 4.4 points, off bases near 57 and 70 percent. Low-type arrivals show the mirror image. They do not raise the manager’s shift count, they push her average arrival time about fifteen minutes later, and they leave co-presence unchanged.

Table 12: Managerial presence and manager–worker co-presence (arrival design)

[\[JS: This is great. Would love to see it visually in a form like the arrival event-study figure.\]](#)

	(1) GM shifts /month	(2) GM arrival hour	(3) GM % weekend	(4) Store % night	(5) Co-presence (worker-hrs)	(6) Any GM overlap
HT manager arrival	1.6958*** (0.4725)	0.0371 (0.0909)	−0.0068* (0.0039)	−0.0019 (0.0026)	0.0387*** (0.0120)	0.0437*** (0.0135)
LT manager arrival	−0.0497 (0.8728)	0.2479** (0.1131)	0.0055 (0.0037)	0.0034 (0.0034)	−0.0092 (0.0181)	−0.0047 (0.0231)
Dep. mean	23.48	8.64	0.296	0.368	0.574	0.700

Each cell is the $\text{Post}_{[0,8]}$ coefficient from the arrival design of Equation (35) with store, year, and month fixed effects and standard errors clustered at the store, estimated on store-month measures built from shift-level time-clock records. “GM shifts” counts the general manager’s worked shifts; “GM arrival hour” is her mean first-punch time; “GM % weekend” her share of shifts on Saturday or Sunday; “Store % night” the share of all worker shifts on night hours. “Co-presence” is the mean share of a worker shift’s hours overlapping the manager’s daily presence interval; “Any GM overlap” the share of worker shifts with a positive overlap. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 11 traces the same result as an event study. Manager–worker co-presence is flat in the months before a high-type arrival and steps up on impact, rising by roughly eight to nine percentage points and holding there.⁷ Figure 12 reports the same event study for the remaining presence measures.

⁷Coefficients are relative to the average of the pre-arrival months; the one-month handover window ($t = -1$), when the outgoing manager is already gone and the incoming one not yet in place, is mechanically depressed and is omitted. Referenced to the pre-arrival baseline the increase is about twice the $\text{Post}_{[0,8]}$ coefficient of Table 12, for the same reason discussed in Section 8: the pooled window keeps still-treated months, where co-presence is still elevated, inside the comparison group.

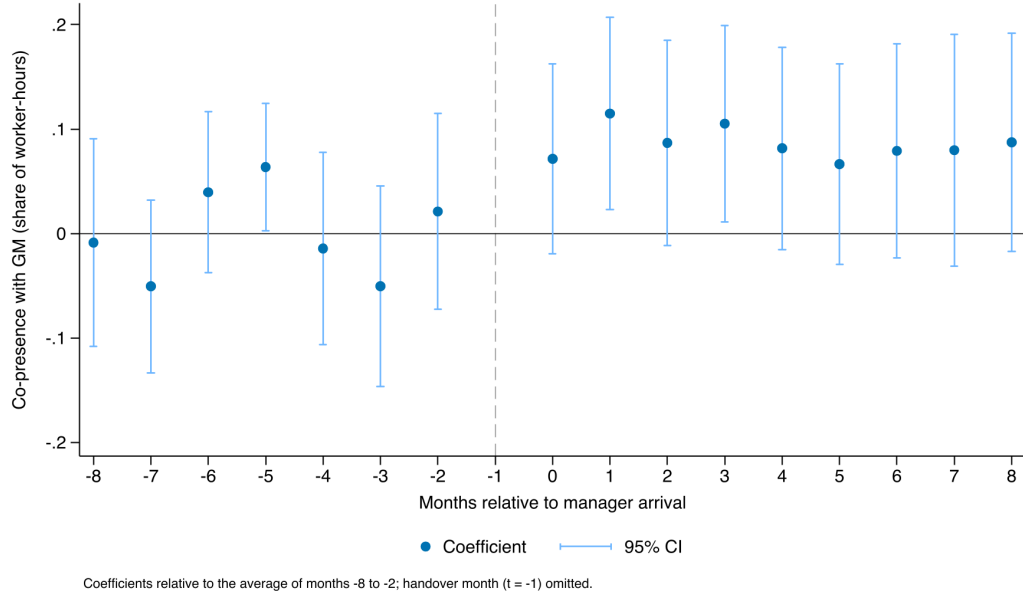
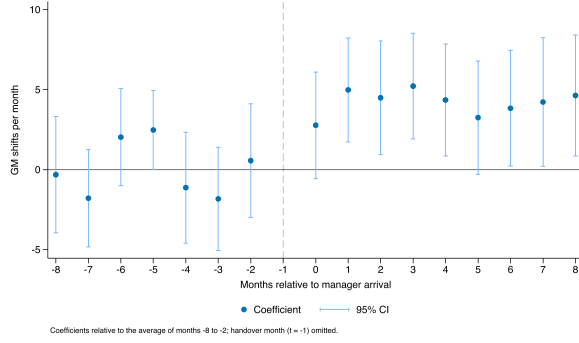
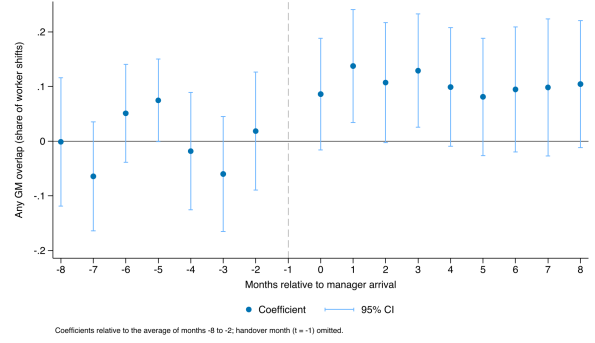


Figure 11: Event study around the arrival of a high-type manager: manager–worker co-presence

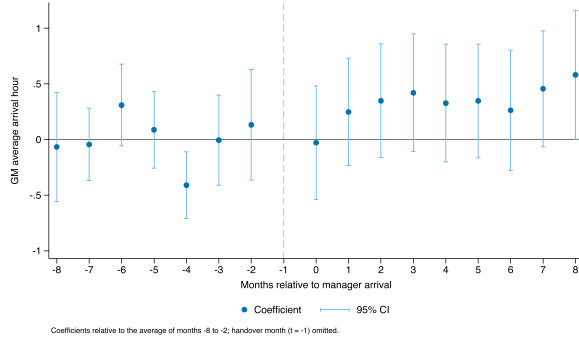
Notes: Store-month manager–worker co-presence (mean share of a worker shift’s hours overlapping the general manager’s presence interval) around a high-type manager’s arrival, in the style of Figure 9. Coefficients are relative to the average of event months -8 to -2 ; the handover month $t = -1$ is omitted. Bands are 95 percent confidence intervals, clustered at the store.



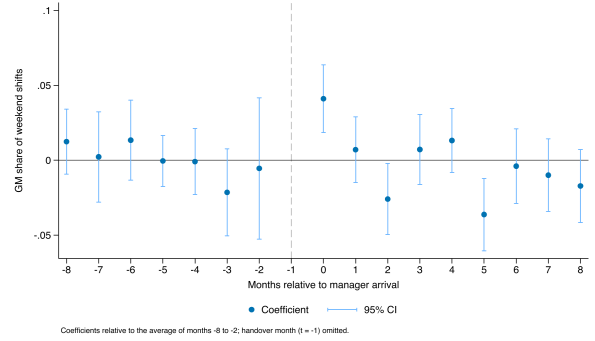
(a) GM shifts per month



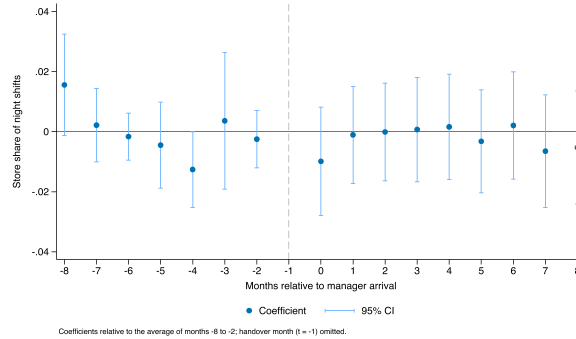
(b) Any GM overlap (share of worker shifts)



(c) GM arrival hour



(d) GM % weekend



(e) Store % night

Figure 12: Event studies for the remaining presence measures of Table 12

Notes: Each panel is the arrival event study for one column of Table 12, in the style of Figure 11: coefficients relative to the average of event months -8 to -2 , the handover month $t = -1$ omitted, with store, year, and month fixed effects and store-clustered 95% confidence intervals. Managerial presence (GM shifts) and co-presence (any GM overlap) rise on arrival and hold; the manager's arrival hour, her weekend share, and the store's night-shift share do not move.

High-type managers do not buy more monitoring—they add no loss-prevention, supervisory, or security staff, as Section 9.2 shows. They *are* present more, and their workers keep better time when they are (Appendix D.4). This reading connects the manager result to the classic evidence that monitoring deters opportunism (Nagin et al., 2002) and that

workers lift effort when observed (Mas and Moretti, 2009). Do these presence margins also switch off when she leaves? Table 13 runs the same six measures under the mirror-image departure design. The arrival effects are large and precise, but the departure effects, estimated from only about two dozen departures, all shrink toward zero and are too imprecise to sign—consistent with the gains fading when she goes, though not decisive on their own. The clearer reversal is on shrinkage itself, which rises again by about 0.065 log points after a high-type manager departs (Section 8). Read together, the presence channel behaves like an attribute the manager carries with her.

Table 13: The presence measures of Table 12 on arrival and on departure

	GM shifts	GM arrival hour	GM % weekend	Store % night	Co-presence	Any GM overlap
High-type arrival	1.696*** (0.473)	0.037 (0.091)	−0.007* (0.004)	−0.002 (0.003)	0.039*** (0.012)	0.044*** (0.014)
High-type departure	0.410 (1.449)	0.181 (0.196)	−0.003 (0.009)	−0.004 (0.007)	−0.006 (0.032)	0.004 (0.038)

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a store-month arrival design (top row) or the mirror-image departure design (bottom row), for the six measures of Table 12, with store, year, and month fixed effects and standard errors clustered at the store. The departure design is identified from roughly two dozen departures, so its estimates are imprecise and read as suggestive. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

9.2 Action 2: Reallocating the Workers She Inherits

A high-type manager could lower shrinkage through two channels this subsection can rule out before returning to the one it cannot. She could churn the workforce into a better team, or she could thicken the store’s apparatus of monitoring. She does neither: she works the people and the floor she already has, and it is the second of these—her own presence—that the rest of the paper turns on.

She reworks the team she inherits; she does not replace it. Table 14 brings each organizational and personnel variable into the arrival design, and Table 15 decomposes the within-store moves by direction. The reallocation is real but modest. Overall rotation rises by 0.33 per hundred workers on a base of 3.6 (a nine percent increase), concentrated in *upward* and lateral moves: reassignments to a higher tier rise by 0.17 per hundred (the largest proportional response), an independent promotions record confirms the upward margin (+0.25 per hundred, $p < 0.05$), and downward moves do not budge. These magnitudes perturb roughly one worker in a typical store in a given month—enough to redirect attention

and reward performance, an order of magnitude too small to be a restructuring. And she does not change *who* the store employs: headcount barely moves, and the composition of the team is untouched. Measuring diversity by the typology of Harrison and Klein (2007)—*separation* (the within-store standard deviation of age, tenure, and education), *variety* (the Blau index of gender, tier, and occupational area), and *disparity* (the coefficient of variation of pay)—Table 16 shows a high-type arrival moves none of them; the lone exception, gender variety, is 0.003 on an index bounded at 0.5, and economically negligible. The shrinkage reduction is therefore not a selection or composition story. The good manager redeploys the incumbents she has, upward and laterally, and keeps the team as it is.

Table 14: What managers change on arrival (within-store arrival design)

	(1) Loss-prev. staff/100	(2) % super- visors	(3) % security staff	(4) Log team size	(5) Promotions /100	(6) Conflict quits/100
HT manager arrival	0.0966 (0.1652)	−0.6352* (0.3761)	−0.1231 (0.1771)	0.0210** (0.0096)	0.2530** (0.1259)	−0.0020 (0.0439)
LT manager arrival	−0.0976 (0.1151)	0.1403 (0.4700)	0.2151* (0.1296)	−0.0037 (0.0101)	−0.0630 (0.1367)	0.0839*** (0.0301)

HT: ≈ 150 stores, 3,578 store-months. LT: ≈ 140 stores.

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a separate store-month arrival regression (Equation 35) with store, year, and month fixed effects and standard errors clustered at the store. Columns 1–3 measure monitoring organization; columns 4–6 team composition and personnel practices. Rate outcomes are per hundred workers. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 15: Internal reallocation is upward, lateral, and across areas

	Position change (tier-based)				Move records	
	All	Upward	Lateral	Downward	Promotions	Demotions
High-type arrival	0.332** (0.154)	0.165** (0.069)	0.160 (0.098)	0.008 (0.063)	0.253** (0.126)	−0.012 (0.018)
Low-type arrival	−0.058 (0.195)	0.024 (0.079)	−0.087 (0.111)	−0.005 (0.071)	−0.063 (0.137)	0.030 (0.025)
Mean of outcome	3.59	1.02	1.92	0.62	2.43	0.12

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a separate store-month arrival regression (Equation 35), rates per hundred workers, with store, year, and month fixed effects and standard errors clustered at the store. The first columns classify a within-store position change by the hierarchical tier of its origin and destination (upward, lateral, downward); the last columns are promotion and demotion counts from the personnel-movement record. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 16: Team diversity does not change on a high-type arrival

	Separation (SD)			Variety (Blau)			Disparity
	Age	Tenure	Educ.	Gender	Tier	Area	Pay (CV)
High-type arrival	0.029 (0.070)	−0.128 (0.451)	−0.007 (0.021)	0.003** (0.001)	−0.001 (0.003)	0.000 (0.003)	0.007 (0.005)
Low-type arrival	−0.045 (0.066)	−0.460 (0.467)	−0.052*** (0.017)	−0.000 (0.002)	−0.003 (0.003)	−0.003 (0.004)	0.005 (0.005)
Mean of outcome	12.4	31.4	2.59	0.477	0.667	0.651	0.500

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a separate store-month arrival regression (Equation 35) with store, year, and month fixed effects and standard errors clustered at the store. Diversity is measured following Harrison and Klein (2007): separation by the within-store standard deviation of age, tenure, and education; variety by the Blau index ($1 - \sum_k p_k^2$) of gender, tier, and occupational area; disparity by the coefficient of variation of monthly pay. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Monitoring, here, is management—not a surveillance apparatus. The first three columns of Table 14 measure the machinery of monitoring a store could add: the share of loss-prevention personnel, of supervisory staff, and of security workers. None responds to a high-type arrival; if anything the supervisory share falls slightly. A good manager does not buy more monitoring. The monitoring that the model’s μ_m requires is therefore not a technology she installs but her own presence on the floor—the shifts she works and the hours she overlaps with her workers (Section 9.1). In the model’s terms, μ_m is the manager watching, not a larger security team.

By ruling out a composition channel and a surveillance-apparatus channel, the reallocation evidence isolates the manager’s own supervisory presence as the technology behind the shrinkage reduction—the empirical content of μ_m the model asks for.

9.3 Otherwise Ordinary

These actions aside, high-type managers are unremarkable on the traits a personnel file would use to explain quality. On the observables a hiring or promotion decision prices—pay and schooling—the two types are statistically indistinguishable; they differ only marginally and along career-path dimensions (a touch older, slightly *less* tenured) that the firm could not read as competence in advance. The shrinkage classification of Section 7 is therefore not recoverable from standard human-resources records: what marks a good manager here is the presence documented above, not a credential the firm already prices. Appendix D.5 reports the full observable comparison, its implication for how the firm prices this quality, and a suggestive test of whether the manager’s vertical distance from her team sharpens her monitoring.

9.4 Validating the Classification

Before drawing on these results, I document that the shrinkage classification captures a broad dimension of managerial quality rather than a narrow specialty, and that the arrival effect is graded in the size of the upgrade the arrival represents.

A fixed effect estimated from shrinkage alone might capture skill at a single task—inventory paperwork, say—rather than managerial quality. I therefore re-estimate the AKM decomposition of Equation (31) separately for five additional outcomes (two alternative shrinkage definitions, turnover, and two non-justified-absence measures) and compare each manager’s fixed effects across outcomes. Figure 13 is a scatterplot matrix of the six manager fixed effects, the core shrinkage-value effect in the first row and column. The three shrinkage-based effects are strongly correlated (0.63–0.85), and the shrinkage-value effect also comoves with the manager effects in turnover (0.38) and, most strongly, in non-justified absence (0.65–0.82). A comparison of mean fixed effects across the two types points the same way: high-type managers carry better (lower) fixed effects in every other outcome—strongly so for the shrinkage normalizations and turnover ($p < 0.001$), more weakly for the two absence measures ($p = 0.06$ – 0.09). HT Managers run stores with less absenteeism and less turnover in general—the classification captures a broad dimension of quality, in line with the manager-specific productivity effects of Lazear et al. (2015) and Bertrand and Schoar (2003), rather than a narrow technical skill.

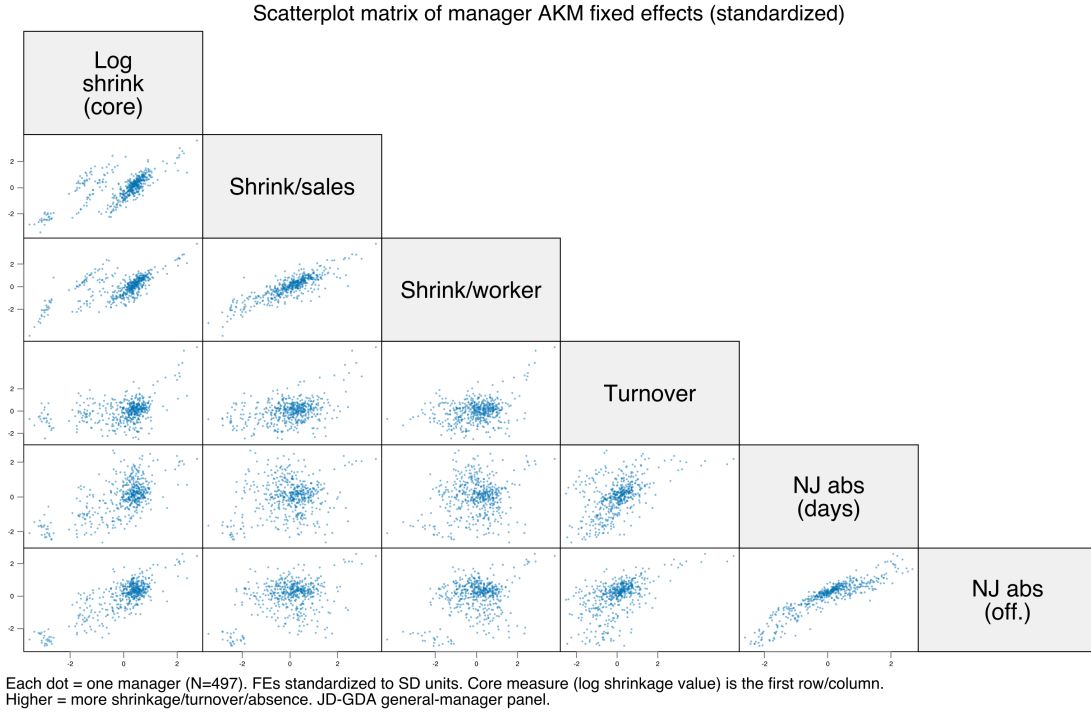


Figure 13: The shrinkage classification captures general managerial quality

Notes: Scatterplot matrix of each manager’s AKM fixed effects across the six outcomes ($N = 497$ general managers), each estimated from a separate AKM decomposition (Equation 31) and standardized to standard-deviation units; the core measure, log shrinkage value, is the first row and column. Higher values mean more shrinkage, turnover, or absence, so the positive comovement in every panel shows that a manager who controls shrinkage well also runs a store with less turnover and absence—the classification captures a broad dimension of quality. Pairwise Pearson correlations range from 0.38 (turnover) to 0.85 (across shrinkage definitions).

The arrival effect is also graded rather than binary: it scales with how much the arriving manager improves on the incumbent she replaces, and vanishes for arrivals that are not upgrades—a dose response and a novelty placebo I report in Appendix D.1. In sum, the good manager’s technology for shrinkage is her own supervisory presence—the μ_m of the model—not a structure she installs. Section 10 tests whether that enforcement and the wage premium reinforce each other.

10 Managers and the Wage Premium: Complements in Discipline

Are the wage premium and the manager’s monitoring *complements* or *substitutes* in producing discipline? The model of Section 3 answers it. There the expected penalty for shirking is a single product—the probability that shirking is detected times the rent the worker loses if dismissed—so the two are not separate deterrents but two factors of one threat: a rent that no one monitors deters nothing, and a monitor with no rent to confiscate has no threat to make. The cross-partial is therefore positive, and the model predicts that a good manager cuts shrinkage by more where the wage premium is high. This section brings the two halves measured separately so far—the premium of Section 4 and the monitor of Sections 7–9—into one design and tests whether they multiply.

10.1 Design

To test the prediction, I take the high-type arrival design of Section 8 and ask one question of it: is the shrinkage reduction a high-type manager delivers larger where the store’s wage premium is high?

I classify each store’s municipality, in the month its manager arrives, as a high-wage-premium (HWP) or low-wage-premium (LWP) environment—HWP if the premium the firm pays over the local ENOE market wage lies above that municipality’s own median over the sample, LWP otherwise. Fixing the environment at arrival makes it pre-determined with respect to everything that follows and constant within a store’s event window. I then re-estimate the arrival difference-in-differences of Equation (35), interacting the $\text{Post}_{[0,8]}$ indicator with this environment, separately for high- and low-type arrivals.

The complementarity hypothesis makes one prediction: the high-type shrinkage reduction should be larger—more negative—in high-premium stores than in low-premium stores. If the rent and the monitor were independent deterrents, that difference would be zero; if they are the two factors of one product, it is negative.

10.2 Good Managers Accomplish More Where the Premium Is High

Table 17 reports the estimates. A high-type manager arriving in a high-premium environment cuts log shrinkage value by 12.6 percent, shrinkage per worker by 13.8 percent, and shrinkage over sales by 7.6 percent—roughly three times the reductions the same arrival produces in

a low-premium environment (4.3, 3.7, and 1.5 percent). The difference between the two environments, the complementarity the model signs, is 8.4 percentage points for shrinkage value and 10.1 for shrinkage per worker, significant at the five and one percent levels. Where the premium gives the manager’s dismissal threat something to confiscate, the same manager accomplishes more. Two further features stand out. Non-justified absence does not respond to the manager in either environment (column 4), consistent with Section 8: the manager margin operates on the inventory discipline she polices, not on the attendance a worker internalizes directly through her own rent. And turnover rises significantly after a high-type arrival only where the premium is high (column 5)—a first sign of the enforcement margin the mechanism runs through, the manager exercising her threat most where the job she protects is worth most—though the difference between environments is not itself precisely estimated.

Table 17: High-type manager arrivals by wage environment at arrival

	(1) Log shrink value	(2) Log shrink/ sales	(3) Log shrink/ worker	(4) Log NJ abs. p.e.	(5) Log turnover p.e.
HT arrival, high-premium (HWP)	−0.126*** (0.032)	−0.076*** (0.027)	−0.138*** (0.035)	−0.049 (0.053)	0.309*** (0.115)
HT arrival, low-premium (LWP)	−0.043** (0.018)	−0.015 (0.021)	−0.037** (0.018)	0.018 (0.030)	0.115 (0.080)
Difference (HWP − LWP)	−0.084** (0.037)	−0.061* (0.033)	−0.101*** (0.038)	−0.066 (0.064)	0.194 (0.126)
Dep. mean	12.43	0.84	8.31	−0.24	−3.32

$\text{Post}_{[0,8]} = 1$ for event times 0 through 8, interacted with the wage environment of the store’s municipality in the arrival month (HWP = premium above the municipality’s median; LWP otherwise). The first two rows report the arrival effect in each environment; the third row their difference (the complementarity). Store, calendar-month, and year fixed effects; standard errors clustered at the store; ever-treated stores only. HT classification, event definition, and outcomes as in Table 10. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 14 shows the same split as event studies.

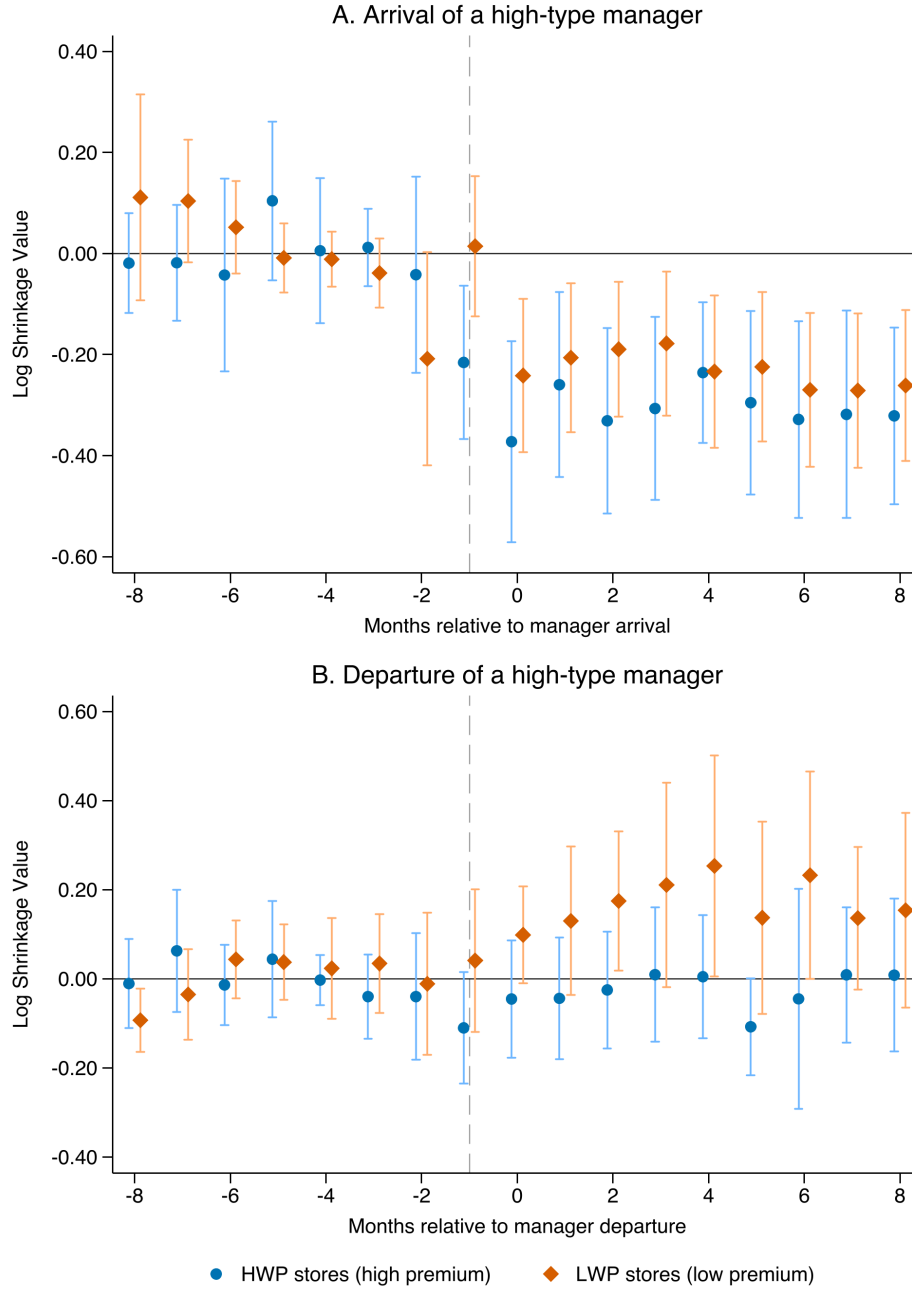


Figure 14: Arrival and departure of a high-type manager, by wage environment

Notes: Event studies of log shrinkage value around a high-type manager's arrival (panel A) and departure (panel B), estimated separately for high-premium (HWP) and low-premium (LWP) store environments, with the environment fixed at the event month as in Table 17. Coefficients are relative to the average of event months -8 to -2 ; the handover month $t = -1$ is omitted. Bands are 95 percent confidence intervals, clustered at the store. Arrival: 38 HWP and 121 LWP events. Departure: about a dozen events per environment, so the departure split is descriptive only.

The binary split has a continuous companion. Figure 15 replaces the high/low dichotomy

with the standardized premium itself, plotting the marginal effect of a high-type arrival on log shrinkage as a smooth function of the premium the store’s municipality pays at the moment the manager arrives. The line slopes steadily downward: the arrival effect is a modest -4 percent where the premium is low and reaches -11 percent where it is high, tracing out as a gradient the same contrast the table reports as two cells. The complementarity is not an artifact of where the median falls.

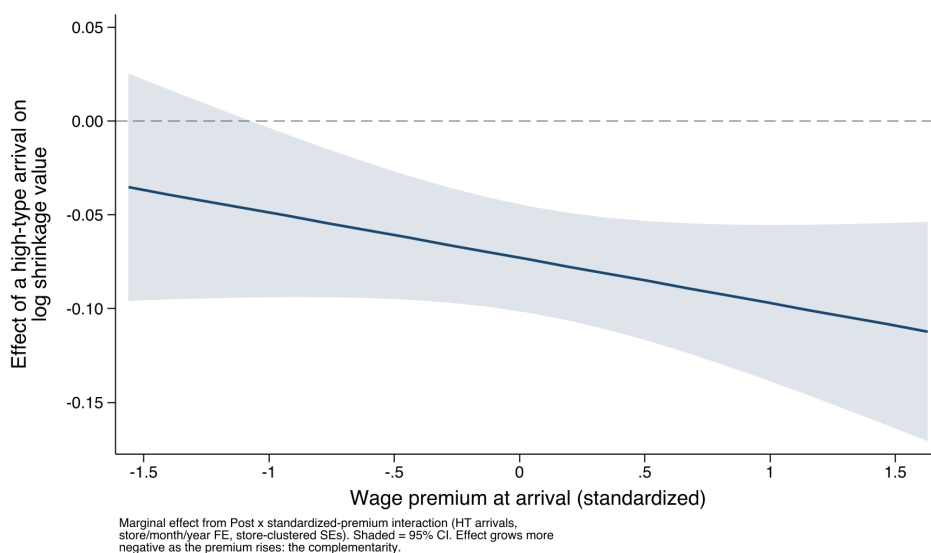


Figure 15: The high-type arrival effect on shrinkage grows more negative as the wage premium rises

[JS: (circulo) Replace with the event study I mention just above.]

Notes: Marginal effect of a high-type arrival ($\text{Post}_{[0,8]}$) on log shrinkage value as a function of the standardized wage premium of the store’s municipality at arrival, from the interaction of $\text{Post}_{[0,8]}$ with the continuous premium (store, calendar-month, and year fixed effects; standard errors clustered at the store). Shaded band is the 95 percent confidence interval; the plotted range spans the 5th–95th percentiles of the arrival premium.

The high-type result shows that the premium amplifies a good manager. The model makes a prediction for bad managers as well. If the rent disciplines the shrinkage margin only through a monitor who makes detection credible, then under a low-type manager, who supplies little supervision, no premium the firm pays has anything to threaten on that margin. A low-type arrival should therefore raise shrinkage in *both* environments, with no premium scaling—the mirror image of the high-type pattern.

Table 18 confirms it. A low-type arrival is followed by shrinkage *increases* of 5.9 percent in value, 8.0 relative to sales, and 7.5 per worker in low-premium stores, and the estimates in high-premium stores carry the same sign and a similar magnitude, with no significant difference on any shrinkage outcome. A bad manager is bad in every wage environment; the

premium does not rescue the store. This result does not make the premium useless under a bad manager—it continues to discipline the margins the worker controls directly, attendance and retention—but on the margin the manager must police, the rent works only through the monitor. In levels, then, the two inputs are complements: the premium amplifies what a good manager accomplishes and cannot replace one.

Table 18: Low-type manager arrivals by wage environment at arrival

	(1) Log shrink value	(2) Log shrink/ sales	(3) Log shrink/ worker	(4) Log NJ abs. p.e.	(5) Log turnover p.e.
LT arrival, high-premium (HWP)	0.037 (0.033)	0.046 (0.033)	0.030 (0.034)	−0.037 (0.046)	−0.083 (0.105)
LT arrival, low-premium (LWP)	0.059*** (0.021)	0.080*** (0.022)	0.075*** (0.023)	−0.000 (0.025)	0.087 (0.071)
Difference (HWP − LWP)	−0.022 (0.039)	−0.035 (0.037)	−0.045 (0.041)	−0.037 (0.049)	−0.170 (0.126)
Dep. mean	12.46	0.85	8.39	−0.28	−3.44

Same specification and layout as Table 17, for low-type (LT) arrivals. Store, calendar-month, and year fixed effects; standard errors clustered at the store; ever-treated stores only. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Pooling the first arrival of either type into a single regression and estimating the full triple interaction tells the same story, if less precisely. Figure 16 plots the four cells for log shrinkage value: the only cell in which arrivals reduce shrinkage is high-type-in-high-premium (−8.5 percent), low-type arrivals in low-premium stores *raise* shrinkage, and the remaining two cells are indistinguishable from zero. The triple-difference coefficient itself carries the predicted negative sign for shrinkage value and shrinkage per worker but is imprecise because high-premium arrivals are scarce, so the triple difference rests on only about two dozen stores in the pivotal cell. I therefore read the pooled design as directionally consistent with the complementarity and rest the powered test on the per-type interactions of Tables 17 and 18, which use every arrival; the robustness appendix reports the full triple-interaction estimates and their cell breakdown (Table 36).

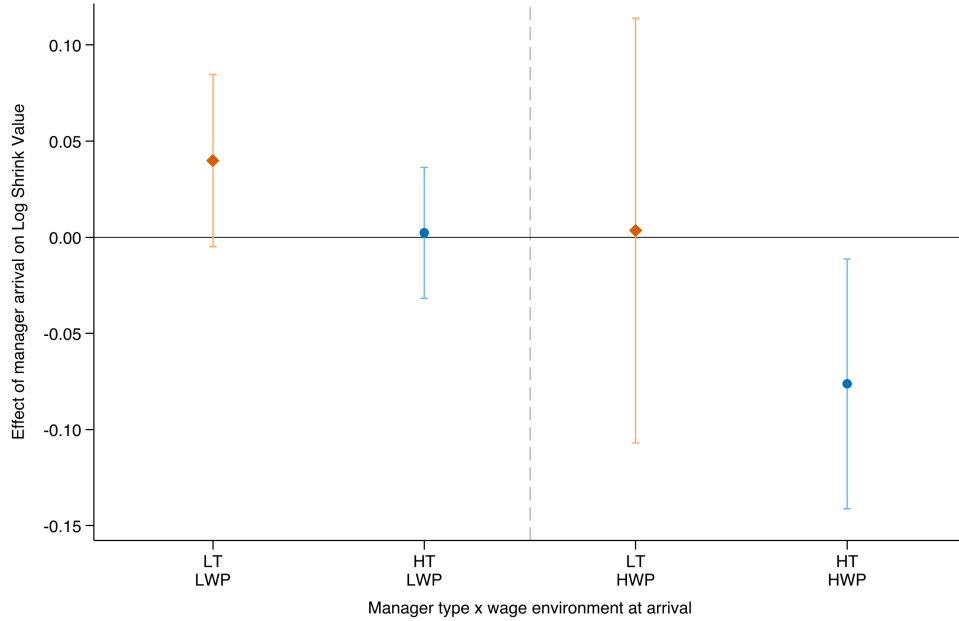


Figure 16: Effect of the first manager arrival on log shrinkage value, by manager type and wage environment at arrival

Notes: Cell effects from the pooled triple-interaction design via linear combinations; 95 percent confidence intervals, standard errors clustered at the store. Sample and definitions as in Table 36. Only the high-type-in-high-premium cell delivers a significant reduction.

10.3 Does the Manager Do More Where the Premium Is High?

Does the manager *do* something different where the premium is high? The complementarity has two readings, and they part on the manager's own conduct. Under the first, a good manager *supplies* more discipline where the rent is large: she spends more of her hours on the floor, reorganizes more aggressively, tightens the store harder—she optimizes her actions against the premium. Under the second, she supplies the *same* actions everywhere, and each one *returns* more where the rent is large, because the threat her presence carries is scaled by the surplus the worker would forfeit. The two readings send the paper's two halves in opposite directions. If the manager's actions scale with the premium, the complementarity is a story about her behavior; if they do not, it is intrinsic to the efficiency-wage mechanism—the rent doing the work, the manager only making it credible. The answer, action by action, is that the action margins are estimated too imprecisely to compare across environments, while her enforcement and her results do respond.

Table 19 crosses each action of Section 9 with the wage environment at arrival (Panel A), alongside the enforcement and shrinkage margins (Panel B). The behavioral-action differentials are individually insignificant, but their confidence intervals are wide—wide enough

to admit high- versus low-premium differences larger than the actions themselves (for promotions the interval runs to more than one-and-a-half times the low-premium effect, for cross-area reassignment to more than twice it)—so I cannot rule out that the manager also does somewhat more where the premium is high; her floor presence and inventory-tightening in fact tilt in that direction ($p = 0.10$ and $p = 0.08$). I therefore do not lean on the action margins. What is precisely estimated, and where the complementarity is significant, is enforcement and results. The disciplinary instrument she controls day to day—the suspension—rises by 0.31 per hundred workers after a high-type arrival in high-premium stores and *falls* by 0.14 in low-premium stores, a swing significant at five percent; and the shrinkage reduction is two to three times larger in high-premium stores on both the value and the per-worker measure ($p = 0.02$ and $p = 0.01$).

Table 19: Does the manager do more where the premium is high? Each action crossed with the wage environment at arrival

	Low-premium (LWP)	High-premium (HWP)	Difference (HWP–LWP)
<i>Panel A. What the manager does</i>			
Presence: manager–worker co-presence	0.020	0.063	0.043
Leaner inventory: log stock value	−0.167	−0.023	0.144*
Promotions (per 100 workers)	0.208	0.131	−0.077
Cross-area reassignment (per 100)	0.173	0.270	0.097
<i>Panel B. Enforcement and return</i>			
Suspensions (per 100 workers)	−0.140	0.311	0.451**
Shrinkage (log value)	−0.043	−0.126	−0.084**
Shrinkage per worker (log)	−0.037	−0.138	−0.101***

Each row is a separate high-type arrival regression (arrival design of Equation 35; store, calendar-month, and year fixed effects; standard errors clustered at the store; ever-treated stores) with the indicated store-month outcome, interacting $\text{Post}_{[0,8]}$ with the wage environment of the store’s municipality at arrival (HWP = premium above the municipality’s median). The first two columns report the arrival effect in low- and high-premium stores; the last column their difference (HWP – LWP). Rates are per hundred workers. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ on the difference.

This finding completes the paper’s argument. Section 4 showed that the premium deters shirking; Sections 7–9 showed that a good manager deters it, and does so through her presence on the floor; and the complementarity between them is now pinned down. It is *not* that good managers work harder where wages are high, but that the wage premium and the manager are the two factors of a single threat. The rent sets what dismissal costs the worker; the manager sets the chance the worker is caught; and neither disciplines the shrinkage margin without the other. The high-type advantage roughly triples where the premium is high not because the manager supplies more supervision there, but because the supervision she

always supplies is worth more—and because it is there that pulling the disciplinary trigger pays. The low-type mirror confirms it: nothing about a bad manager’s arrival responds to the premium, on any margin, because without a monitor to make detection credible the rent has nothing to threaten on the margin the manager polices. The mechanism appendix (Appendix E) collects the fuller battery—presence measured for the shrinkage-prone roles and the low-oversight hours, and a sharper cut at the channel (whether the reshuffling changes composition *within* occupational areas)—and it does not disturb this reading.

10.4 Robustness

I subject the complementarity to three checks, reported in full in Appendix F, and it survives all three. The pooled triple-difference that estimates all four manager-type $\times$ premium cells in a single regression reproduces the per-type message cell by cell—high-type in a high-premium environment is the only cell in which arrivals reduce shrinkage. The complementarity is not an artifact of how the wage environment is defined: re-defining it three additional ways, all fixed at arrival—a cross-sectional split against other municipalities in the same quarter, the standardized continuous premium and a cost-of-job-loss measure—preserves its sign and the ordering of the two environments in every case (Appendix F.2). And the arrival dynamics survive the correction for staggered timing: the imputation event study of Borusyak et al. (2024), estimated separately by wage environment on the genuine (non-left-censored) arrivals, recovers flat pre-trends and a deeper post-arrival reduction for high-premium than for low-premium stores—the same ordering as the binary table under a heterogeneity-robust estimator, though the small high-premium cell leaves the two series’ confidence intervals overlapping, so the powered test of the differential remains the per-type interaction above. Appendix F collects the triple-difference table, the four-definition comparison, and the dual event study.

Taken together, the designs support the reading that managers and efficiency wages are complements *in levels*: the premium roughly triples what a high-type manager accomplishes and cannot substitute for one. This is the store-level counterpart of the worker-level premium evidence of Section 4, and it matches Coviello et al. (2022) from the opposite direction—there a rent matters only where monitoring is intense; here a monitor matters most where the rent is large—completing the levels half of the paper’s central argument: the wage and the manager discipline together, each worth most where the other is present. What remains is to ask whether the statutory movement of the rent in Sections 5–6 does its *marginal* damage through the monitor—the test the closing section takes up.

11 The Monitor at the Moment of the Shock

The squeeze of Section 5 raises border shrinkage with a lag. Whether it does so in every border store, or only where no manager is on the floor to absorb it, is what this closing section settles—where the minimum-wage evidence meets the manager evidence of the rest of the paper. Section 10 established the level form of that dependence—a good manager accomplishes more where the standing premium is high—and the corner logic of Section 3.8 supplies its marginal form: a marginal narrowing of the rent moves behavior only where the care margin is interior—where little monitoring is supplied—while the well-watched store, whose workers already comply, is insured. The operative input is the manager’s presence on the sales floor, the technology Section 9 identifies with good management, so I split border stores by their general manager’s floor presence in the months before each hike, taken above or below the median within format-by-hike cells.

Figure 17 reports the two phases by presence group, and the two responses trace opposite paths. Where the manager is on the floor, the raise pays off on impact—border shrinkage falls 4.9 percent relative to sales ($p = 0.10$)—and the later compression does no measurable damage (+1.9 percent, indistinguishable from zero): the erosion of the premium is, in effect, insured. Where she is not, the raise buys nothing on impact and the compression damage lands in full, a late rise of 4.9 percent ($p = 0.02$). The two late responses differ in the predicted direction, though at the monthly frequency the difference is not sharply estimated (−3.1 points, $p = 0.42$). What carries this is the manager’s presence—what she does—not her estimated type—what she is. The natural consistency check confirms it. Splitting the same border stores by manager *type*—the high- versus low-type AKM label used in Sections 9 and 10, fixed at the general manager in place the December before each hike—and running the identical two-phase design, manager type does not moderate the compression: both groups suffer it (a late rise of 5.0 percent under low-type managers, $p = 0.02$, and 7.6 percent under high-type, $p < 0.01$), and the high-minus-low difference is small, insignificant, and if anything the wrong sign (+2.6 points, $p = 0.47$). The split is clean where the continuous fixed effect is not: on the net-of-sales outcome the pre-hike coefficients are flat in both type groups ($p = 0.44$ and 0.65), so this is a genuine null, not a contaminated one. The moderation the data support therefore runs through what the manager does, her presence on the floor, and not through her estimated quality; Appendix G reaches the same verdict for the continuous fixed effect, which additionally fails its own pre-trend test at the border.

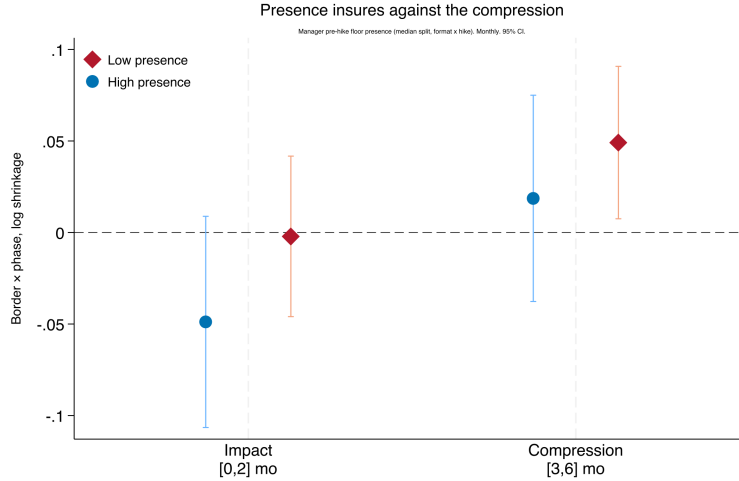


Figure 17: The compression response by the manager’s floor presence at the hike. Where presence is high, border shrinkage falls on impact and the later compression does no measurable damage; where it is low, the impact response is absent and the full compression rise lands (+4.9 percent, $p = 0.02$). Border stores split at the median presence within format-by-hike cells; 95 percent confidence intervals.

The border squeeze thus falls inside the paper’s central condition. A dated, exogenous narrowing of the wage premium raises shrinkage with the lag the model predicts, and the increase concentrates in stores whose managers do not monitor the floor. It is the marginal counterpart of the level complementarity of Section 10: the wage premium and the manager deter shirking as a product, so a store that loses part of its rent keeps its discipline only where the other factor—the monitor—is still on the floor. Withdrawing the premium does its damage where no one is watching.

Non-shirking channels. A statutory hike can move shrinkage through channels that have nothing to do with the worker’s care, and the border design must face them. Four are the natural candidates: the firm could thin the floor—headcount, hours, guards—to absorb the higher wage bill; the workforce could churn and rehire at the new floor, replacing experienced workers with novices; border prices and incomes could shift the pressure of customer theft on an unwatched aisle; and the peso value of a lost unit, or the sales denominator beneath the ratio, could move with no change in physical losses. The payroll rejects the first two directly. In a companion of the border design collapsed to early and late phases, border headcount is flat through the compression window (-0.2 percent, $p = 0.89$), the count and share of security personnel are flat ($p > 0.8$), and total shifts worked *rise* by 4 percent ($p = 0.007$); hiring and separations both fall, and the share of workers with under six months of tenure falls by 3.2 percentage points ($p = 0.001$). The border store after the hike is better staffed,

more stable, and more experienced, not less: whatever raised losses, it was not fewer eyes or newer hands.

The accounting channels earn a more careful answer, because the denominator did move: border sales fall by about 2 percent in the compression phase, so of the 4.1 percent rise in the shrinkage ratio over the same phase in that companion specification ($p = 0.007$), roughly half is sales, not losses—and the falling sales themselves dispose of the demand half of the income story: whatever the hike did to border incomes, it did not arrive at the register as more traffic to steal under. The other half survives every accounting test. The peso value of losses rises while sales fall (2.3 percent, $p = 0.11$); the value per lost unit is flat ($p = 0.39$), so nothing was revalued; the border’s implied sale prices in fact *decline* slightly (-0.6 percent, $p = 0.02$), so price pass-through runs against the result rather than toward it; and losses in physical units move up (1.6 percent, $p = 0.31$) while unit sales move down—a store with mechanically less merchandise flowing through it should lose less by sheer flow, and it lost more. The raise phase shows the same accounting in reverse: on impact, border losses fall by six percent in value and units alike ($p < 0.001$), the store-level image of the swept worker’s reciprocity documented in Section 6. And where income does touch the incentive to take merchandise, its sign runs *against* the finding: the hike raises income precisely at the bottom—for the border’s covered households, and for the firm’s own swept workers, whose pay and whose rent rise together—so the need that motivates taking merchandise weakens exactly as the job a caught worker forfeits becomes more valuable (Dube et al., 2019). An income channel, where it operates, pushes losses down; the design finds them going up, which makes the estimated rise a conservative reading.

What the store-level comparison cannot do alone is separate *whose* behavior deteriorated—the worker who supplies less care, or an environment that presses harder on an unwatched floor. Both hikes land in January, so the compression phase is always a border spring, and no store-level estimate can exclude a border-specific seasonal shock on its own. The worker-level design of Section 6 is built for exactly this gap: with store-by-month-by-hike fixed effects, everything that reaches a border store in a given month—traffic, prices, external pressure, staffing, the stock on the shelf—is absorbed, and what remains is the conduct of identified workers on margins only they control. There, the workers whose premium the floor erodes fall behind their swept coworkers in the same store and month, by more where the erosion is deeper, on the clock of the squeeze—while certified absence, the margin a worker cannot manipulate, stays flat. No channel that operates on the store can produce that gradient; a channel that operates on the worker’s rent does. The store-level rise in losses and the worker-level erosion of discipline are one fact read at two levels, and only the shirking account has access to the second.

12 Conclusion

Every large employer faces the problem this paper studies: work that cannot be perfectly watched, performed by workers who earn more or less than their outside option, under managers who watch more or less closely. Economics has long had a theory for that problem—pay a rent and threaten to take it away—but testing it requires seeing things firms rarely let researchers see: what each worker earns against her own outside market, what she does when no one checks, and who is watching. This paper assembled all three inside one large Mexican retailer, using inventory shrinkage—a hundred-billion-dollar blind spot of the retail industry—as the audited residue of shirking. Both halves of the theory turned out to matter. Workers with larger premia over their outside option shirk less; stores run by better managers lose less; and each instrument raises the return to the other. The complementarity has two faces, and the paper shows both. Adding an instrument pays most where the other is strong: a good manager accomplishes roughly three times more where the premium is high, and a premium that no one watches deters little. Losing an instrument, by the same product logic, hurts most where the other is weak: when the border hikes compressed the premium, shrinkage rose with a delay—but only in stores whose managers were seldom on the floor, while stores with a present manager rode out the squeeze insured. These are not two findings but one: discipline is the product of the rent and the monitor, so a store that loses one factor keeps its discipline only if the other is still there. The statute revealed, store by store, who was doing the disciplining—the premium, the manager, or no one.

For firms, pay policy and supervision are not separable levers. A wage ladder set without asking who monitors misprices both instruments: the same rent buys discipline in a watched store and nothing in an unwatched one, so compression shocks carry hidden costs concentrated precisely where management is thin, and the allocation of managers across stores functions as insurance against them. The business case for paying above the market runs through the monitoring capacity that converts rents into discipline.

For policy, minimum-wage effects operate inside the wage ladder, not only at the floor. Part of the cost of a hike arrives months later, on hidden margins—inventory losses—that standard datasets do not record, and its incidence depends on management quality. Small average effects in the minimum-wage literature are consistent with substantial heterogeneity across establishments that differ in how they are run.

The evidence comes from one firm in formal retail, and presence is a proxy for monitoring, not a direct measure of it. These boundaries point to the open questions: whether firms can create presence through the selection, training, and assignment of managers; how workers select into rents; how turnover evolves after a squeeze; and whether the complementarity

between rents and monitoring generalizes to other hidden-action settings.

A Appendix: Two-Way Fixed-Effects Evidence

The store-level correlations of Section 4 exploit the wide dispersion of local market wages across Mexico. Figure 18 maps this dispersion across municipalities; because the firm sets wages centrally by position, this geographic variation in the outside option is the primary source of variation in the store wage premium.

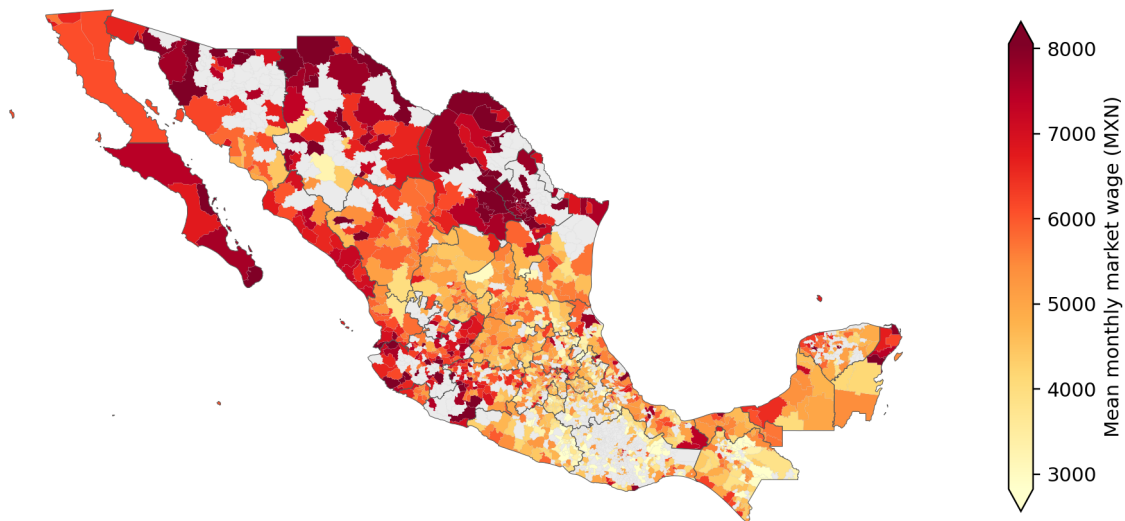


Figure 18: Dispersion of local market wages across Mexico. Each municipality is shaded by its mean monthly market wage, computed from the ENOE labor-force survey, 2021–2024; municipalities not sampled by ENOE are shown in grey and the color scale is clipped at the 5th and 95th percentiles. The middle 90 percent of municipalities span about USD 150 to USD 440, so the same firm position commands a substantially different wage premium depending on where the store is located.

Table 20 reports the full store-level estimates for attendance, turnover, and effort summarized in Section 4. The gradients keep the same sign and significance across the inclusion of store format, log sales, and the choice and aggregation of time fixed effects (columns 2–4), which is why the body reports only their summary.

Table 20: Store-level TWFE: attendance, turnover, and effort (log wage premium)

<i>Outcome (Log)</i>	(1)	(2)	(3)	(4)	Mean	Median
Absenteeism	−0.036*** (0.010)	−0.038*** (0.011)	−0.033*** (0.010)	−0.039*** (0.011)	0.68	0.71
Non-justified absenteeism	−0.129*** (0.023)	−0.131*** (0.023)	−0.125*** (0.022)	−0.127*** (0.021)	0.00	0.06
Turnover (total)	−0.158*** (0.025)	−0.158*** (0.025)	−0.162*** (0.025)	−0.140*** (0.021)	2.55	2.69
Turnover (voluntary)	−0.153*** (0.023)	−0.153*** (0.023)	−0.158*** (0.023)	−0.138*** (0.020)	2.40	2.54
Monthly shifts	+0.032*** (0.009)	+0.032*** (0.009)	+0.032*** (0.009)	+0.030*** (0.009)	2.98	2.99
Daily worked hours	+0.012*** (0.002)	+0.011*** (0.001)	+0.010*** (0.001)	+0.011*** (0.002)	2.10	2.10
Store format	No	Yes	Yes	No		
Log sales	No	No	Yes	No		
Municipality FE	Yes	Yes	Yes	Yes		
Year×Qtr FE	Yes	Yes	Yes	No		
Year + Quarter FE	No	No	No	Yes		

Store-quarter level. Each cell is the coefficient on the log monthly wage premium from a separate regression; SE below in parentheses. Municipality and year×quarter fixed effects throughout. Mean and median are of the log outcome. Roughly 2,400 store-quarters (all outcomes except turnover, roughly 2,200). Turnover is a rate per hundred workers. SE clustered at the municipality level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.1 Worker-Level Results

The store-level estimates in Section 4 aggregate the responses of each store’s workforce. Here I follow individual workers over time. Where the store-level specification compares stores that pay different premia, the worker-level specification compares the same worker with herself as her own premium moves, absorbing with a worker fixed effect all permanent differences in attendance and effort across workers. Identification rests on within-worker movements in the premium as the local outside wage shifts, and the store-level estimates can be read as the aggregate of these individual responses.

Table 21 places the store (S) and worker (W) coefficients side by side for the four outcomes observed at both levels; the two levels tell a consistent story—a larger wage premium is associated with less absenteeism and more time on the job, whether I aggregate to the store or follow individual workers.

Table 21: Comparative TWFE: effect of the log wage premium (baseline)

	Absenteeism		NJ Absenteeism		Daily Hours		Monthly Shifts	
<i>Wage premium (Log)</i>	(S)	(W)	(S)	(W)	(S)	(W)	(S)	(W)
Coefficient	-0.036***	-0.048***	-0.129***	-0.046***	+0.012***	+0.009***	+0.032***	+0.046***
SE	(0.010)	(0.003)	(0.023)	(0.002)	(0.002)	(0.0003)	(0.009)	(0.001)
Dependent mean (Log)	0.68	0.61	0.00	0.22	2.10	2.09	2.98	2.77
Dependent median (Log)	0.71	0.69	0.06	0.00	2.10	2.07	2.99	3.11

(S) = store-quarter level (municipality and year×quarter FE). (W) = worker-quarter level (worker, store, and year×quarter FE). No controls. Clustered standard errors in parentheses (municipality for (S), store for (W)). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

At the worker level I estimate

$$\log Y_{jt} = \delta \log W_{jt}^p + Z_{jt} \theta + \eta_j + \tau_t + \nu_{jt}, \quad (36)$$

where Y_{jt} is a shirking or effort outcome (total absence days, non-justified absence days, monthly shifts, or average daily hours), $\log W_{jt}^p$ is the log worker-level wage premium of Equation 1, η_j is a worker fixed effect, τ_t is a year-by-quarter fixed effect, and Z_{jt} is an optional vector of controls (managerial tier, tenure, and age group). The specification additionally absorbs store fixed effects, and standard errors are clustered at the store level. The worker fixed effect absorbs all permanent differences in absenteeism or productivity across workers, so δ is identified from within-worker movements in the premium as the local outside wage shifts.

A larger wage premium is associated with less absenteeism and more time on the job. In the baseline specification a ten percent increase in the wage premium is associated with a 0.48 percent reduction in total absence days and a 0.46 percent reduction in non-justified absence days, both significant at the one percent level. The same increase is associated with a 0.46 percent rise in monthly shifts and a 0.09 percent rise in average daily hours, and the estimates are stable across the inclusion of managerial tier, tenure, and age controls and across alternative time fixed effects (Table 22).

Table 22: Worker-level TWFE: continuous outcomes (log wage premium)

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Total absence days</i>					
Wage premium (Log)	−0.048***	−0.050***	−0.049***	−0.045***	−0.015***
SE	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Mean; median (log)	0.61; 0.69	(level 2.03; 1.00)			
<i>Panel B. Non-justified absence days</i>					
Wage premium (Log)	−0.046***	−0.047***	−0.047***	−0.038***	−0.032***
SE	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Mean; median (log)	0.22; 0.00	(level 1.09; 0.33)			
<i>Panel C. Monthly shifts</i>					
Wage premium (Log)	+0.046***	+0.046***	+0.046***	+0.045***	+0.039***
SE	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Mean; median (log)	2.77; 3.11	(level 19.28; 22.33)			
<i>Panel D. Average daily hours</i>					
Wage premium (Log)	+0.0093***	+0.0069***	+0.0069***	+0.0086***	+0.0081***
SE	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
Mean; median (log)	2.09; 2.07	(level 8.21; 7.91)			
Tier control	No	Yes	Yes	No	No
Tenure control	No	Yes	Yes	No	No
Age group	No	No	Yes	No	No
Worker FE	Yes	Yes	Yes	Yes	Yes
Store FE	Yes	Yes	Yes	Yes	Yes
Year×Qtr FE	Yes	Yes	Yes	No	No
Year FE	No	No	No	Yes	No
Quarter FE	No	No	No	No	Yes

Worker-quarter level; roughly 131,000 worker-quarters (columns 1–2, 4–5) and roughly 131,000 with tenure/age controls. The regressor is the log monthly wage premium (mean USD 93). Standard errors clustered at the store level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 23 reports the logit estimates for the discrete worker-level outcomes summarized in Section 4. A higher wage premium is associated with lower odds of any absence, any non-justified absence, and separation (both total and voluntary), all significant at the one percent level.

Table 23: Worker-level logit: discrete outcomes (log wage premium)

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Any absence (=1)</i>					
Wage premium (Log)	-0.201***	-0.182***	-0.182***	-0.148***	-0.149***
SE	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
Mean; median	0.70; 1				
<i>Panel B. Any non-justified absence (=1)</i>					
Wage premium (Log)	-0.436***	-0.414***	-0.380***	-0.264***	-0.283***
SE	(0.010)	(0.010)	(0.010)	(0.010)	(0.011)
Mean; median	0.57; 1				
<i>Panel C. Separation — total (=1)</i>					
Wage premium (Log)	-0.370***	-0.369***	-0.323***	-0.314***	-0.322***
SE	(0.010)	(0.010)	(0.010)	(0.011)	(0.011)
Mean; median	0.15; 0				
<i>Panel D. Separation — voluntary (=1)</i>					
Wage premium (Log)	-0.380***	-0.380***	-0.335***	-0.324***	-0.332***
SE	(0.009)	(0.009)	(0.010)	(0.010)	(0.011)
Mean; median	0.13; 0				
Education control	No	Yes	Yes	Yes	Yes
Tenure control	No	No	Yes	Yes	Yes
Tier control	No	No	No	Yes	Yes
Time FE	No	No	No	No	Yes

Worker-quarter level; logit coefficients (log-odds). N ranges from roughly 149,000 to 151,000 worker-quarters. Standard errors clustered at the store level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B Appendix: Limited-Mobility Bias in the AKM Decomposition

The variance decomposition reported in Section 7.3 is a plug-in estimate: it uses the estimated fixed effects in place of the true ones. Like every AKM decomposition, it is subject to limited-mobility bias. Fixed effects estimated from finite mover counts are noisy, and this noise mechanically inflates the estimated variances of $\hat{\theta}$ and $\hat{\psi}$ while biasing their correlation downward (Andrews et al., 2008). The manager effect, identified from only a handful of stores per manager, is the noisiest object and the one most exposed to the bias. I therefore assess it with two corrections on the largest connected set, where the effects are best identified: the homoskedastic correction of Andrews et al. (2008) and the heteroskedasticity-robust leave-out estimator of Kline et al. (2020) (KSS).

Table 24 reports the corrected components, and three patterns stand out. First, the store-effect variance is essentially unchanged: for log shrinkage value, 87 percent of the baseline $\text{Var}(\hat{\psi})$ survives the KSS correction (0.251 versus 0.287), as expected when each store is observed for many months. Second, the manager-effect variance shrinks—the manager effects are the noisiest object, since each manager appears in only a handful of stores—but about *one third* of the baseline $\text{Var}(\hat{\theta})$ remains (0.016 versus 0.049): the corrected manager component is more modest than the plug-in decomposition implies, yet clearly non-zero. Third, the corrected correlation between manager and store effects is close to zero for the headline definition (-0.038 under KSS), so the firm does not systematically assign low-shrinkage managers to intrinsically low-shrinkage stores.⁸ I read the corrected numbers as confirming that the manager component of shrinkage is real rather than an artifact of estimation noise, while noting that its magnitude is best bounded from below by the leave-out estimate.

⁸For the two ratio definitions the corrected correlations exceed one in absolute value. This pattern is a known artifact of bias-corrected estimators when the corrected manager variance approaches zero, and I read those columns as uninformative about sorting rather than as evidence of extreme assortative placement.

Table 24: AKM variance decomposition with limited-mobility-bias corrections

	log(Shrink Value)			log(Shrink/Sales)			log(Shrink/Worker)		
	<i>Baseline</i>	<i>Andrews</i>	<i>KSS</i>	<i>Baseline</i>	<i>Andrews</i>	<i>KSS</i>	<i>Baseline</i>	<i>Andrews</i>	<i>KSS</i>
$\widehat{\text{Var}}(\theta)$	0.049	0.009	0.016	0.038	0.003	0.006	0.044	0.002	0.006
$\widehat{\text{Var}}(\psi)$	0.287	0.243	0.251	0.075	0.037	0.040	0.073	0.027	0.032
$\widehat{\text{Cov}}(\theta, \psi)$	-0.014	0.002	-0.002	-0.029	-0.015	-0.018	-0.024	-0.007	-0.010
$\widehat{\text{Corr}}(\theta, \psi)$	-0.120	0.044	-0.038	-0.553	-1.399	-1.145	-0.424	-0.913	-0.739
N	$\approx 5,900$			$\approx 5,900$			$\approx 5,900$		

Largest connected set of the general-manager panel. θ = manager FE, ψ = store FE; observation-weighted variances and covariances. *Baseline*: OLS variance of the estimated fixed effects. *Andrews*: homoskedastic correction of Andrews et al. (2008). *KSS*: leave-out estimator of Kline et al. (2020). Correlations exceeding one in absolute value reflect near-zero corrected manager variance and are a known feature of these estimators.

C Portability: Alternative Estimators for the Staggered Arrival Design

Arrivals of high- and low-type managers are staggered across stores, and the two-way fixed-effects (TWFE) estimator then aggregates comparisons that include already-treated stores as controls; when treatment effects vary over time or across cohorts, such comparisons can bias the estimate and even flip its sign (Goodman-Bacon, 2021). I therefore re-estimate the pooled arrival effect with three heterogeneity-robust estimators that identify from the timing of arrivals: the average of dynamic effects of de Chaisemartin and D’Haultfœuille (2020) (dCdH), the interaction-weighted estimator of Sun and Abraham (2021) (SA), and the imputation estimator of Borusyak et al. (2024) (BJS).

Why I do not report Callaway–Sant’Anna. The remaining leading estimator, Callaway and Sant’Anna (2021) (CS), is not applicable in this panel, and reporting it would mislead. CS builds its doubly-robust comparison from stores that are never treated (or not yet treated) by the manager type in question. Here that control group is not exogenous: the never-treated stores are *selected on manager type*—they are precisely the stores that only ever operate under low-type managers—so their shrinkage path is not a valid counterfactual for a store that receives a high-type manager. When forced onto that group, CS flips to a positive coefficient for the shrinkage outcomes and produces an implausibly large negative effect for non-justified absence that no other estimator reproduces—both symptoms of the invalid comparison rather than of the treatment. The three timing-based estimators below do not lean on the selected never-treated group, and all three confirm the reduction, so I rest the robustness on them.

High-type arrivals. Table 11 of the main text reports the comparison, and the three heterogeneity-robust estimators agree with the two-way baseline and, where it matters, strengthen it. For log shrinkage value, TWFE gives -6.8 percent and SA -6.2 percent, while the two estimators that fully purge already-treated comparisons—dCdH and BJS—give -17.6 and -17.4 percent, all significant at the one-percent level, with the same ordering for the ratio definitions. The larger corrected magnitudes are consistent with the attenuation logic of Section 8.2: under TWFE, store-months far after arrival enter the comparison group and absorb part of the effect. Turnover is positive only under the pooled TWFE and insignificant or negative elsewhere; restricting the same regression to the event window flips it negative (Table 11), marking the positive reading as an artifact of the contaminated comparison.

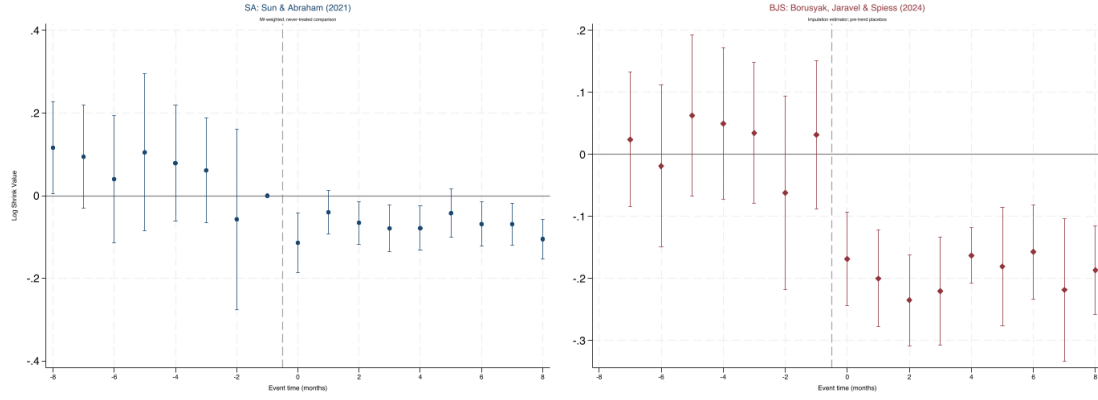
Low-type arrivals. Table 25 repeats the exercise for low-type arrivals, and here the estimators are far less conclusive. Only TWFE and BJS deliver positive and significant shrinkage effects (5 to 13 percent); dCdH and SA are statistically indistinguishable from zero throughout. This instability is why the main text quantifies the portability of manager effects with high-type arrivals and treats the low-type contrast as qualitative.

Table 25: Low-type arrival effect across staggered-DiD estimators

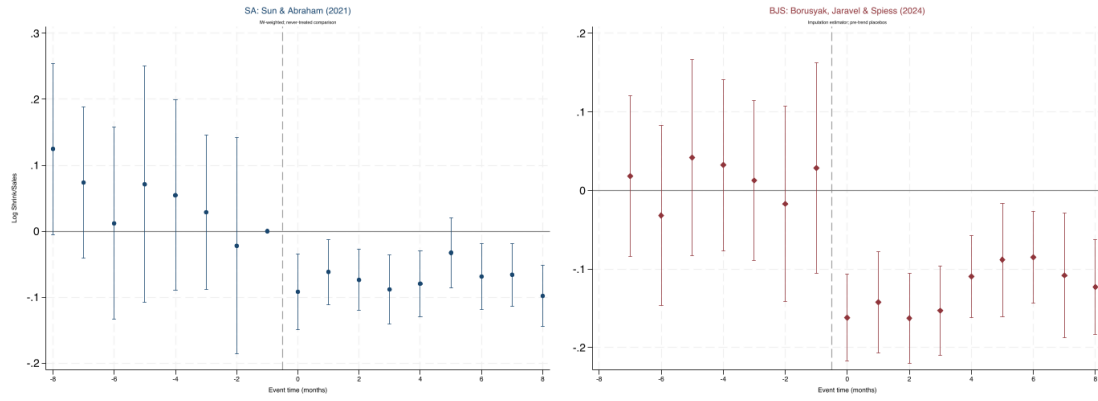
Outcome	TWFE	dCdH	SA	BJS
Log Shrink Value	0.0489*** (0.0176)	0.0929 (0.0946)	0.0237 (0.0185)	0.1065*** (0.0331)
Log Shrink/Sales	0.0640*** (0.0193)	0.1127 (0.0726)	0.0208 (0.0176)	0.1322*** (0.0302)
Log Shrink/Worker	0.0585*** (0.0185)	0.1193 (0.1162)	0.0283 (0.0194)	0.0934*** (0.0325)
Log NJ Abs p.e.	−0.0121 (0.0235)	−0.1598 (0.0990)	−0.0151 (0.0230)	−0.0915*** (0.0292)
Log Turnover p.e.	0.0245 (0.0610)	0.3401 (0.5470)	−0.0239 (0.0635)	−0.2813*** (0.0732)

Same design and estimators as Table 11, with low-type arrivals as the treatment. *** $p < .01$, ** $p < .05$, * $p < .10$.

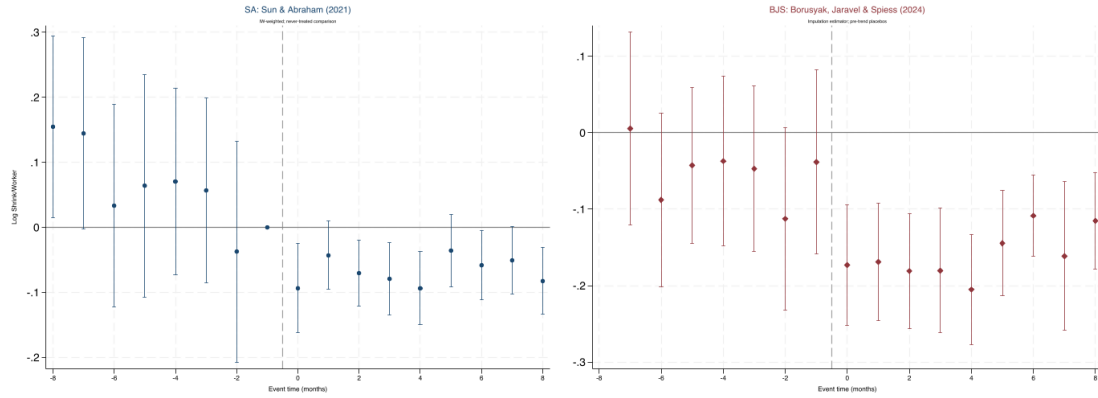
Event studies under Sun–Abraham and Borusyak–Jaravel–Spiess. Figure 19 plots the period-by-period estimates from the two estimators that produce full event-study paths, for high-type arrivals and the three shrinkage definitions, together with the BJS pre-trend placebos. The corrected paths reproduce the shape of the main-text event study (Figure 9): no systematic pre-trends and a persistent post-arrival reduction.



(a) Log shrinkage value



(b) Log shrinkage-to-sales



(c) Log shrinkage per worker

Figure 19: High-type arrival event studies under Sun–Abraham and Borusyak–Jaravel–Spiess

Notes: Interaction-weighted (Sun and Abraham, 2021) and imputation (Borusyak et al., 2024) estimates with pre-trend placebos; 95 percent confidence intervals, standard errors clustered at the store. Because nearly all stores are eventually treated, the per-horizon imputation samples are small and the horizon-specific BJS estimates should be read as descriptive; the pooled estimates in Table 11 are the reliable summary.

D Characterization: Dose Response and Cross-Area Reallocation

This appendix collects exercises referenced in Section 9: the dose response and novelty placebo of the arrival effect, the cross-area component of the personnel reallocation of Section 9.2, the targeted-presence and within-shift discipline results, and supporting material on ordinary managers and the comovement of the estimated fixed effects.

D.1 Dose Response and a Novelty Placebo

The arrival design classifies managers by a median split, but the shrinkage effect should be graded, not binary: a store that trades up by a wide margin should improve more than one that trades up narrowly. To test this I rank every arrival of a new general manager by the *improvement* it represents—the incumbent management’s mean shrinkage fixed effect minus the arriving manager’s—and estimate the arrival effect separately for the largest upgrades. Because the ranking uses the continuous fixed effect rather than the high-type indicator, it does not depend on the classification cutoff. Table 26 shows a dose response: the shrinkage reduction grows from 4.3 percent in the top half of upgrades to 7.3 percent in the top quartile and 9.3 percent in the top decile.

The same table delivers a novelty placebo. If workers simply exerted more effort for any new boss, arrivals that are *not* upgrades would also lower shrinkage. They do not. When the arriving manager is no better than the incumbent she replaces (the bottom half of the improvement distribution), shrinkage *rises* by 4.9 percent, and pooling all arrivals of new management regardless of quality yields no effect at all (+0.002, not significant). What moves shrinkage is the quality of the arriving manager, not the disruption of a new face—a distinction the high-type effect shares with the manager-quality effects of Lazear et al. (2015).

Table 26: Dose response and a novelty placebo (log shrinkage value)

Treatment definition	Effect $\text{Post}_{[0,8]}$	Stores
Top 10% upgrade (arriving $\gg$ incumbent)	-0.0928^{***} (0.0220)	≈ 25
Top 25% upgrade	-0.0729^{***} (0.0159)	≈ 65
Top 50% upgrade	-0.0430^{***} (0.0105)	≈ 130
Bottom 50% (novelty placebo)	$+0.0490^{***}$ (0.0096)	≈ 130
Any arrival of new management	$+0.0017$ (0.0077)	≈ 270

Each row is the $\text{Post}_{[0,8]}$ coefficient from a separate arrival difference-in-differences on log shrinkage value. “Improvement” is the store’s baseline incumbent shrinkage fixed effect minus the arriving manager’s; upgrade tiers are the top 10/25/50 percent of that distribution. “Novelty placebo” restricts to arrivals in the bottom half (arriving manager no better than the incumbent); the last row pools all arrivals of new management. Store, year, and month fixed effects; standard errors clustered at the store; store counts rounded.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D.2 Reallocation Across Functional Areas

Section 9.2 reports that high-type managers rotate the workers they inherit; Table 27 shows that part of this rotation moves workers across the store’s functional areas, not only up and down its tiers. On a high-type arrival, the rate at which a worker’s occupational area changes from one month to the next rises by about 0.21 per hundred workers, and the increase is concentrated in moves into client-facing roles (sales, service, and food; +0.09 per hundred, $p < 0.01$), with moves into back-office roles (logistics and administration) positive but imprecise (+0.05). Moves into the loss-prevention chain do not rise significantly. Low-type arrivals move none of these margins. The pattern is consistent with the redeployment of incumbents documented for good managers in the sister setting (Adhvaryu et al., 2026), and, like the upward mobility of the main text, it describes a manager who reshuffles the people she has rather than replacing them.

Table 27: High-type managers reassign workers across functional areas

	Area change (any)	Into client- facing	Into back- office	Into loss- prevention
High-type arrival	0.208** (0.092)	0.088*** (0.031)	0.054 (0.035)	0.026 (0.025)
Low-type arrival	0.029 (0.113)	0.020 (0.031)	−0.021 (0.032)	0.001 (0.024)
Mean of outcome	1.44	0.32	0.25	0.14

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a separate store-month arrival regression (Equation 35), rates per hundred workers, with store, year, and month fixed effects and standard errors clustered at the store. “Area change” is any month-over-month change in a worker’s occupational area; “client-facing” comprises sales, service, and food roles, “back-office” logistics and administration, and “loss-prevention” the receiving–tagging–protection chain. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D.3 Presence Is Broad, Not Targeted

A further question is whether the manager directs this added presence toward the workers who matter most for shrinkage. Table 28 recomputes co-presence separately for four high-risk groups—the loss-prevention chain that receives, tags, and protects merchandise; security staff; cashiers; and merchandise organizers—and re-runs the arrival design on each. A high-type arrival raises co-presence by a nearly identical three to four percentage points for *every* group, including the average worker; the increase is broad, not concentrated on any one role. High-type managers therefore supervise everyone more rather than reallocating a fixed budget of attention toward shrinkage-prone tasks—the opposite of the targeted-monitoring pattern of Bandiera et al. (2007), and consistent with a margin of generalized presence rather than a surveillance technology. The pattern holds across hours as well: the rise in co-presence is, if anything, slightly larger on night and weekend shifts than on average, so high-type managers extend their presence into precisely the low-oversight hours when shrinkage risk is highest. [\[JS: This is interesting but is it central? I think we need to cut table and streamline discussion.\]](#)

Table 28: The rise in co-presence is broad, not targeted to risk groups

	(1)	(2)	(3)	(4)	(5)
	All workers	Loss- prevention	Security	Cashiers	Perishables
HT manager arrival	0.0387*** (0.0120)	0.0370*** (0.0140)	0.0326*** (0.0107)	0.0385*** (0.0121)	0.0368** (0.0144)
LT manager arrival	-0.0092 (0.0181)	-0.0110 (0.0183)	-0.0067 (0.0156)	-0.0076 (0.0186)	-0.0136 (0.0194)
Dep. mean	0.574	0.611	0.485	0.568	0.609

Each cell is the $\text{Post}_{[0,8]}$ coefficient from the arrival design of Equation (35), with the outcome being the mean manager-worker co-presence share computed over the worker shifts in the indicated risk group: the loss-prevention chain that receives, tags, and protects merchandise; security staff; cashier positions; and the perishables category. Store, year, and month fixed effects; standard errors clustered at the store. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D.4 Within-Shift Discipline

The added supervision should in turn change how workers behave, and the minute-level records allow a test of this claim at the level of the individual shift. Table 29 regresses, over millions of worker shifts, three deviations from each worker’s own norm—how late she clocks in, how far her break runs over, and how early she leaves—on whether the general manager is co-present that shift, absorbing worker, store, month, hour-of-day, and day-of-week fixed effects. The timing fixed effects are essential: because the manager works days, a raw co-presence comparison would confound supervision with the difference between day and night shifts; with them, identification is the *same* worker, at the *same* time of day and weekday within a store-month, on days the manager is on the clock versus days she is not. The robust margin is punctuality: when the manager is co-present, the worker arrives about one to two minutes less late than her own norm, and the sign and significance are identical whether co-presence is measured by a binary indicator or by the continuous overlap share. The other two margins—break over-run and early departure—are not robustly signed: they switch sign between the binary and the continuous specification, which indicates that they remain entangled with the structure of the shift, so I do not interpret them. I read the tardiness result alone as the clean finding: a within-worker response to being watched, at daily frequency. [\[JS: This seems interesting but I don't think I fully understand how it's calculated. Also, is it central to your main story about shrinkage?\]](#)

Table 29: Workers arrive on time when the manager is present**[JS: Could any of these be turned into measures of shirking to be used in the earlier wage premium**

	(1)	(2)	(3)
	Late arrival (minutes)	Break over-run (minutes)	Early departure (minutes)
analysis?] GM co-present (0/1)	−1.988*** (0.627)	1.846*** (0.267)	−12.414*** (0.881)
GM overlap share	−1.007*** (0.259)	—	—
Worker-shifts	8,862,484		

Worker-shift level. Each outcome is the deviation of that shift from the worker’s own median (arrival minute, break minutes, and departure minute; early departure is the median departure minus the actual, so a negative coefficient means leaving *less* early). “GM co-present” is an indicator that the manager’s store-day presence interval overlaps the shift; “GM overlap share” its continuous version. Worker, store, month, hour-of-day, and day-of-week fixed effects; standard errors clustered at the store. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D.5 Managers Are Otherwise Ordinary

These actions aside, high-type managers are unremarkable on the traits a personnel file would use to explain quality. Table 30 compares the two types. They earn the same salary—a log difference of 0.015, far from significant—and are equally educated; high-type managers are about a year older, a difference of under three percent that clears significance only at the ten-percent level. Where they differ is in career path rather than credentials: high-type managers have been at the firm about three and a half months *less* (a gap of roughly nine percent that runs against a seniority reading of quality), rotate through slightly fewer stores, and hold each posting marginally longer. None of these is a dimension the firm could read as competence in advance; on pay and schooling—the observables a hiring or promotion decision would price—the two types are indistinguishable.

Table 30: Manager observables by type (spell averages)

	HTM (N≈500)	LTM (N≈500)	Diff. (HTM–LTM)	<i>p</i> -value
Log salary (monthly, MXN USD)	10.203 7.285	10.188 7.270	0.015	0.408
Tenure at the firm (months)	36.51	40.08	−3.57**	0.010
Female (= 1)	0.239	0.214	0.024	0.362
Age (years)	38.31	37.33	0.98*	0.070
Education (years)	13.40	13.24	0.16	0.393
Spell length (months)	7.83	7.06	0.77*	0.056
Stores worked in sample	2.49	2.65	−0.17**	0.039

Manager characteristics averaged over the spell; each observation is a manager×store spell. HTM = shrinkage-value fixed effect below the format-by-border cell median. *p*-values from two-sample *t*-tests. ****p*<0.01, ***p*<0.05, **p*<0.1.

Two implications follow. First, the wage-setting process does not price the quality I estimate: a manager whose arrival reduces shrinkage costs the firm exactly what one whose arrival does not costs. This mirrors the finding of Hoffman and Tadelis (2021) that the people-management skills of supervisors predict performance but not pay, and suggests the firm either cannot observe the quality dimension I recover or does not act on it. Second, because the two types are observationally identical on the traits a personnel file records, the fixed-effect classification of Section 7 is not recoverable from standard human-resources observables.

One determinant of how much that presence disciplines is suggestive but worth recording: the social distance between the manager and the team she watches. Marx et al. (2021), in a Kenyan field experiment with random supervisor assignment, find that vertically homogeneous supervisor–worker pairs exert lower effort—social proximity slackens monitoring, so vertical *distance* sharpens it. My setting has no ethnic dimension, but it records the manager’s and the team’s gender, age, tenure, and education, so I can interact the arrival effect with the manager–team distance on each (Table 31). The evidence is at most weakly supportive, and I record it as such. The *tenure* dimension—the gap in firm experience, the dimension that most directly marks hierarchical distance—moves the enforcement margin in the predicted direction: a manager whose tenure stands farther from her team’s suspends workers more (+0.19 per hundred, *p* < 0.05). But its interaction with shrinkage per worker, negative as the mechanism predicts, is not significant (−0.028, s.e. 0.020), and the gender-distance interaction is significant with the opposite sign for shrinkage value. Because the assignment is quasi-experimental and the pattern is confined to the enforcement margin, I read this as no more than a suggestive echo of the enforcement channel, not as corroboration on the shrinkage margin itself.

Table 31: Manager–team vertical distance and the disciplining effect (suggestive)

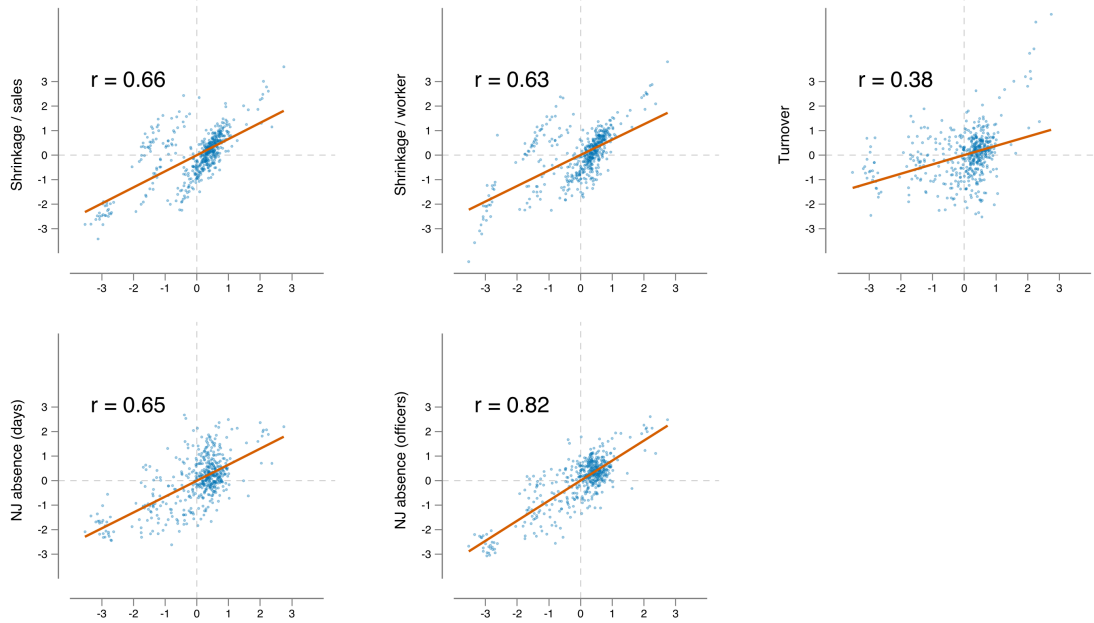
	Shrink value	Shrink/ worker	Suspensions per 100
Gender distance	0.027** (0.013)	0.014 (0.015)	−0.037 (0.070)
Age distance	−0.002 (0.009)	−0.002 (0.010)	−0.035 (0.083)
Tenure distance	−0.022 (0.020)	−0.028 (0.020)	0.191** (0.079)
Education distance	−0.006 (0.011)	−0.005 (0.013)	−0.066 (0.077)

Each cell is the $\text{Post}_{[0,8]} \times$ (standardized manager–team distance) interaction from a separate high-type arrival regression (arrival design of Equation 35; store, year, and month fixed effects; standard errors clustered at the store), with the distance fixed at arrival. The gender distance is the share of the team of the opposite gender to the manager; the others are the absolute manager-minus-team-mean gap, z -scored. A negative interaction on shrinkage, or a positive one on suspensions, indicates that a manager more distant from her team disciplines more (Marx et al., 2021). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D.6 Fixed-Effect Comovement: Core-Focused View

Figure 13 presents the comovement of manager fixed effects as a full scatterplot matrix. Figure 20 gives the same evidence in a core-focused layout—the shrinkage-value fixed effect against each other outcome, one panel apiece, with the pairwise correlation shown in each.

Manager quality comoves across outcomes: core shrinkage FE vs. each alternative



Manager fixed effect: log shrinkage value (standardized, core measure)

Each dot = one manager ($N=497$). Axes are standardized AKM manager fixed effects (SD units); higher = more shrinkage/turnover/absence. Line = OLS fit; r = Pearson correlation of raw FEs (matches Table A Panel A). JD-GDA general-manager panel.

Figure 20: Manager fixed effects: core shrinkage effect against each other outcome

Notes: Each panel plots the manager's standardized AKM fixed effect for log shrinkage value (horizontal) against her fixed effect in one other outcome (vertical); $N = 497$ general managers. The line is the OLS fit and r the Pearson correlation of the raw fixed effects (matching Figure 13).

E The Mechanism of the Complementarity: The Full Battery

Section 10.3 adjudicates between two readings of the wage-premium complementarity by asking whether the manager’s conduct scales with the rent, and reports a compact summary of the evidence. This appendix collects the full battery behind that summary, and adds a sharper cut at the composition channel. Both exercises use the high-type classification of Section 8 and the arrival design of Equation 35, replacing the shrinkage outcome with each mechanism in turn and interacting the $\text{Post}_{[0,8]}$ indicator with the wage environment of the store’s municipality at arrival.

E.1 Every Mechanism, Crossed with the Premium

Table 32 takes each margin examined in Section 9 and estimates its high-type arrival effect separately in low- and high-premium stores, alongside the difference between the two and its p -value. The table groups the margins into six panels: the organization of monitoring (Panel A), the composition and Harrison–Klein diversity of the team (Panel B), the internal reallocation of workers (Panel C), the manager’s own presence (Panel D), the enforcement she exercises (Panel E), and the shrinkage return (Panel F). The first four panels collect the manager’s inputs; the last two collect the enforcement and the return.

The pattern is one-directional and confirms the reading of the main text. The manager’s inputs do not scale with the premium. The organization of monitoring is flat—the loss-prevention, supervisory, and security shares of the workforce move by statistically indistinguishable amounts in the two environments, with the single exception of the supervisory share, whose differential is modest and isolated among the two dozen input margins in the table. Team composition and diversity are flat throughout Panel B: neither the size, the gender or occupational mix, nor any Harrison–Klein construct responds to the premium. The internal reallocation of Panel C—rotation, upward moves, and cross-area moves—rises on a high-type arrival, as Section 9.2 documents, but the rise is no larger where the premium is high. Most tellingly, the manager’s own presence in Panel D does not scale: overall co-presence, the presence targeted at the shrinkage-prone roles—the loss-prevention chain, cashiers, perishables, and security—and presence in the low-oversight night and weekend hours all move by amounts that are statistically indistinguishable across the premium. The faint upward tilts that survive—overall co-presence and manager shifts drift a little higher where the premium is high—do not reach the sharpness of the enforcement and return margins, and they run against, not with, the reallocation of attention that the supply reading

would require. The manager deploys the same supervisory technology regardless of the rent.

What scales is the enforcement and the return. The disciplinary instrument the manager controls day to day—the suspension—rises after a high-type arrival only where the premium is high (+0.31 per hundred workers in high-premium stores against -0.14 in low-premium, a differential of 0.45 significant at five percent), while the dismissal, which the manager does not control as directly, does not respond. The shrinkage reduction is two to three times larger in high-premium stores on all three measures. I read the contrast between the flat inputs of Panels A–D and the scaling of Panels E–F as direct evidence for the second reading: the complementarity is the return to the manager’s presence rising with the rent, together with the more frequent use of the sanction that presence makes credible, not a reallocation of the manager’s attention toward high-rent stores.

Table 32: The full channel of the complementarity: the manager's inputs do not scale with the premium, the enforcement and the return do (Part 1: organization, composition, reallocation)

	Low-premium (LWP)	High-premium (HWP)	Difference (HWP–LWP)	$p(\text{diff})$
<i>Panel A. Monitoring organization</i>				
Loss-prevention staff /100	0.340** (0.161)	0.182 (0.187)	-0.158 (0.230)	0.492
Supervisory share (%)	-0.503 (0.407)	0.931 (0.629)	1.434** (0.636)	0.024
Security-staff share (%)	0.117 (0.183)	0.038 (0.289)	-0.079 (0.295)	0.789
<i>Panel B. Team composition and Harrison–Klein diversity</i>				
Log team size	0.001 (0.011)	0.023 (0.019)	0.022 (0.019)	0.237
Female share (%)	-1.277*** (0.456)	-0.471 (0.560)	0.805 (0.631)	0.202
Sales-floor share (%)	-0.079 (0.377)	-0.189 (0.676)	-0.110 (0.699)	0.874
Gender variety (Blau)	0.003** (0.001)	0.001 (0.002)	-0.002 (0.003)	0.362
Tier variety (Blau)	-0.002 (0.003)	0.005 (0.005)	0.007 (0.005)	0.137
Occupational-area variety (Blau)	0.000 (0.003)	0.006 (0.005)	0.006 (0.006)	0.325
Age dispersion (SD)	0.069 (0.079)	0.093 (0.144)	0.024 (0.157)	0.880
Tenure dispersion (SD)	-0.363 (0.504)	-0.315 (0.689)	0.048 (0.761)	0.950
Education dispersion (SD)	-0.038 (0.026)	-0.013 (0.024)	0.025 (0.032)	0.435
Pay dispersion (CV)	0.010 (0.007)	0.000 (0.009)	-0.010 (0.011)	0.368
<i>Panel C. Internal reallocation</i>				
Internal rotation /100	0.132 (0.207)	0.349* (0.204)	0.217 (0.289)	0.453
Upward moves /100	0.208** (0.084)	0.131 (0.082)	-0.077 (0.115)	0.504
Cross-area moves /100	0.173 (0.112)	0.270* (0.155)	0.097 (0.190)	0.608

Part 1 of 2; see the notes to Part 2 (Table 33).

Table 33: The full channel of the complementarity: the manager’s inputs do not scale with the premium, the enforcement and the return do (Part 2: presence, enforcement, and the return)

	Low-premium (LWP)	High-premium (HWP)	Difference (HWP–LWP)	$p(\text{diff})$
<i>Panel D. The manager’s own presence</i>				
Manager–worker co-presence (share)	0.020 (0.014)	0.063*** (0.022)	0.043 (0.026)	0.102
Any manager overlap (share)	0.023 (0.016)	0.077*** (0.025)	0.054* (0.031)	0.078
Manager shifts per month	0.916 (0.585)	3.041*** (0.913)	2.126* (1.126)	0.059
Co-presence: loss-prevention chain	0.022 (0.017)	0.045* (0.024)	0.023 (0.028)	0.414
Co-presence: cashiers	0.021 (0.015)	0.060*** (0.022)	0.040 (0.027)	0.135
Co-presence: perishables	0.012 (0.016)	0.069** (0.027)	0.057* (0.031)	0.064
Co-presence: security	0.018 (0.012)	0.049** (0.021)	0.031 (0.024)	0.197
Manager presence, night hours (%)	0.017 (0.013)	-0.003 (0.017)	-0.020 (0.020)	0.309
Manager presence, weekend hours (%)	-0.000 (0.004)	-0.007 (0.006)	-0.006 (0.007)	0.379
<i>Panel E. Enforcement</i>				
Suspensions /100	-0.140 (0.139)	0.311** (0.153)	0.451** (0.181)	0.013
Dismissals /100	-0.008 (0.090)	0.019 (0.124)	0.027 (0.145)	0.852
<i>Panel F. Shrinkage (the return)</i>				
Log shrinkage value	-0.043** (0.017)	-0.126*** (0.032)	-0.084** (0.037)	0.022
Log shrinkage / sales	-0.015 (0.021)	-0.076*** (0.027)	-0.061* (0.033)	0.065
Log shrinkage / worker	-0.037** (0.018)	-0.138*** (0.035)	-0.101*** (0.038)	0.008

Each row is a separate high-type arrival regression with the indicated store-month outcome, following the arrival design of Equation 35 (store, calendar-month, and year fixed effects; standard errors clustered at the store; ever-treated stores). The wage environment is fixed at arrival (high-premium, HWP, = a wage premium above the municipality’s median). The first two columns report the arrival effect in low- and high-premium stores, with stars reflecting each effect’s own significance; the third column is the difference between them and the last is the p -value that the two differ—the premium-scaling test. Rates are per hundred workers. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

E.2 Within- and Between-Area Diversity

The evidence that the manager’s inputs are flat across the premium closes one door that the reallocation results of Section 9.2 might seem to leave open. High-type managers reassign workers across occupational areas, and a reshuffle across areas can in principle change team diversity *within* each area even while leaving it unchanged at the store level: mixing areas raises within-area heterogeneity and lowers between-area heterogeneity, with the store total held fixed (Harrison and Klein, 2007). Table 34 tests this directly. I rebuild each Harrison–Klein construct *within* occupational area—the separation measures (the standard deviation of age, tenure, and education), the variety measures (the Blau index of gender and of tier), and the disparity measure (the coefficient of variation of pay)—as the headcount-weighted mean across areas, together with the corresponding between-area dispersion, and run each through the arrival design. This exercise asks only whether the reshuffling changes composition inside each area, so it is not crossed with the premium.

A high-type arrival moves none of them. Within-area age, tenure, and education dispersion, within-area gender and tier variety, within-area pay dispersion, and their between-area counterparts are all unchanged, and the low-type arrival, reported for contrast, moves them no more. The only movement to approach conventional significance is within-area education dispersion, which edges down on either arrival by a similar amount and so does not distinguish the high-type manager. The composition of the workforce is intact not only at the store level but inside each team the manager runs. I interpret this as evidence that the manager deploys the workers she inherits rather than recomposing them at any level of aggregation, consistent with the flat composition inputs of Table 32.

Table 34: Within- and between-area diversity do not change on a high-type arrival

Diversity measure	HT arrival	LT arrival
Within-area age SD	0.0208 (0.0956)	-0.0550 (0.0866)
Within-area tenure SD	-0.1331 (0.4156)	-0.4963 (0.4598)
Within-area education SD	-0.0483* (0.0247)	-0.0470** (0.0218)
Within-area gender variety (Blau)	0.0032 (0.0035)	0.0017 (0.0034)
Within-area tier variety (Blau)	0.0015 (0.0035)	0.0018 (0.0035)
Within-area pay CV	0.0034 (0.0050)	-0.0054 (0.0036)
Between-area age SD	0.0969 (0.1606)	-0.0927 (0.1409)
Between-area tenure SD	-0.9678 (0.6275)	0.1094 (0.4877)
Between-area pay SD	102.4159 (166.3813)	218.5141 (161.2184)

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a separate store-month arrival regression following Equation 35 (store, calendar-month, and year fixed effects; standard errors clustered at the store; ever-treated stores). Within-area measures are the headcount-weighted mean across occupational areas of each within-area Harrison–Klein construct (Harrison and Klein, 2007); between-area measures are the standard deviation of area means across areas. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

E.3 The Complementarity at the Worker-Shift Level

The complementarity so far lives at the level of the store’s aggregate losses. If it is the reinforcement that Shapiro and Stiglitz (1984)- and Coviello et al. (2022)-style models describe—the manager’s supervision and the worker’s rent combining in the production of *effort*—it should also be visible in how individual workers respond to being watched, at the daily frequency the store-month results aggregate over. Appendix D.4 isolated one clean worker-level response of that kind: punctuality—the same worker arrives closer to her own norm on days the general manager is on the clock. Here I ask whether that punctuality response to supervision is itself stronger where the premium is high. I return to the shift-level discipline panel and interact the manager’s co-presence with the store’s high-premium status, absorbing worker, store $\times$ month, arrival-hour, and day-of-week fixed effects, so that identification

is the same worker, at the same time of day, on days the manager is present versus absent, compared across high- and low-premium environments.

Table 35 reports the probe, and the evidence is suggestive rather than decisive. On the clean punctuality margin the response to supervision is about twice as large in high-premium stores: co-presence pulls arrivals 1.4 minutes closer to the norm where the premium is low and 2.8 minutes where it is high, the direction the complementarity predicts. The interaction itself, however, is imprecise ($p \approx 0.11$), and the continuous-premium version is not consistently signed, so the gradient is a hint rather than a result. (The table reports the early-departure column for completeness; as in Appendix D.4 that margin is entangled with shift structure and carries little weight.) I therefore read the effort-margin evidence as consistent with, not independent confirmation of, the complementarity: the clean version of the result lives at the store level (Table 17), and the daily worker-shift data can only hint that the same reinforcement operates on the effort margin where the theory locates it. Establishing it would require richer premium variation at the worker level or a higher-frequency effort measure than punch-clock timing. **[JS: I think we need to have a unified set of core measures of shirking and present results for all of them in parallel. Fine if you want to have late arrival as one of the measures but then we should include it throughout.]**

Table 35: The wage $\times$ supervision complementarity at the worker-shift level (suggestive)

	Late arrival (min)	Early departure (min)
GM co-present, low-premium store	−1.36* (0.74)	−13.10*** (1.04)
× high premium (HWP)	−1.42 (0.88)	1.20 (0.99)
GM co-present, high-premium store (total)	−2.78*** (0.81)	−11.90*** (1.02)
GM co-present $\times$ continuous premium (z)	1.78 (1.15)	−1.49 (1.28)
Worker-shifts	≈ 8.7 million	

Worker-shift regressions on the discipline panel of Appendix D.4. “GM co-present” indicates the manager’s presence interval overlaps the shift; outcomes are deviations (minutes) from the worker’s own median arrival and departure. The $\times$ HWP row is the additional effect in high-premium stores and the total their sum; the last row uses the continuous premium z -score. Worker, store $\times$ month, arrival-hour, and day-of-week fixed effects; standard errors clustered at the store. This probe does not depend on the manager classification. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

F Wage-Premium Complementarity: Robustness

This appendix reports in full the two robustness exercises summarized in Section 10.4: the pooled triple-difference that estimates all four manager-type $\times$ premium cells jointly, and the comparison of the high-type interaction across four definitions of the wage environment.

F.1 The Pooled Triple-Difference

Pooling the first arrival of either type into a single regression and estimating the full triple interaction,

$$Y_{jt} = \beta_1 \text{Post}_{jt} + \beta_2 \text{Post}_{jt} \times \text{HT}_j + \beta_3 \text{Post}_{jt} \times \text{HWP}_j + \beta_4 \text{Post}_{jt} \times \text{HT}_j \times \text{HWP}_j + \varphi_j + \lambda_{m(t)} + \xi_{y(t)} + \varepsilon_{jt}, \quad (37)$$

reproduces the per-type message cell by cell (Table 36): the only cell in which arrivals reduce shrinkage is high-type-in-high-premium, low-type arrivals in low-premium stores raise it, and the remaining cells are indistinguishable from zero. The triple-difference coefficient β_4 carries the predicted negative sign for shrinkage value and shrinkage per worker but is imprecise, because high-premium arrivals are scarce: the pivotal high-premium cells rest on only about two dozen stores each. I therefore rest the powered test on the per-type interactions of the main text, which use every arrival.

Table 36: Pooled first arrivals: manager type $\times$ wage environment (DDD)

	(1) Log shrink value	(2) Log shrink/ sales	(3) Log shrink/ worker	(4) Log NJ abs. p.e.	(5) Log turnover p.e.
Arrival $\times$ HT $\times$ HWP (DDD)	−0.060 (0.065)	0.007 (0.056)	−0.045 (0.069)	−0.037 (0.108)	0.244 (0.203)
<i>Cell effects (linear combinations):</i>					
HT arrival in HWP	−0.085** (0.035)	−0.016 (0.032)	−0.093** (0.039)	−0.087 (0.087)	0.441*** (0.135)
HT arrival in LWP	0.005 (0.017)	0.043** (0.019)	0.008 (0.020)	0.033 (0.030)	0.356*** (0.078)
LT arrival in HWP	0.006 (0.053)	0.012 (0.040)	−0.008 (0.057)	−0.044 (0.063)	0.104 (0.148)
LT arrival in LWP	0.036 (0.023)	0.077*** (0.022)	0.048** (0.024)	0.039 (0.028)	0.262*** (0.087)

Pooled first arrival of either type; HT = type of the first arriving spell; HWP = wage environment at arrival. Top row: the triple-interaction coefficient β_4 of Equation (37); lower rows: the total effect in each cell (linear combinations). Store, calendar-month, and year fixed effects; standard errors clustered at the store. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

F.2 Alternative Definitions of the Wage Environment

The complementarity should be a property of the economics, not of any one way of defining the wage environment. I therefore re-estimate the high-type interaction under three additional definitions, all fixed at the arrival month (Table 37), each using a different slice of the premium’s variation. The first is a cross-sectional split—whether the store’s municipality lies above the same-quarter median premium of *other* municipalities—which by construction carries no time trend. The second uses the standardized premium itself as a continuous moderator, in the spirit of the continuous relative wage of Cappelli and Chauvin (1991). The third replaces the premium with a cost-of-job-loss measure in the tradition of Schor and Bowles (1987), scaling the outside wage by the reemployment probability implied by the municipal unemployment rate. The complementarity holds under all four definitions. Under the cross-sectional split the high-type effect on shrinkage value is -8.0 percent in high-premium against -5.2 percent in low-premium stores; each standard deviation of the arrival premium amplifies the effect on shrinkage-to-sales by about 3.4 percentage points; and each standard deviation of the cost of job loss by about 4.7. The magnitudes move with how much of the premium’s variation each definition uses, as they should; the sign and the ordering of the two environments—the complementarity itself—are invariant across every definition, which is the mark of a result that belongs to the setting rather than to a specification.

Table 37: High-type arrivals under four definitions of the wage environment

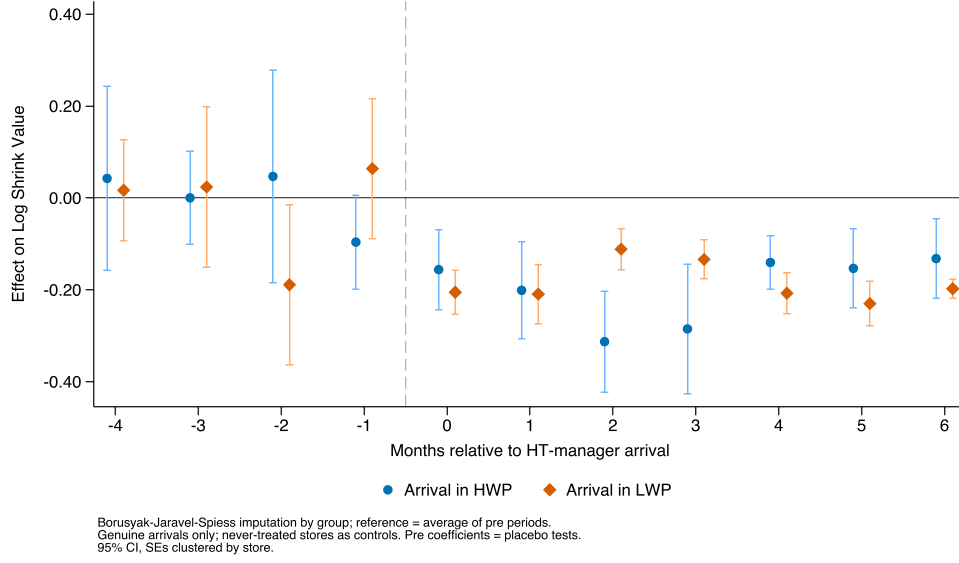
	(1) Log shrink value	(2) Log shrink/ sales	(3) Log shrink/ worker	(4) Log NJ abs. p.e.	(5) Log turnover p.e.
<i>A. Binary, within-municipality median (main definition)</i>					
HT arrival in HWP	−0.126*** (0.032)	−0.076*** (0.027)	−0.138*** (0.035)	−0.049 (0.053)	0.309*** (0.115)
Difference (HWP − LWP)	−0.084** (0.037)	−0.061* (0.033)	−0.101*** (0.038)	−0.066 (0.064)	0.194 (0.126)
<i>B. Binary, cross-sectional (vs. same-quarter median of other municipalities)</i>					
HT arrival in HWP	−0.080*** (0.015)	−0.049** (0.019)	−0.086*** (0.018)	−0.017 (0.030)	0.209** (0.089)
Difference (HWP − LWP)	−0.029 (0.026)	−0.036 (0.027)	−0.045 (0.029)	−0.035 (0.038)	0.086 (0.092)
<i>C. Continuous premium (standardized)</i>					
HT arrival at mean premium ($z = 0$)	−0.073*** (0.015)	−0.041** (0.017)	−0.074*** (0.017)	−0.003 (0.026)	0.190** (0.076)
Gradient per +1 SD of premium	−0.024 (0.017)	−0.036** (0.018)	−0.030 (0.019)	−0.001 (0.033)	0.078 (0.071)
<i>D. Continuous cost of job loss (standardized)</i>					
HT arrival at mean cost ($z = 0$)	−0.081*** (0.017)	−0.059*** (0.020)	−0.083*** (0.022)	−0.025 (0.033)	0.193** (0.085)
Gradient per +1 SD of cost	−0.029 (0.025)	−0.055** (0.025)	−0.035 (0.029)	−0.039 (0.045)	0.053 (0.092)

Notes: All moderators fixed at the arrival month; high-type classification as in Section 8. Panel A repeats the main table. Panel B: municipality above the same-quarter median of other municipalities. Panel C: standardized premium. Panel D: outside wage scaled by the reemployment probability implied by the municipal unemployment rate (smaller sample, as the unemployment rate is missing for some municipality-quarters). Store, calendar-month, and year fixed effects; standard errors clustered at the store. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

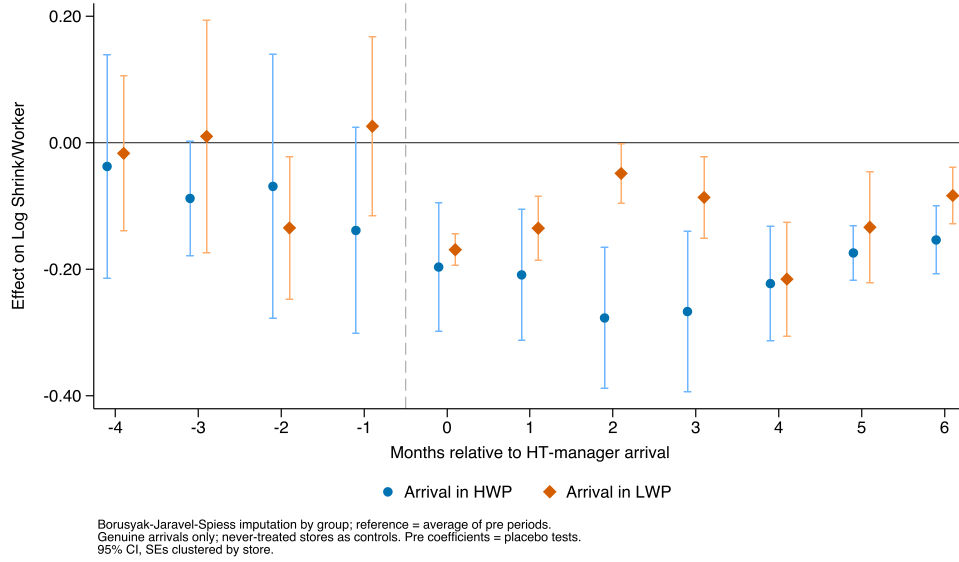
F.3 Staggered-Robust Dynamics by Wage Environment

The main-text interaction is a two-way fixed-effects design, and the same concern about staggered timing that motivates the estimator comparison of Appendix C applies here. I therefore re-estimate the arrival dynamics separately for high-type arrivals in high- and low-premium environments with the imputation estimator of Borusyak et al. (2024), on the genuine (non-left-censored) arrivals with never-treated stores as controls. Figure 21 plots the two paths, and two features matter. The pre-arrival placebos are flat and centered on zero in both environments, so the corrected design is valid; and on impact and through the post-window the high-premium path lies below the low-premium path—the same ordering the binary table reports, now under a heterogeneity-robust estimator. I do not read the gap as separately identified: with only about two dozen genuine high-premium arrivals, the two series’ confidence intervals overlap, so the powered test of the differential remains the

full-sample per-type interaction of the main text (Table 17). What the figure adds is that the direction of the complementarity survives the correction for staggered timing.



(a) Log shrinkage value



(b) Log shrinkage per worker

Figure 21: High-type arrival dynamics by wage environment (Borusyak–Jaravel–Spiess imputation)

Notes: Imputation event study (Borusyak et al., 2024) estimated separately by wage environment at arrival; reference = average of pre-treatment periods; never-treated stores serve as controls; genuine (non-left-censored) arrivals only. Pre-arrival coefficients are placebo tests. 95 percent confidence intervals; standard errors clustered at the store. The high-premium and low-premium series overlap, so the differential is not separately identified here; the powered test is Table 17.

G Manager Type Does Not Moderate the Compression

Section 11 moderates the compression by the manager’s floor presence and reports that her estimated *type* does not moderate it. This appendix gives that result in full, in the two ways one can measure type. The first is the discrete high/low-type label the paper uses throughout—the exact parallel of the presence split—which delivers a clean null. The second is the AKM shrinkage fixed effect of Section 7 used continuously, where a right-signed gradient appears but fails its own pre-trend test and so cannot be read causally. Both reach the same conclusion: what moderates the squeeze is what the manager does, not the type she is estimated to be.

The discrete type split. I split the border stores exactly as the presence design does, but on manager type: high- versus low-type by the AKM shrinkage fixed effect, classified within format-by-border cells—the paper’s `htm2` label, the same high/low split the manager sections use—and fixed at the general manager in place the December before each hike. The two-phase specification is otherwise identical. Figure 22 reports it. The compression lands on both type groups—a late rise of 5.0 percent for low-type stores ($p = 0.02$) and 7.6 percent for high-type ($p < 0.01$)—and the high-minus-low difference is +2.6 points (s.e. 3.6, $p = 0.47$), small and, if anything, the wrong sign for a protective role. The split is clean: on the net-of-sales outcome the pre-hike coefficients are jointly flat in both groups (low-type $p = 0.44$, high-type $p = 0.65$), the same check the presence groups pass. Manager type, measured the way the rest of the paper measures it, simply does not blunt the compression.

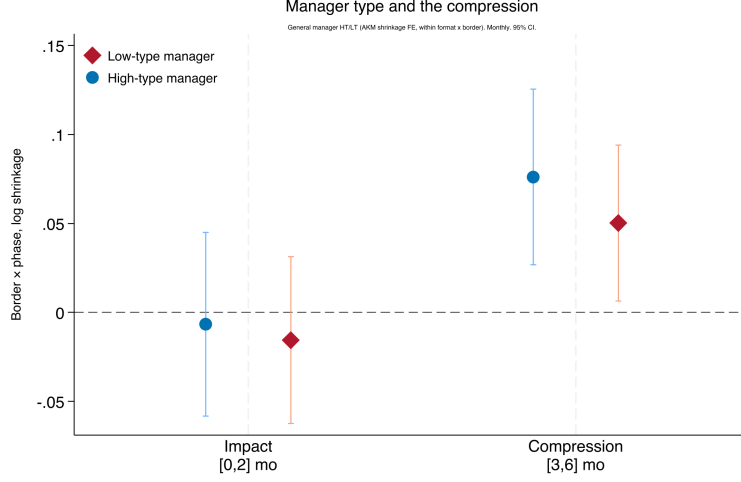


Figure 22: The compression response by the manager’s estimated type at the hike. Border stores split at the median AKM shrinkage fixed effect within format-by-border cells (the `htm2` high/low-type label), the general manager fixed in the December before each hike. Both type groups show a significant compression (+5.0 percent low-type, +7.6 percent high-type), and their difference is insignificant (+2.6 points, $p = 0.47$); pre-hike coefficients are jointly flat in both groups on the net-of-sales outcome. 95 percent confidence intervals.

The continuous fixed effect. I take the monthly stacked border design of Section 5 unchanged—store-by-hike and format-by-calendar-month fixed effects, the November–December reference, and the impact $[0, 2]$ and compression $[3, 6]$ phases—and add the continuous moderator. Let Q_k be the standardized quality of store k ’s general manager in the December before the hike, defined as the negative of her AKM shrinkage fixed effect, so that a higher Q marks a better manager, one whose stores lose less inventory.⁹ The triple interaction $\text{Border} \times \text{Compression} \times Q$ measures the gradient of the late-phase border effect in manager quality; the monitoring story predicts a negative sign—a better manager should blunt the compression damage.

Result. At face value the gradient obliges: it is negative and statistically strong for every shrinkage measure (-0.096 , s.e. 0.025 , for shrinkage net of sales; similar for value and units), the sales placebo is flat, and a tercile version shows the late border rise concentrated under the worst-ranked managers. But the design flunks the check that gives such gradients their meaning: interacting the *pre-hike* event-time indicators with $\text{Border} \times Q$

⁹The fixed effect is the per-manager effect from the log-shrinkage-value decomposition of Section 7; a median split on it recovers the high/low-type label used in the manager sections up to the border dimension, and managers labeled high-type carry the lower shrinkage effect, confirming the sign. The manager is the general manager in place in the December before each hike, predetermined with respect to the January shock; Q is defined for 122 border store-episodes, more than the binary split retains.

yields jointly significant coefficients (net-of-sales $p = 0.004$, value $p = 0.001$)—border stores run by managers with different fixed effects were already on systematically different paths before the hikes arrived. The contamination is worst in the 2024 episode, whose distant pre-window is problematic even unconditionally (joint $p < 0.001$), and remains marginal in the clean 2023 episode ($p = 0.083$). A moderation that rides on differential pre-paths cannot be read causally: with the fixed effect estimated from the same shrinkage histories that generate those paths, the apparent buffering is indistinguishable from mean reversion in the label itself. This failure is the border contamination of the AKM person effect that motivates the paper’s within-format-by-border classification (Section 7), surfacing here as a failed identification check rather than a usable result.

Reading. The contrast with the presence moderator is the point. The presence split is predetermined, measured from time-clock data rather than from outcome histories, and passes the same pre-trend test in both groups (net-of-sales joint $p = 0.41$ and $p = 0.36$ for low- and high-presence stores). The manager’s estimated *quality* cannot be separated from the outcome paths it was estimated on; what she *does*—her presence on the floor, characterized in Section 9—can, and it is the moderator the paper rests on.

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